

ASSESSMENT BRIEF - COVER SHEET

Course: BSc Computing (All pathways)		Year: 1	CSY1019 - Software Engineering 1																				
PJ1		Title: System Specification, Analysis and Design																					
Date due out: 22/11/2021	Date due in:	Extension date:	Extension agreed by:																				
Group Name: <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th style="width: 60%;">Group Member Names</th> <th style="width: 40%;">Student ID's</th> </tr> <tr><td>Sushila Timalina</td><td>21422893</td></tr> <tr><td>Rabindra Prasad Joshi</td><td>21422047</td></tr> <tr><td>Rajiv Shrestha</td><td>21422063</td></tr> <tr><td>Amir Tamang</td><td>21422018</td></tr> <tr><td>Sudarshan Thapa</td><td>21422045</td></tr> <tr><td>Pawan Bhusal</td><td>21422143</td></tr> <tr><td>Sudarshan Oli</td><td>21422056</td></tr> <tr><td>Krishna Oli</td><td>21422057</td></tr> <tr><td>Siddharth Yonzon Tamang</td><td>21422134</td></tr> </table>		Group Member Names	Student ID's	Sushila Timalina	21422893	Rabindra Prasad Joshi	21422047	Rajiv Shrestha	21422063	Amir Tamang	21422018	Sudarshan Thapa	21422045	Pawan Bhusal	21422143	Sudarshan Oli	21422056	Krishna Oli	21422057	Siddharth Yonzon Tamang	21422134	Tutor: Mark Johnson/Bijayashri Barun	
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ASSESSMENT FEEDBACK:

RATING SCALE	A+ - A-	B+ - B-	C+ - C-	D+ - D-	F+ - F-
Formulated Aims and Objectives (10%)					
Problem Domain Elicitation Activities and Presented Findings (10%)					
Requirement Specification Document (20%)					
System Interface Design Documentation (20%)					
Systems Architecture Analysis and Design (20%)					
Client Presentation (10%)					
Report Quality – Presentation, format and use of English (10%)					
Specific aspects of your assignment that the marker likes:		Specific aspects of your assignment that need more work:			
Tutor's Signature:		Date:		Grade:	

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1. Introduction

Claybrook Zoo is a little family versed zoo situated within the Northwest of Britain. Since its inception in 1965, this zoo has entertained residents of the area while also providing educational assistance to students and others in need. Not only that, it has been popular for its astounding visitor-friendly environment and top-level administrations that are well recognized by everybody and the number of visitors has always looked healthy until recently.

All the animals in the zoo are managed and kept in a proper way. All animals in the zoo are currently in some specific areas. These are:

- The Aviary (Where all the birds are housed)
- The Hothouse (Where all reptiles and amphibians reside)
- The Aquarium (For all fish within the zoo)
- The Cages/Compounds. (Housing all mammals).

Matthew Jones is the current manager of the zoo, who is monitoring the decline in reputation and seriously investigating the issue. He has recognized certain big obstacles that are preventing the zoo from performing its best, with the only goal of rising up like the good old days.

1.1 Project Background

This project aims to develop a computerized solution for Claybrook Zoo, which currently uses a clerical system to manage the zoo's information. The process will be client-driven, with a strong emphasis on customer satisfaction. It will include information on the animal database, such as species, gender, description, and location, as well as many attributes important to specific animal kinds, such as mammals, birds, fish, and reptiles. Visitors to the zoo will be able to get this information via the system.

The essential background information is provided in the form of a requirements document produced by the client. Required features provided by customer-provided documentation include the ability to add, delete, or search animal records at the zoo, as well as the ability for visitors to retrieve related information. It also includes ability to search using multiple criteria, the inclusion of an animal sponsorship scheme, the inclusion of location, internet-enabled animal catalogue pages for visitors, and related security features. All of these elements will be included in the final result, which will be an online web application.

The current zoo manager has made contact with our software development company to learn more about the possibility of creating a software suite product, including:

- A zoo website that provides information to visitors.
- Technologies for interacting with visitors
- A zoo records administration/content management system that will allow the zoo to successfully handle essential business records.

So, to develop a software as a customer requirement we formed 10 members team named TOKYO_MANJI to complete the task and provide our best wireframe design to our customer. As our client has proposed the prerequisites for the site. However, we need further information

regarding the zoo so we scheduled an interview with several of the zoo's clients to obtain appropriate information about the zoo.

1.2 Project Aims and Objective:

The main aim of this project was to discuss the features and requirements provided by the clients and stakeholders at Workshop and to develop the online web application, mobile application and to transform the existing clerical system into a computerized system. To achieve this requirement, the goal can be set as follows:

- Examine and understand the information provided, such as briefings and documentation
- A clear understanding of the problem domain through the various requirements engineering processes.
- Using workshops, gather the essential issue domain information and gain a better understanding of the needs.
- Create a list of appropriate questions for key stakeholders to ask to better understand the problem area and possible solution for the system.
- Create a specification document that includes all of the knowledge and information gained through requirements specification documentation.
- Conduct design workshops involving relevant stakeholders and proceed the prototype to the next stage of development.
- Implement all of the basic and essential functionalities to meet the client requirements through a process of consultations and confirmations.
- Test and evaluate the system including clients if necessary.
- Check the final system and present it to the customer.

The project's goal is to ensure that the final version of the website to have all of the features and functionalities that the client requires. Since the website is designed to be user-friendly, it should also have the following features:

- Responsive and Instructive
- Simple to use and accessible to users of all ages
- Perfectly Function like (effective navigation and search box)

1.3 Project Development Methodology

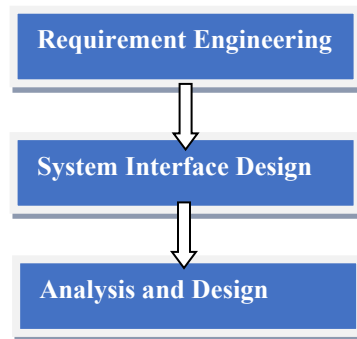
Following the objectives described previously in this report, our team will adapt multiple approaches to develop a strong and user-friendly system for this project. Through numerous researches, we will first get a complete understanding of the problem area. Then we will take a closer look at the other systems that are similar to the one we're developing in order to obtain some inspiration. This will allow us to gain a better understanding of the expected functional characteristics.

We'll also describes the laws that apply to animals, software, and data. Then we'll try to figure out the classes and objects in the system and attempts to create a trusted data system for them. Following that, we'll design the wireframe, and then design the UI based on feedback from stakeholders, with suitable functional behaviour and consistent performance. After completion

all those things, we will test our result in order to be sure that whatever we were told is made and is functioning properly.

Then finally we will be validating all our works to increase its standard. Validating it with all laws and after validating, we will be handling our whole solution to our clients.

Our project process is broken down into phases as shown in the chart below:



2. Requirement Engineering

Requirement Engineering is the step where we investigate and explain problem areas and requirements, following designs and documents the properties of the solution system that meets these requirements. Different procedures have been provided to assist us in achieving our goal of obtaining requirements. Elicitation activities include the sub-sections, which describes interview strategies, interview outcomes, problem domain description, similar software system review, and formulation of appropriate laws. It also describes functional requirements, performance requirements, design constraints, and commercial constraints.

The goal of requirement engineering is to develop and maintain sophisticated and descriptive 'System Requirements Specification' document. To create the system, requirements engineering includes a number of activities. The four step of Requirement Engineering process are:

- 1) Feasibility study
- 2) Requirement gathering (Elicitation)
- 3) Software requirement specification
- 4) Software requirement validation

Feasibility study:

- When the client approaches the organization for getting the desired product developed, it comes up with a rough idea about what all functions the software must perform and which all features are expected from software.
- Referencing to this information, the analysts do a detailed study about whether the desired system and its functionality are feasible to develop.

Requirement gathering:

- If the feasibility report is positive towards undertaking the project, next phase starts with gathering requirements from the user. Analysts and engineers communicate with the client and end-users to know their ideas on what the software should provide and which features they want the software to include.

Software requirement specification:

- Software requirement specification is a document created by system analyst after the requirements are collected from the various stakeholders.
- SRS defines how the intended software will interact with hardware, external interfaces, speed of operation, response time of system, portability of software across various platforms, maintainability and speed of recovery after crashing. Security, quality limitations, etc.
- The requirements received from the client are written in natural language. It is the responsibility of the system analyst to document the requirements in technical language so that they can be comprehended and used by the software development team.

■ Features for SRS:

Software requirement specification should come up with the following features:

- User requirements are expressed in natural language.
- Technical requirements are expressed in structured language, which is used inside the organization.
- Design the description should be written in pseudo code.
- Format of Forms and GUI screen prints.
- Conditional and mathematical notations for data flow diagrams etc.

Software requirement validation:

- After requirements specifications are developed, the requirements method in this document are validated. User might ask for illegal, impractical solutions or experts may interpret the requirements inaccurately. This result in huge increase in cost if not nipped in the bud. Requirements can be checked against following conditions:
- If they can be practically implemented.
- If they are valid and as per functionality and domain of the software.
- If they are ambiguities.
- If they are complete.
- If they can be demonstrated.

2.1 Elicitation Activities

Elicitation Activities is concerned about collecting information. It is additionally known as requirement gathering or requirements acquisition. In this stage, we obtain information from the client about the problem and the problem domain and create a requirements document and share this information on to the rest of the development group members.

The records management component of the system must have a sufficient, high-quality, intuitive, and user-friendly graphical interface as part of the minimum prototype criteria. The software solution must have the following features: -

- Features for adding or removing animals from the zoo on the website.
- It needs to appropriate the map location and filtered search results like (animal species, habitats, characteristics, etc.).
- The ability to search the zoo catalogue, and display the appropriate search results, for a variety of user-initiated searches (e.g., animals by category, by type, by habitat, etc).

Each animal in the system will require a unique identity, which is thought to be based on the animal classification. Furthermore, we are required to elicit the following background supplement requirement. The electronic catalogue will need to maintain track of the following key data for all animal location in the zoo: -

- The Aviary (Where all the birds are housed)
- The Hothouse (Where all reptiles and amphibians reside)
- The Aquarium (For all fish within the zoo)
- The Cages/Compounds. (Housing all mammals).

The electronic catalogue will need to maintain track of the following key data for all animal location in the zoo: -

- Species Name.
- Animal “given” Name.
- Date of Birth of Animal.
- Gender.
- Average Life Span (e.g., months/years).
- Species Category/Classification.
- Dietary Requirements.
- Natural Habitat Description.
- Global Population Distribution.
- Date animal joined the zoo.
- Typical animal dimensions (e.g., height, weight).

In addition, certain animal species require specific information. This includes:

Birds

- Nest Construction Method.
- Clutch size.
- Wing span.
- Ability to fly (Yes/No).
- Plumage colour variants (possibly by gender).

Fish

- Average body temperature.

- Water type (e.g., Salt/Fresh).
- Colour variants (possibly by gender).

Reptiles and Amphibians

- Reproduction type (e.g., egg layer/livebearer).
- Average Clutch Size (Where reptile/amphibian is an egg layer).
- Average number of offspring (Where reptile/amphibian is a livebearer).

Mammals

- Gestational Period.
- Mammal Category (e.g., Prototheria/Metatheria/Eutherian.)
- Average body temperature.

2.1.1 Interview Plans

We have interviewed with the following Stakeholders:

- Matthew Jones:** The current zoo manager and expert on existing organisational procedures and practice.
- Thomas Smith:** An existing, regular visitor to the zoo, who has a good insight into the current visiting experience available for general members of the public.
- Jonathan Rodgers:** A senior administrator within the zoo, who is responsible for running the animal sponsorship scheme.
- Phillip Brown:** A local small business owner who is interested in the animal sponsorship scheme if he can perceive potential benefits by way of advertising/positive publicity available through the scheme.

Matthew Jones (Zoo Manager):

1. Firstly, is there any documentation, staff manuals, or business-related items we might review to get a quick overview of the zoo's current operations? If so, could you please provide us these documents?
2. What is the major aim and expectation of developing new software?
3. What motivated you to go from a handwritten to a computerized accounting system?
4. Do you have a vision for your new program or any additional features that you'd want to include that your competitors don't?
5. Why did you choose to collaborate with us on software development?
6. Although English is a default language would you like to add another language?
7. Which features of accessibility, data gathering, and interactivity should the present software package support?
8. Would you prefer kiosk system to be installed? If yes, what type of features would you prefer?

9. How do you keep track of visitors on a daily basis, categorizing them as children, adults, or the elderly? Would you like new software to assist you keep track if it's difficult to do so?
10. How do you maintain track of your employees' and animals' records, accounting, and data?
11. In navigation would you like to have a virtual tour for visitors?
12. How do you like to highlight the events?
13. When would you like your system to be updated, and how often would you like it done?
14. Is there a probability that this software will increase the number of visitors?
15. What is the software's estimated budget and completion date, and when is it expected to be finished?
16. Is there anything you think we should have discussed that we didn't?

Thomas Smith (Visitor):

1. What differences have you observed since you began going, how frequently do you go, and how long have you been going?
2. As a frequent visitor, are you satisfied with the present system? What changes would you like to see that would improve your visit experience?
3. What type of website layout do you prefer, and what kind of content do you believe the zoo's new official website should include?
4. Would you like to keep some of the current system ideas?
5. What changes should be made to the navigation, website, and new technologies to provide visitors the best possible experience?
6. What are the main attractions of the zoo?
7. What are your thoughts on ticket prices, availability, and how long it takes to get a ticket? Would you be interested if the tickets could be purchased online?
8. How would you feel if our website had a section dedicated to animals and birds, as well as some animal and bird information?
9. Do you have a zoo membership, and if so, how helpful do you think it is to you?
10. Well, what are your thoughts on the zoo's systems and handling of animals and birds?
11. What kind of benefits and discounts could the zoo provide to regular visitors or members in order to attract more visitors?
12. Is there anything more that you would like to add?

Jonathan Rodgers (Senior Administrator):

1. What is the current sponsoring program like? What kind of new sponsorship program do you wish to implement for successful sponsorship?
2. What kind of advantage do you wish to provide sponsors in order to get them to invest in the zoo?
3. What is your experience in running sponsorship programs, and what, in your opinion, are the advantages of becoming a zoo sponsor?
4. How would you like the advertisement to be handled? Could you delegate it to your administration or sponsors with restricted access to the new software system?

5. How has the present advertisement aided in increasing the number of visits to your website? Would you like to see further progress in advertising with the aid of new software?
6. What are the programs that you have available for sponsor?
7. Which animal does sponsor prefer most for the sponsorship?
8. What sort of sponsorship payment option do you prefer? Would you like a system to manage it or will you do it yourself?
9. What is the progress report on the sponsorship scheme as compared to last year?
10. Is there any volunteer work available? How do we go about applying for it?
11. Is there a limit on the number of species that may be sponsored and how long a typical sponsorship plan lasts?
12. In general, how do you inform sponsors about the new plan, and do you want a new mechanism to assist in notifying sponsors through email?
13. Is your existing sponsorship program sufficient to attract more sponsors? Would you like to build or add a new sponsorship program for effective sponsorship?
14. What are the conditions for rejecting a sponsorship if there is any disagreement?
15. Is there anything else you'd like me to know about sponsorship?

Phillip Brown (Business Owner):

1. Give us more details about your business and how long you've been supporting the animals?
2. What do you believe this zoo lacks in terms of sponsorship? Is there any type of restriction? If so, what can be done about it?
3. What is your current impression of the current advertisement, does not meet your approval?
4. How regularly do you get the sponsorship?
5. Can you tell us about some similar companies/zoos you've sponsored and what kind of similar plan you'd want to see from this zoo based on your experience?
6. Do you have any doubts about making a payment through the internet? If so, what type of security mechanism do you expect as a sponsor from the new system?
7. Do you invest money for animal welfare?
8. How do you want your company's name and logo to appear in advertisements?
9. How long have you been investing money in this zoo?
10. What type of promotion or offer encourages you to invest in sponsorship?
11. Overall, what is your view regarding the sponsor's advertisement and what change should be made in new computerized system?
12. Is there anything more you'd want to say?

2.1.2 Interview Findings

Interview Title: Initial interview with Key Stakeholder-**Matthew Jones**

Interview Date:

Duration: 1 hour

Person in attendance:

- Sushila Timalsina
- Rabindra Prasad Joshi
- Rajiv Shrestha
- Amir Tamang
- Siddharth Yonzon Tamang
- Pawan Bhusal
- Sudarshan Thapa
- Prabesh Rimal
- Krishna Oli
- Sudarshan Oli

Stakeholder	Question	Questions and Client responses
Matthew Jones (<u>zoo manager</u>):	Rabindra 01	Firstly, is there any documentation, staff manuals, or business-related items we might review to get a quick overview of the zoo's current operations? If so, could you please provide us these documents? Answer: Well, concerning the operation, you will receive a packet containing sample papers, forms, and other resources, as well as brief information about how the zoo operates. You may find some relevant documents useful.

	Rabindra 02	What is the major aim and expectation of developing new software? Answer: The main goal of designing new software is to create a website, a mobile application, a quality-of-service system, and a record-keeping system. The website should have all of the data and information on the animals as well as the zoo's peripherals, as well as all of the online ticketing sections and the gallery, as well as any other information relevant to the zoo. Aside from that, the record management system should have all of the most up-to-date information about the animals, workers, and types of activities conducted within the zoo. I'd want to have a display of certain animals and navigation bars for the zoo peripheral on the QOS system.
	Rabindra 03	What motivated you to go from a handwritten to a computerized accounting system? Answer: Since it is a confidential query, the zoo is in financial difficulty. 2million visitor per year which is about 55 thousand per day, general 36 million per year food sold profit is 1million profit and sponsorship is 1.4 million so it is difficult to manually store and preserve all of the data, we are shifting to a computerized system to keep track of everything. The main goal is to create an interactive electronic zoo system that makes it easy to manage the database and all animals, visitors, employees and sponsors. Ability to secure a safe place to change display content, display business information, perform a lot of paperwork electronically, and log in to check the sponsorship system throughout the year.
	Rabindra 04	Do you have a vision for your new program or any additional features that you'd want to include that your competitors don't? Answer:

		I want to have some chat board facilities as well as feedback forms which are not available in competitors like Buster's Zoo or Zester Zoo.
	Rabindra 05	Why did you choose to collaborate with us on software development? Answer: I looked into the company and reviews of software development firms since I needed to construct a program where everything (i.e., tracking, accounting, etc.) was currently manually inputted at our company. During that procedure, I realized that this company's concept and employees appear to be superior, and the method of understanding, asking, and answering questions is more straightforward. As a consequence, we're excited to collaborate with this firm on software development.
	Rabindra 06	Although English is a default language would you like to add another language? Answer: Currently it is only focused for UK peoples. So, I don't think it is necessary for now but for the future enhancement it can be added.
	Rabindra 07	Which features of accessibility, data gathering, and interactivity should the present software package support? Answer: Observing to the current scenario to find the animal species we need to see after the paper-based catalog. So, when looking for a certain animal, such as a lion, the new system should display what kind of lion it is, and where it came from. Theses should be accessible to the user on the website. When it comes to data collection, while a certain number of visitors are monitored by the administrator, all visitor records, such as what type of visitor they are and what age visited the zoo, should be automated.

		When the zoo has an event on any given day, it should send a message or update to the website announcing the event and a gift basket. Everything inside the zoo should be updated on the website so that visitors can learn about the environment and what kinds of amenities (such as food, personnel, and games) it offers.
	Rabindra 08	Would you prefer kiosk system to be installed? If yes, what type of features would you prefer? Answer: Yes, I would like to have a kiosk system to be installed. I want the kiosk system to provide information about animals, along with quizzes, maps and guides, complex searches, and more.
	Sushila 09	How do you keep track of visitors on a daily basis, categorizing them as children, adults, or the elderly? Would you like new software to assist you keep track if it's difficult to do so? Answer: Currently, we are keeping track of visitors manually, which has made things a little more difficult and necessitated the hiring of more personnel. Yes, it would be preferable if software could assist us in keeping track of daily visitor numbers as they rise or drop.
	Sushila 10	How do you maintain track of your employees' and animals' records, accounting, and data? Answer: As I previously stated, preserving and tracking staff details, animal records, accounting, and data have all been manually documented.
	Sushila 11	In navigation would you like to have a virtual tour for visitors? Answer: Yes, I want to have a virtual tour for visitor in navigation as we discussed in the above questions.

	Sushila 12	How do you like to highlight the events? Answer: The website design and looks should be simple and elegant. The menu bars and font should be eye catchy and attractive.
	Sushila 13	When would you like your system to be updated, and how often would you like it done? Answer: I need the system update once in a month in every year.
	Sushila 14	Is there a probability that this software will increase the number of visitors? Answer: Yes, I am quite confident about it. Since the number of viewers will increase after the website development and they can easily know the features and facilities of the zoo which will enhance in the increase in the number of the visitors.
	Sushila 15	What is the software's estimated budget and completion date, and when is it expected to be finished? Answer: Approximately the budget for this project is 2.5million pound. It should be done before December 5 2021.

	Sushila 16	Is there anything you think we should have discussed that we didn't? Answer: You've addressed all of your questions, and I've answered all of your questions as well, but as a reminder. In general, I'd like to stress that kiosk systems must be secure and should have data protection in it and that we also want hands-on and app-based systems (i.e., laptops and PCs) (mobile application). The website/app should be mobile-friendly, with an emphasis on individuals of all ages (i.e.: child, adults, old aged, etc.). We also want a map with a 3D tour and thorough information on each species, as well as a map connection to a suitable audio guide. We also want our website to feature a game for all website visitors that is related to animals, plants, birds, and other similar topics, so that people may have fun while learning about animals and birds.
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Interview Title: Initial interview with Key Stakeholder-**Thomas Smith**

Interview Date:

Duration: 1 hour

Person in attendance:

- Sushila Timalina
- Rabindra Prasad Joshi
- Rajiv Shrestha
- Amir Tamang
- Siddharth Yonzon Tamang
- Pawan Bhusal
- Sudarshan Thapa
- Prabesh Rimal
- Krishna Oli
- Sudarshan Oli

Thomas Smith <u>(visitor)</u>	Krishna 01	What differences have you observed since you began going, how frequently do you go, and how long have you been going? Answer: Since I was a child, I've been a regular visitor to the zoo, and it's always had a particular place in my heart. Because it used to be in my neighborhood, I've known this zoo since I was a child and have many childhood memories of it. I've been going to the zoo on a regular basis since I'm interested in animals and birds. During this period, I've seen that the zoo hasn't altered and is still the same as it was. The system has stayed constant, despite providing exceptional animal care. As a consequence, I am hopeful that the company will be able to improve the system and make things better and more enjoyable.
	Krishna 02	As a frequent visitor, are you satisfied with the present system? What changes would you like to see that would improve your visit experience? Answer: In terms of whether I am content or not, I believe I am dissatisfied with the current state of the zoo. It's because I have to wait in line for the ticket, and the animals' records aren't up to date. Obtaining information about any occurrences can occasionally fail to reach the intended recipient.
	Krishna 03	What type of website layout do you prefer, and what kind of content do you believe the zoo's new official website should include? Answer: For example, animal of the month would be more intriguing if you highlighted prominent creatures. Contact information, a map, directions to the zoo, travel plans, a games area, and any special offers that may be included in the pricing structure, such as a discount. A website sponsorship section is a must. The new official websites should include the information like online ticketing or informing about the visiting hours of the zoo, information about events.

	Krishna 04	Would you like to keep some of the current system ideas? Answer: Well, I don't find any of the system currently being useful, so helpful. Just like I mentioned earlier if you could make the improvements like online ticketing or informing about the visiting hours of the zoo then that would be beneficial but the current practice needs to be improved.
	Krishna 05	What changes should be made to the navigation, website, and new technologies to provide visitors the best possible experience? Answer: Talking about the changes I think if you could provide QR on the website so that the users can just scan the QR and know about the visiting hours, holidays or different type of events going to take place on the zoo, that would be helpful. Along with that if you could implement systems like online ticketing, that would also improve the overall visiting experience of the visitors.
	Krishna 06	What are the main attractions of the zoo? Answer: Different animals, the rare species that are kept in this zoo are the major set of attractions.
	Siddharth 07	What are your thoughts on ticket prices, availability, and how long it takes to get a ticket? Would you be interested if the tickets could be purchased online? Answer: When it comes to admission prices, they are fairly inexpensive, but the amount of time it takes is the key cause for the decline in tourists. So, if you could implement online ticketing, I believe it would benefit a large number of tourists by allowing them to avoid wasting time waiting in lines.

	Siddharth 08	How would you feel if website had a section dedicated to animals and birds, as well as some animal and bird information? Answer: Yes, it would be much better if we had a section dedicated to animals and birds, so that all visitors who want to learn about animals and birds at home may benefit from the material provided.
	Siddharth 09	Do you have a zoo membership, and if so, how helpful do you think it is to you? Answer: Well, when it comes to zoo membership, I had a subscription, but due to the manual approach for receiving event information, it doesn't always reach me in time. Some discounts have been offered, although they are not as generous as those offered by zoo's rivals to their members.
	Siddharth 10	Well, what are your thoughts on the zoo's systems and handling of animals and birds? Answer: As the number of visitors declines, the manual management system is no longer providing effective and efficient service to the organization. When it comes to dealing with animals and birds, It's a nice zoo that's nicely run. The zoo was built in accordance with the government's specifications. As a result, it's stunning.
	Siddharth 11	What kind of benefits and discounts could the zoo provide to regular visitors or members in order to attract more visitors? Answer: The major focus should be on making ticketing easier when it comes to the change. So, if you could make online ticketing pre-book or buy tickets online, I believe it would benefit a large number of visitors because they would not have to waste time waiting in lines, and for those who are unable to use mobile banking, you could scan and pay using the QR system in front of the ticket counter. Obtaining information via the website and email would be ideal. Games would be a fantastic

		addition as well. The website should be appealing, simple, and attractive. There should also be an interesting award, as well as a package for the family, to motivate them to take a trip. Members of the zoo should be able to meet, dine, and tour the zoo, and certain percentage discounts should be available.
	Siddharth 12	Is there anything more that you would like to add? Answer: You could set up a feedback system so visitors can provide input and can make improvements based on that. You may include a talking system so that visitors can converse and engage. The zoo's administration should use newspapers, television, and social media to promote the zoo and upcoming events. If we could manage basic information like visiting hour time and closing hour time it would be much better and helpful for the visitors.

Interview Title: Initial interview with Key Stakeholder-**Jonathan Rodgers**

Interview Date:

Duration: 1 hour

Person in attendance:

- Sushila Timalisina
- Rabindra Prasad Joshi
- Rajiv Shrestha
- Amir Tamang
- Siddharth Yonzon Tamang
- Pawan Bhusal
- Sudarshan Thapa
- Prabesh Rimal
- Krishna Oli
- Sudarshan Oli

Jonathan Rodgers (<u>senior administrator</u>):	Rajiv 01	What is the current sponsoring program like? What kind of new sponsorship program do you wish to implement for successful sponsorship? Answer:
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		Our sponsorship program runs from January 1st to December 31st, and whomever supports the animals is responsible for the animals' maintenance. Because we need to complete all of the paperwork, the real process of signing up for sponsorship begins around the middle of the year. We have a lot of paperwork in our current system, and we want to eliminate it. Also, sending documents to sponsors through fliers is dangerous, therefore I'd want this method to assist in providing information to sponsors and members by mail. When it comes to advertising, we promote the sponsor with a banner, which is pricey, thus a small piece or part of the website should be set aside for the sponsor. In order to register for the sponsorship, I'd want for the system to assist people in registering through the internet by offering them with an online form to avoid paperwork, save time, and so on additionally giving the sponsor access to modify, remove, and insert their advertising.
	Rajiv 02	What kind of advantage do you provide sponsors in order to get them to invest in the zoo? Answer: We'd want to put everything online, as indicated, to make it easier for people to support, as well as a system where sponsors can check on the condition of the animals and the zoo they've funded on a frequent basis. We give exclusive incentives as a benefit of being a sponsor. For example, we host special activities during Christmas and New Year's, and we also offer family packages that include an overnight stay at the zoo for two nights and three days, all provided by the zoo. We also provide free special awareness and education programs. We provide unique hotel and transportation rates. These are the primary advantages offered to sponsors.

	Rajiv 03	What is your experience in running sponsorship programs? Answer: I have three years of expertise administering a sponsorship program. Because the entire sponsorship program is dependent on paper, it's impossible to keep track of everything.
	Rajiv 04	How would you like the advertisement to be handled? Could you delegate it to your administration or sponsors with restricted access to the new software system? Answer: As I previously stated, the sponsor's advertising should be handled by them, however there should be certain restrictions on access for the sponsor.
	Sudarshan Thapa05	How has the present advertisement aided in increasing the number of visits to your website? Would you like to see further progress in advertising with the aid of new software? Answer: Yes, advertising has had a significant role in increasing the number of visitors. It has risen at a rate of around 5% to 10% every year. We used to advertise the zoo in our local newspaper and on television, as well as all of the new creatures that had been brought in. It has made a significant contribution to raising the number of visitors. moreover, in order to enhance the number of visitors inside the zoo. Yes, I would like that modern software could assist us in making the sponsorship advertisement known worldwide, so that we can attract tourists and out-of-town residents to visit us as well as learn about us and the sponsor.

	Sudarshan Thapa06	What are the programs that you have available for sponsor? Answer: We've run a variety of programs, ranging from instructive to awareness-building. Festivals and festivities occur on a regular basis. We can also arrange your special savings on food and housing, as well as a private zoo visit.
	Sudarshan Thapa 07	Which animal does sponsor prefer most for the sponsorship? Answer: There are five different categories of organisms. As a result, the major sponsors are rare creatures and birds that cannot be found anywhere else. The most amount ever sponsored is €2500, while the smallest is €500.
	Sudarshan Thapa 08	What sort of sponsorship payment option do you prefer? Would you like a system to manage it or will you do it yourself? Answer: In the existing system, we have been dealing payments with cheque can lead to issues such as signature mismatching, which can take a week or two to resolve. I'd want to see an online system where sponsors may make payments via internet transfer.
	Amir 09	What is the progress report on the sponsorship scheme as compared to last year? Answer: The number of sponsors has reduced by 1.5 percent as compared to the previous period.

	Amir 10	Is there any volunteer work available? How do we go about applying for it? Answer: The majority of visitors come during the summer vacation, and we need more staff at the zoo during that time because of the various program parties we have these stalls and everything, so we recruit students from nearby colleges and institutions, as well as high school students, for staff. We don't pay them as staff, but we do give them some cash and prizes for their assistance. We primarily aim to provide this job to students who have direct interaction with colleges and universities.
	Amir 11	Is there a limit on the number of species that may be sponsored and how long a typical sponsorship plan lasts? Answer: Yes, there is. The maximum numbers of animals one can sponsor is 4 animals. But for an organization it's maximum limits is 10 animals. Typically, sponsorship last for 1 years of time but if the sponsors want to extend the time it maybe for more than 1 year.
	Amir 12	In general, how do you inform sponsors about the new plan, and do you want a new mechanism to assist in notifying sponsors through email? Answer: They either visit the zoo or learn about it via television or the papers. It would be simpler if we could send them direct messages by mail or other means. Yes, if the system could aid with email notifications, that would be beneficial.

	Prabesh 13	Is your existing sponsorship program sufficient to attract more sponsors? Would you like to build or add a new sponsorship program for effective sponsorship? Answer: No, manually it is hard to satisfy the sponsor so we would like to convert all the possible documentation payment to be Electronized for saving time, avoiding the issues and so on.
	Prabesh 14	What are the conditions for rejecting a sponsorship if there is any disagreement? Answer: There is no condition for rejecting a sponsorship in between the period. If they want to reject sponsorship, they have to notify us before accepting the sponsorship. Sponsorship is legal process so if they reject the sponsorship, we have to take legal action and also, we won't be refunding their money.
	Prabesh 15	Is there anything else you'd like me to know about sponsorship? Answer: Yes. I'd like to add that there should be a specific section of the website where our sponsors may provide us with feedback that is only accessible to our sponsors. I'd also want to mention that people that support will receive a personalized sign board at the zoo.

Interview Title: Initial interview with Key Stakeholder-**Phillip Brown**

Interview Date:

Duration: 1 hour

Person in attendance:

- Sushila Timalina
- Rabindra Prasad Joshi
- Rajiv Shrestha
- Amir Tamang
- Siddharth Yonzon Tamang

- Pawan Bhusal
- Sudarshan Thapa
- Prabesh Rimal
- Krishna Oli
- Sudarshan Oli

Phillip Brown <u>(business owner)</u>	Pawan 01	Give us more details about your business and how long you've been supporting the animals? Answer: In terms of his business, he owns a gas company that produces gas from biodegradable materials. Since the beginning of his company, he has been an animal advocate.
	Pawan 02	What do you believe this zoo lacks in terms of sponsorship? Is there any type of restriction? If so, what can be done about it? Answer: Yes, he believes the zoo is behind the times in terms of technology. For the sake of the company's success, Zoo should embrace modern technologies. He claims that as a sponsor, the zoo has limited some of the species he wants to support, but that this is not a negative thing because it is a zoo rule, and that he hopes that they will expand their animal and bird sponsorship options soon.
	Pawan 03	What is your current impression of the current advertisement, does not meet your approval? Answer: It is satisfactory according to the manual system that the zoo has been using recently, but since the world has become more digital, it is no longer the best. As a sponsor, I hope that the zoo can progress to computerization in order to become the finest firm. From my perspective, the Zoo should endeavor to keep up with its competition in every way possible, including computerization with greater functions and services and more imaginative advertising ideas.

	Pawan 04	How regularly do you get the sponsorship? Answer: I've received sponsorship three times in the last five years, and I'm hoping that there will be enough sponsors in the future to safeguard animals as well.
	Pawan 05	Can you tell us about some similar companies/zoos you've sponsored and what kind of similar plan you'd want to see from this zoo based on your experience? Answer: Previously he had worked with the bank. Everything is digitized, with paperwork, plays, and policies explicitly stating the laws and regulations. The program's emails are correct. The advertisement is done globally, and the section for the firm has been created with the necessary information for the sponsored company specified and displayed appropriately and satisfactorily. I was really pleased with that technique, and I believe this zoo may use it as well.
	Pawan 06	Do you have any doubts about making a payment through the internet? If so, what type of security mechanism do you expect as a sponsor from the new system? Answer: If we had an online payment system, it would be much easier than handing over cash or a check. Because we are dealing with funds with the zoo, there should be a high level of security.

	Sudarshan Oli07	Do you invest money for animal welfare? Answer: I was animal lover since my childhood and used to feed street dogs and birds. I always wanted to help and protect animals so I always invested for their welfare as much as I could. I also invest 25% of my salary to help and preserve animals.
	Sudarshan Oli 08	How do you want your company's name and logo to appear in advertisements? Answer: For sponsorship a few banners and pamphlet will be placed in various locations for reference, and a large number of people will attend, with most of the couples who own their own home coming to the zoo to learn more about my business. And I would be happier if my company name and product appear on the website in an era of rapid digital marketing.
	Sudarshan Oli 09	How long have you been investing money in this zoo? Answer: I have been investing money for 5 years in this zoo. And I would like to invest the more money for the welfare of animals also.

	Sudarshan Oli 10	What type of promotion or offer encourages you to invest in sponsorship? Answer: During a visit, I learned about Claybrook Zoo's sponsorship program. We had been looking for a venue to market our items as a business. Claybrook Zoo is a well-known zoo with many visitors each day, so we decided it would be a profitable investment as well as a good act. As a result, we agreed to sponsor their animal. I was also interested in this sponsorship program after hearing the testimony and getting feedback from the people.
	Sudarshan Oli 11	Overall, what is your view regarding the sponsor's advertisement and what change should be made in computerized system? Answer: The number of billboards in Claybrook Zoo is currently limited, and they are only positioned in one location. Furthermore, the concept of banners and billboards is outdated. Instead, incorporating videos or short clips would be appealing. I believe that as soon as a website launched, the advertising will improve on its own as it joins the digital market. Paying for animal sponsors online is a very good idea. I used to pay by check, but this can be very time consuming. Going to the bank and paying in person isn't always feasible. As a result, online payment methods will be greatly appreciated.

	Sudarshan Oli 12	Is there anything more you'd want to say? Answer: Although I'm not sure what your client's requirements are, we'd love it if you could include a function that allows us sponsors to update our information ourselves.
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2.1.3 Other problems Domain and Research

This phase contains research on existing equivalent systems and other sources such as related legislation that will help us better grasp the problem domain. Research is a systematic method of gathering facts and elaborating on a specific issue while also discovering fresh information relevant to the topic.

2.1.3.1 Comparable Software System Review

We referenced to some of the websites in this part and compared the general design as well as the functionalities for a Zoo-based website. We did a lot of research on several websites, but these websites drew our attention the most.

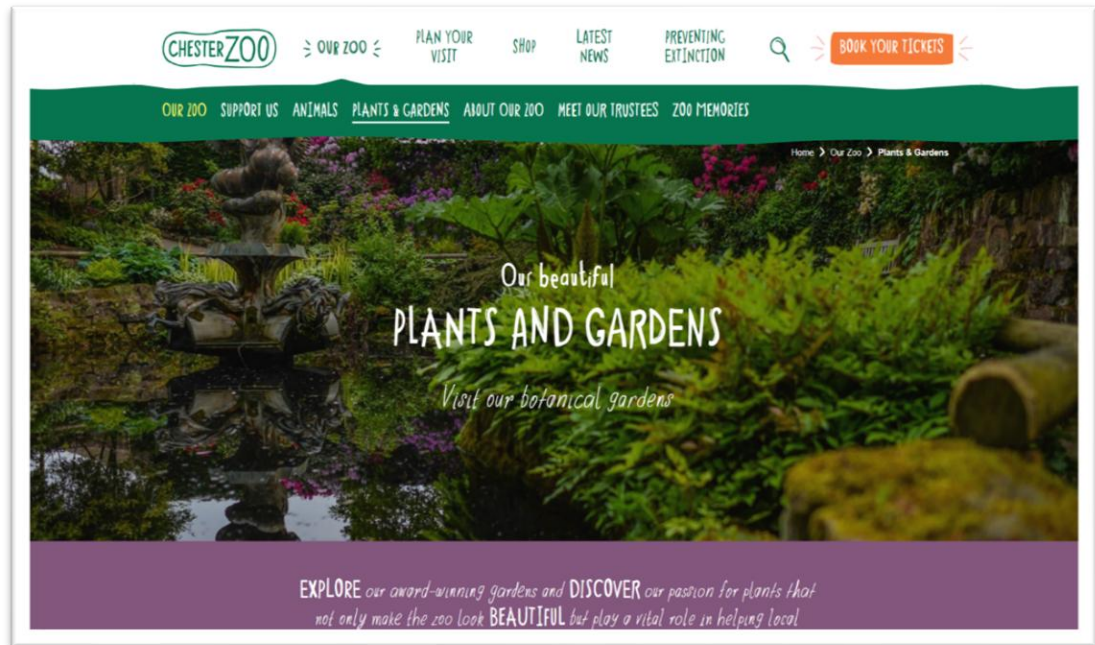
2.1.3.1.1 Record Management System

1) Chester Zoo:

We had taken a Chester Zoo as our motivational website for storing the data for our project. This zoo has maintained the data properly in their sites with proper information's.

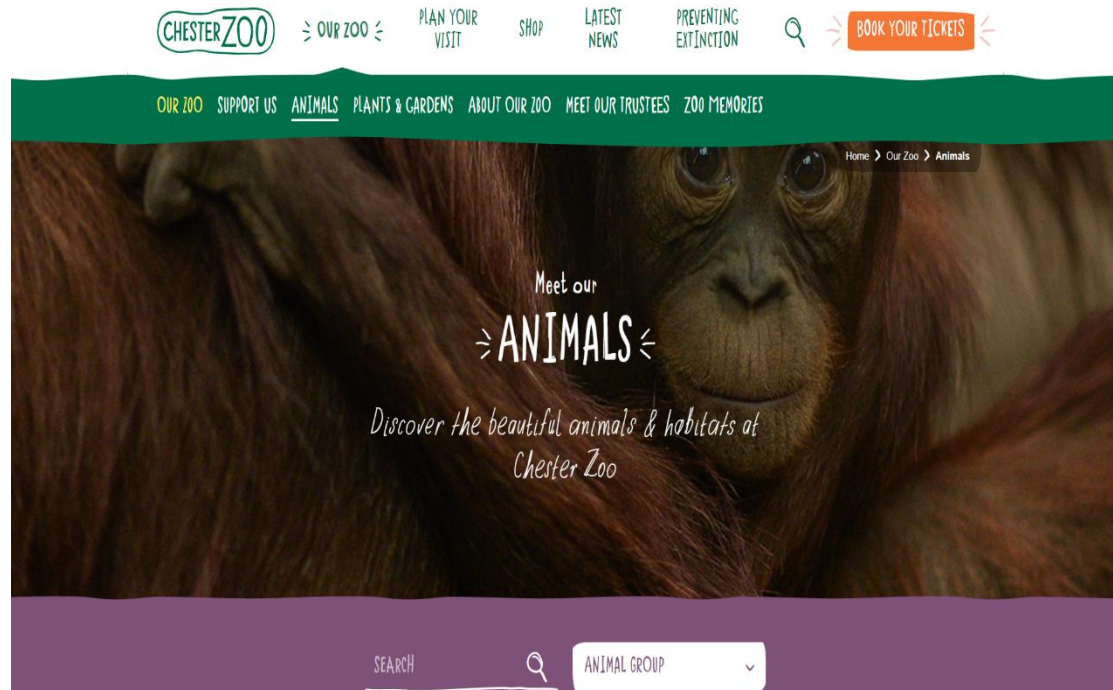
Home Page View:

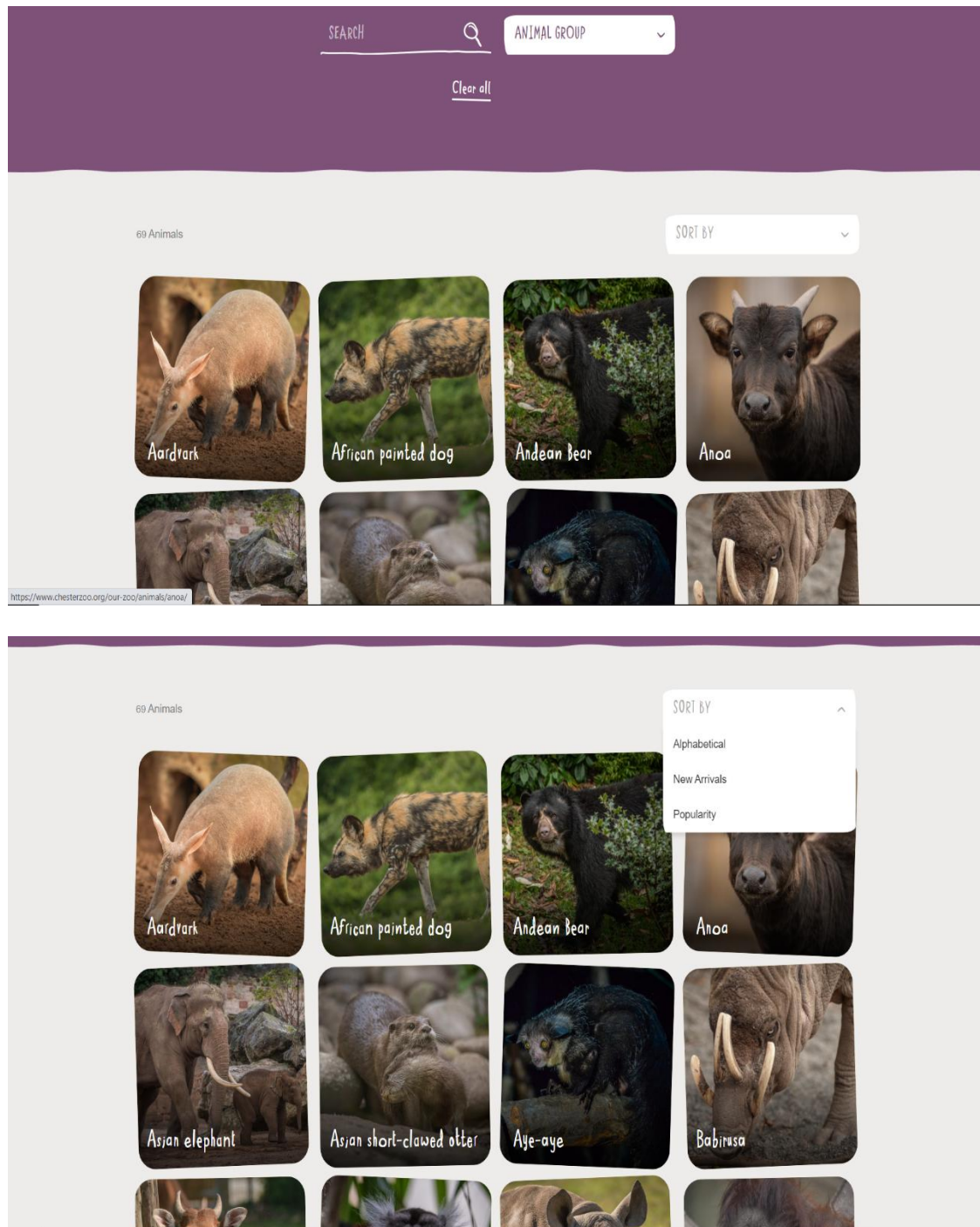
This is the Chester Zoo website's home page, features such as a navigation bar, booking tickets, and search box. Through the navigation bar we can acquire more information about animals, plants, etc.



Animal record system view:

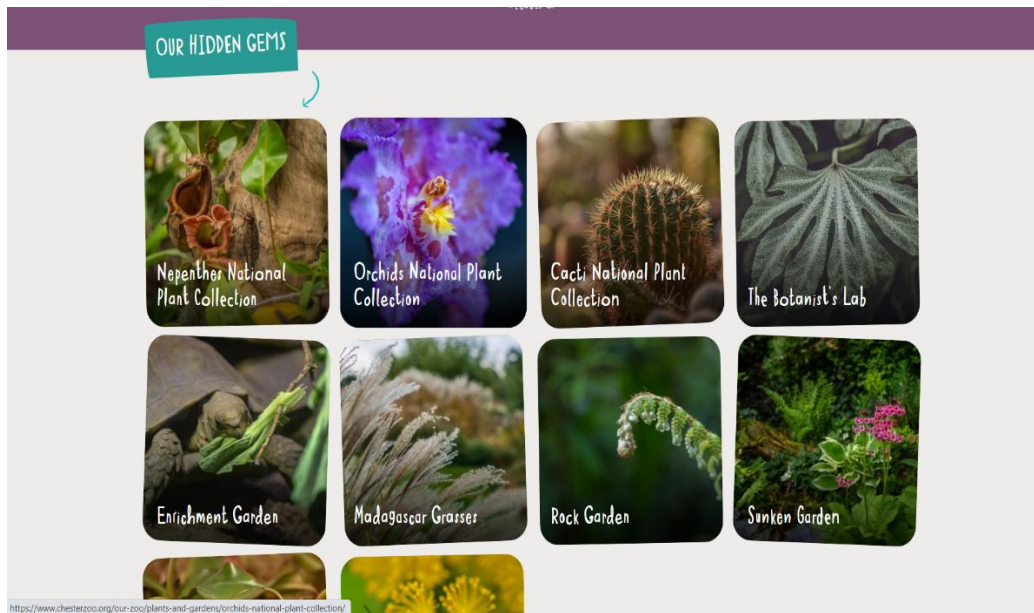
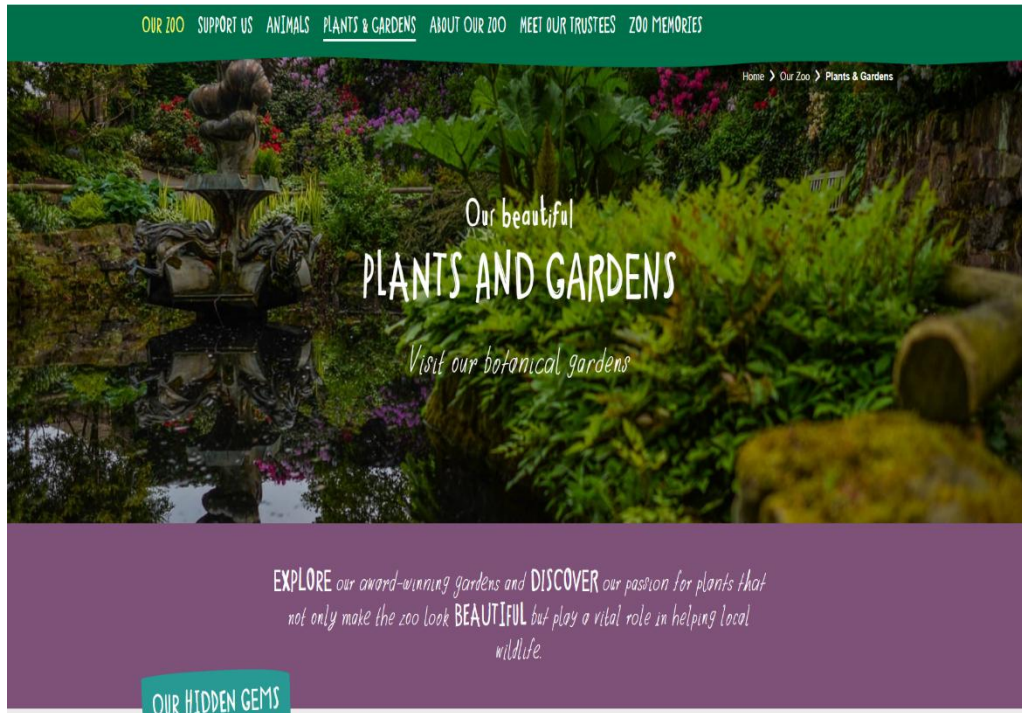
This is how the animal records were preserved via electronic system at the zoo's web page, as seen in the image below. Animal records are updated on a daily basis, and their information are arranged alphabetically with respective images of animals and birds. If you wish to learn more about a particular animal, the search box can assist you in locating information about any species.





Plant record system view:

Despite the fact that the botanical garden has a small number of plants, this is how the plant records have been kept. Some uncommon and gorgeous flower species have also been planted to improve the zoo's aesthetic while also being beneficial for study.



Trustees' information:

Some information about the trustees' primary responsibilities has been included, as well as what they have been working on and how they have been carrying out their duties. For the appreciation and continual cooperation of the aid which they have been offering to the zoo, the trustees' names and some details information have been presented.

Meet Our Trustees

The key responsibilities of the Trustees

With other Trustees to hold the Society "in Trust" for current and future beneficiaries by:

- Ensuring that the Society has a clear mission and strategic direction and is focused on achieving these;
- Being responsible for the performance of the Society and for its "corporate" behaviour;
- Ensuring that the Society complies with all legal and regulatory requirements;
- Acting as guardians of the Society's assets, both tangible and intangible, taking all due care over their security, deployment and proper application; and
- Ensuring that the Society's governance is of the highest possible standard.



Chair of Trustees – Malcolm Aidson

Malcolm is a company managing director providing engineering and construction services, and Fellow of the Institution of Mechanical Engineers. After studying Engineering Science at Oxford University, he has worked in both public and private sectors ranging from construction of nuclear power stations, road and rail projects, industrial and commercial buildings; consultant to nuclear and manufacturing sectors in UK and Europe on commercial dispute resolution including Expert Witness to Court (Technology & Construction Division).



Deputy Chair of Trustees – Penny Cudde BSc (Hons), MBA

Former Managing Director at the Macmillan Health Centre, Director of Adult and Non-Executive Director of the Cooperative Society Specialist Business, Penny has held senior roles in retail, aviation, manufacturing and financial services. Penny is now keen to allow her to focus on Non-Executive Director and Trustee roles.



2) Colchester Zoo:

We had taken a Chester Zoo as our motivational website for storing the data for our project. This zoo has maintained the data properly in their sites with proper information's.

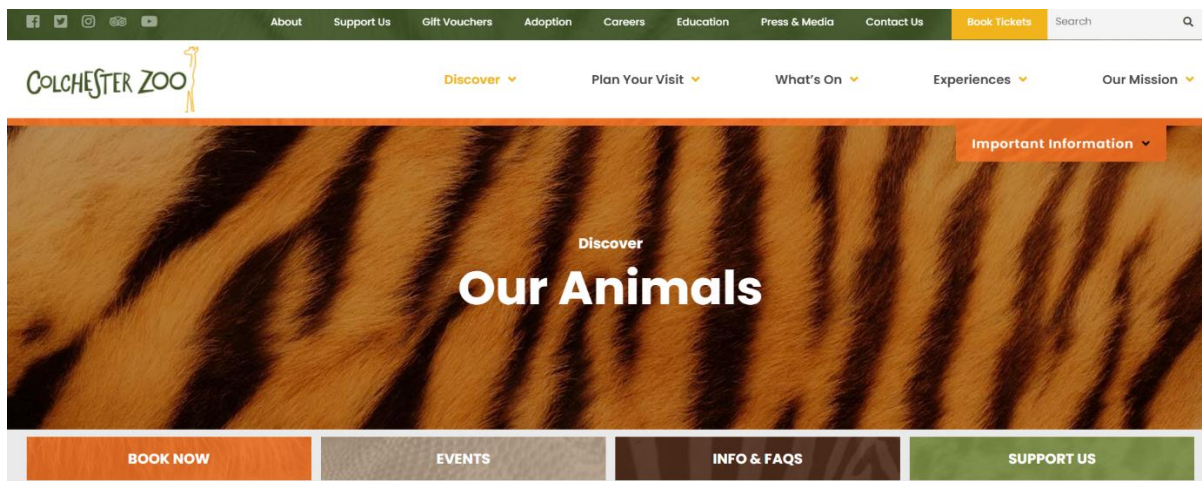
Home page view:

The navigation provides greater information detail and more search choices for different sorts of information, making it user friendly even for inexperienced users, as shown in this image. Its typography, style, and design look to be quite good and pleasant, and it uses an appealing arrangement to promote each event and program.



Animal Record view:

We can see 200 different animal species on this website image of Colchester Zoo. We may search for animals, birds, and other species by their kind and country of origin. Each and every animal's thorough information has been provided.



Discover over 200 species!

[Privacy & Cookies Policy](#)





[Discover](#)
[Plan Your Visit](#)
[What's On](#)
[Experiences](#)
[Our Mission](#)

As a preventative measure following a Housing Order by the Secretary of State regarding Avian Flu, the majority of our bird species are currently off-show.

[Important Information](#)

Filter Animals by Type
Filter Animals by Location

Africa

Asia

Australia

Caribbean Islands

Europe

Fiji

North America

Oceania

South America

United Kingdom



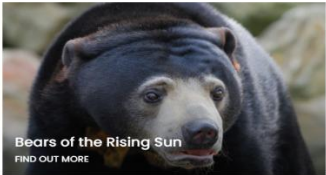
Aardvark Burrow

FIND OUT MORE



Australian Rainbows

FIND OUT MORE



Bears of the Rising Sun

FIND OUT MORE

[Privacy & Cookies Policy](#)
[TOP](#)

Activities Record view:

The zoo's daily activities and events, such as animal feeding and care, games and festivals, and other indoor and outdoor activities, are updated daily on the website page.



[Discover](#)
[Plan Your Visit](#)
[What's On](#)
[Experiences](#)
[Our Mission](#)

Discover

Important Information

Activities

BOOK NOW

EVENTS

INFO & FAQs

SUPPORT US

Activities at Colchester Zoo

As you journey around the Zoo, you'll find there's more to explore with our range of activities!

Discover new heights on our many adventure play areas or take a ride on our Lost Madagascar Express Train to the Lemur Walkthrough, where you'll be greeted by our friendly troop of lemurs! You can even transform into your favourite animal at our Face Painting Station, try bug handling at our Discovery Centre and check out our interactive sand pit!

[Privacy & Cookies Policy](#)





Conservation Programme Record:

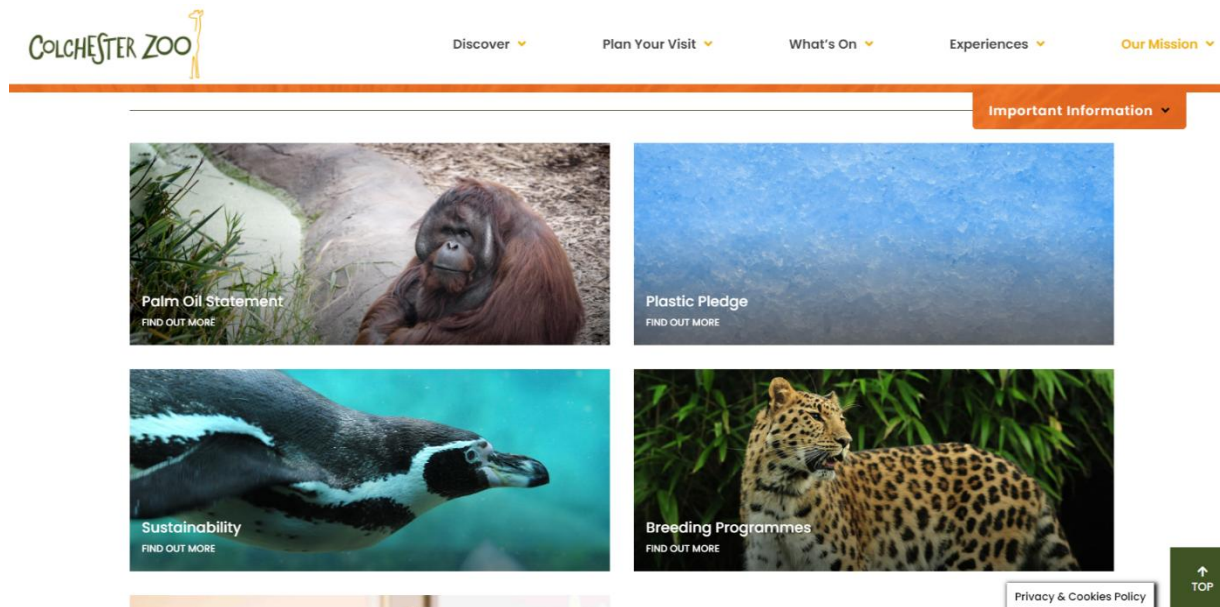
When there is a program established in the zoo itself, conservation programs such as breeding, animal sustainability, chat programs for rare species conversation, and also safeguarding measures by preventing animal health are updated daily.

Conservation at The Zoo

[BOOK NOW](#)
[EVENTS](#)
[INFO & FAQs](#)
[SUPPORT US](#)

Conservation at The Zoo

Find out more about just some of the conservation work we undertake and support at Colchester Zoo and through our charity Action for the Wild.



2.1.3.1.2 Zoo Website

System	Description
 Chester Zoo https://www.chesterzoo.org/ (Web Application)	Chester Zoo is a zoo situated at Upton-by-Chester, Cheshire, England. George Mottershead and his family founded Chester Zoo in 1931. More than 20,000 animals and 128 acres of zoological gardens can be found in Chester Zoo to explore. The Chester Zoo is run by the North of England Zoological Society, a registered charity established in 1934. The zoo is not supported financially by the government. This is the UK's most visited wildlife

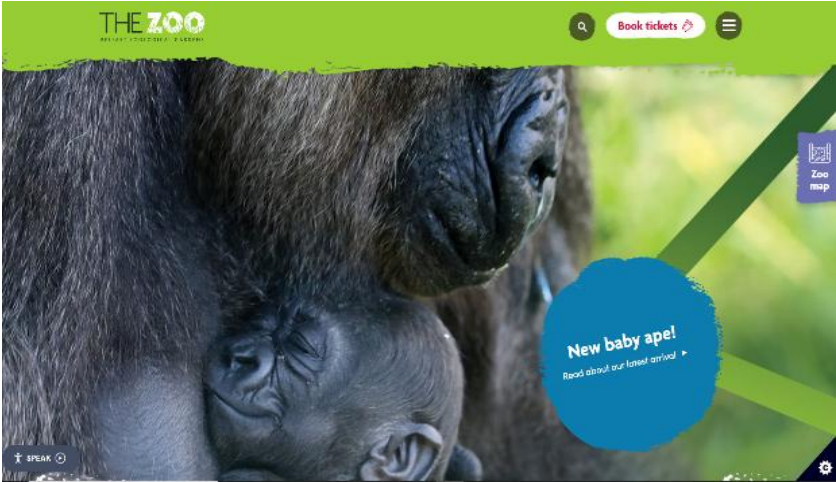
attraction with over 2 million visitors in 2019.



Aadvantage	Disadvantage
In the background, there are high-quality photos which make website more attractive.	It doesnot contains a part of navigational data such as stopping, location and internal zoo maps.
In terms of convenience, the framework is exceptionally responsive and works exceptionally well.	Though it contains a lot of information in home page but it is not fruitful for visitor and had made it a bit boring.
There is a lot of supplementary information in the system, such as entertaining facts, a quiz, and so on.	The image size of animals and plants are so small that its hard to find its beauty.
The information about tickets and zoo membership is very clearly displayed.	

System	Description						
 Colchester Zoo https://www.colchester-zoo.com/ (Web Application)	Colchester Zoo is a zoo near Colchester, England. The zoo first opened its doors in 1963, and on June 2, 2013, it celebrated its 50th anniversary. It is home to many uncommon and endangered species that lives there, including big cats, monkeys, and birds, as well as a variety of invertebrates and fish. The zoo has its own charity Action for the Wild, to support projects around the world. It aims to provide both financial and technical support, with the goal of increasing local awareness of community conservation programs and supporting conservation research around the world.  Fig: Colchester Zoo Website <table> <tr> <th>Advantage</th><th>Disadvantage</th></tr> <tr> <td>It consists a map of the zoo's several locations which helps to find your favourite animals.</td><td>While trying to navigate the webpage, the huge navigation bar blocks a lot of material.</td></tr> <tr> <td>It has a features like audio talks from the Animal Care Team that tells tell you more about selected species and the animals they care for.</td><td></td></tr> </table>	Advantage	Disadvantage	It consists a map of the zoo's several locations which helps to find your favourite animals.	While trying to navigate the webpage, the huge navigation bar blocks a lot of material.	It has a features like audio talks from the Animal Care Team that tells tell you more about selected species and the animals they care for.	
Advantage	Disadvantage						
It consists a map of the zoo's several locations which helps to find your favourite animals.	While trying to navigate the webpage, the huge navigation bar blocks a lot of material.						
It has a features like audio talks from the Animal Care Team that tells tell you more about selected species and the animals they care for.							

	The information about tickets is exceptionally clearly displayed.	

System	Description						
 Belfast Zoo https://www.belfastcity.gov.uk/zoo/home (Web Application)	The zoo was established in 1934 at Belfast, United Kingdom. The zoo carries out vital preservation work and participates in more than 90 European and worldwide breeding programs that help ensure the survival of many endangered species. Belfast City Council is the owner of Belfast Zoo. The council spends £ 1.5 million annually on running and promoting the zoo. The zoo is a member of the British and Irish Association of Zoos and Aquariums (BIAZA), the European Association of Zoos and Aquaria (EAZA), and the World Association of Zoos and Aquariums (WAZA).  <table border="1"> <thead> <tr> <th>Advantage</th><th>Disadvantage</th></tr> </thead> <tbody> <tr> <td>The design is simple and attractive, making it ideal for grabbing a lot of attention.</td><td></td></tr> <tr> <td>Clear and proper guide for navigational information about animals' location and inner zoo maps.</td><td></td></tr> </tbody> </table>	Advantage	Disadvantage	The design is simple and attractive, making it ideal for grabbing a lot of attention.		Clear and proper guide for navigational information about animals' location and inner zoo maps.	
Advantage	Disadvantage						
The design is simple and attractive, making it ideal for grabbing a lot of attention.							
Clear and proper guide for navigational information about animals' location and inner zoo maps.							

	The website is very responsive and smooth and works very well.	
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2.1.3.1.3 Visitor Kiosk/Information Systems

1) Chester Zoo:

Online Ticketing System:

The visitor's time is saved by using online ticketing, so they don't have to visit the zoo only to get a ticket, no matter how quickly or when they want to go.

The screenshot displays the Chester Zoo website's online ticketing system. The top navigation bar is orange with a 'Back to chesterzoo.org' link, the 'CHESTERZOO' logo, and links for 'Register', 'Basket', and 'Login'. Below the navigation bar, a green banner reads 'EXPLORE THE CHESTER ZOO SHOP!'. Three main promotional tiles are shown: 'Zoo Admission' with a 'BUY TICKETS' button, 'Become a member' featuring a group of lemurs, and 'Adopt an animal' featuring flamingos. A green banner below these tiles says 'FOLLOW, SHARE AND WATCH!'. The bottom section is titled 'BOOK YOUR TICKETS' and includes a welcome message: 'We can't wait to welcome you to the zoo!'. It provides instructions on how to book tickets, including a note about temporary closures due to avian influenza. A calendar for December 2021 is shown, with the 1st, 2nd, 3rd, 4th, and 5th highlighted. A green 'BOOKING SUMMARY' box on the right shows 'YOUR SELECTED DATE' as Tuesday, 28 December 2021, with a price of £0.00.

Online Membership Form:

The visitor is presented with an online membership form for becoming a member of the zoo, as well as offers and a zoo event.

[← Back to chesterzoo.org](#)

CHESTERZOO

[Register](#)
[Basket](#)
[Login](#)

⇒ BECOME A MEMBER ⇐

Pay in full today and save 10% if you choose to include an automatic renewal*

*With an automatic renewal in place, your membership will renew onto a 12 monthly direct debit payment in your second year. You will receive a renewal reminder 4 to 6 weeks prior to this date.

Membership	Quantity		Standard price	Auto Renewal*	
Adult		0		£96.00	£86.40
Junior (3-17)		0		£75.00	£67.50
Little Monkey (0-2)		0		FREE	£0.00

VIEW MEMBER BENEFITS

Share this page

Online Gift Shop:

For any special event, whether it's for children, adults, or anybody else. The zoo website provides exquisite homeware, as well as exciting books for study, gifts, and presents, and everything is updated regularly, so visitors and members don't have to go shopping for another shop to give gifts or study books.

KIDS BOOKS

Filter

All products

Sort

Featured

Jingle bells SOMETHING SMELLS! Board Book
£7.99

Pip And Posy The Christmas Tree £6.99

The Gruffalo Paperback Book £6.99

The Gruffalos Child Paperback £6.99

FOOD & DRINK

Filter

All products

Sort

Featured

Chester Zoo Gin 50CL - Limit of 2 per transaction
£34.95

Chester Zoo Winter Gin 50CL - Limit of 2 per transaction
£34.95

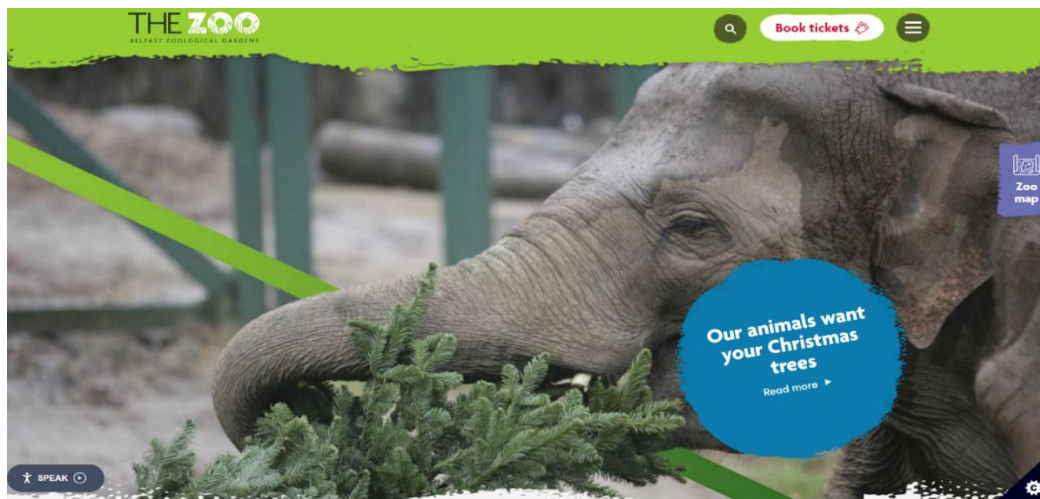
Chester Zoo Milk Chocolate Bar
£2.50

Coco Pazzo Gin & Tonic Fudge 150G
£3.99

2) Belfast Zoo:

Online ticketing:

The ticketing system form can be seen in the image below, asking visitors for their full data, as well as the cost and a privacy statement, so that when purchasing an online ticket, you know what to avoid.



Kids go free - one free child ticket for every paying adult. Valid for visits between Wednesday 1 December - Friday 31 December 2021.

Adult (17+) £13.50

Child* (under 4 years old) free

Child* (4 - 16 years old) £6.75

Student** £6.75

Senior** £6.75

Family day ticket (up to 2 adults and 3 children) £38.00

* Children under 17 must be accompanied by an adult.

** Only valid with acceptable identification.

*** Please choose the date you would like to visit**

dd/mm/yyyy 

*** Please select your arrival time?**

☐ 10am - 11am
☐ 11am - 12pm
☐ 12pm - 1pm
☐ 1pm - 2pm
☐ 2pm - 2.30pm

When you enter the zoo, you are free to enjoy your visit until we close at 4pm.

Your details

*** Name**

*** Enter postcode with a space to find the address eg BT1 1GS**

YOUR DETAILS

*** Name**

*** Enter postcode with a space to find the address eg BT1 1GS**

☐ Outside Northern Ireland

*** Address**

*** Phone number**

*** Email**

Privacy notice

Belfast City Council is the Data Controller under Data Protection legislation for the personal data gathered on this form.

Belfast City Council will use the information that you provide for the purpose of processing your booking and online ticket and any associated communications. By completing this form, you are providing your personal data on the basis of consent.

Personal data may be shared between Council staff who are involved in providing this service and between Council Departments

ZOO Map:

The audio tour of the zoo comes with an online map so that visitors can figure out which way they should go. The map also includes animal photos, allowing viewers to follow the audio tour to the correct species

Overview

Covid restrictions

Facilities in the zoo

Accessibility in the zoo

Getting around

Your zoo map

Getting to the zoo

Daily talks and feeding times

Zoo audio tour

Zoomobile

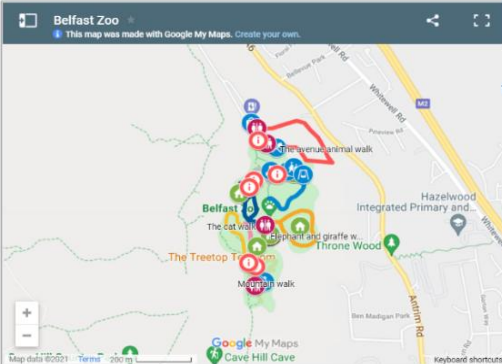
Zoovenir shop

Group visits

More zoomobile information

Check out our interactive map

Use our map to guide yourself around all your favourite areas of the zoo.



 **SPEAK** 





2.1.3.2 Development Relevant Legislation

The computerized system proposed by Claybrook Zoo contains information about zoo staff, sponsors, and potential members. The data should be used wisely and only kept when absolutely necessary. There are rules given by UK's usage of the General Data Protection Regulation (GDPR) as the Information Assurance Act 2018. The following is the rule regarding the use of private data stored within the system (GOV.UK, 2018).

- Private data stored within the system should be used in a fair, legal, and transparent manner.
- The system's private data should only be utilized for certain, clear objectives.
- The system ought to be utilized in a way that's satisfactory, important and limited to only what is required.
- Data stored on the system should be correct and up to date as needed.
- Do not preserve data stored on the system longer than necessary.

- Must be managed in a secure manner, including protection against illegal or unauthorized processing, access, loss, destruction or damage.

The pages should be accessible, navigable, and understandable regardless of any limitations, according to W3C accessibility evaluation guidelines. Web accessibility evaluation tools may and must be used to test individual pages for accessibility.

2.2 Requirements Specification

The software's purpose, nature and requirement are outlined in it, which ensures that the required features are met in accordance with the client's requirements. It also covers the software's yield and cost, as well as product development and implementation. The act of gathering user and system requirements is compiled into a document that is clear, intelligible, complete, and consistent, as well as a functional and non-functional description of the solution system.

2.2.1 Problem Domain Description

The limitations of the present clerical management system, as well as its procedures with certain information, have been described in the list below with the assistance of a flowchart.

2.2.1.1 Existing Business Operation

The term "business operation" refers to what a company does on a daily basis to keep it functioning and profitable, such activities might vary greatly from one organization to the next. The business process may alter from time to time for a variety of reasons, according to company policy. A section on operations is common in business plans to ensure that company founders are aware of the systems, equipment, people, and processes needed to run the business, as well as the rules and responsibilities of the various faculties and the business criteria established to ensure that the organization runs smoothly. When all of the company's plans and policies are in place, including the system, markets, vital items and equipment, and day-to-day operations, it will be simpler for the founder to comprehend the company's function, which will aid in firm expansion and profit. The company's operations department supervises all procedures and does market research in order to provide exceptional service while also keeping track of earnings.

As a company expands, its operations will change, and management should plan for these changes to avoid system malfunctions. As a small firm expands, for example, it must be prepared to deal with new obstacles such as legal, marketing, and capacity issues. Errors and omissions will occur if the business does not develop to keep up with the changing business operations. However, the following components must be considered in order to make business operations more effective for most businesses:

- i. Keep track of performance and keep up to date with industry trends:
The business milestones of a firm should be measured in a realistic and actionable manner. The goal-setting stage of assessing performance is when you keep up with what's going on in the business to get ideas on how to outperform the competition. The company's management should set reasonable goals with defined deadlines and stay on top of current events, such as new technology, changing state and federal laws, or changes in the local economy.
- ii. Allocating duties and tasks to employees(staffing):

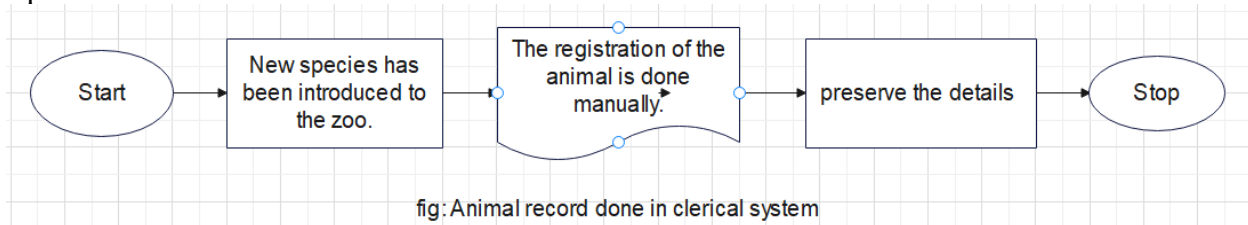
Processes dictate how many people are needed. Who will be responsible for the tasks stated in the work procedures, and how many will be required? Roles and duties are delegated and distributed by business operations professionals to other members of their team. When it comes to recruiting new staff, it's evident that each candidate has a unique combination of skills. HR is included in a company's operations in order to cover all of its critical procedures. Firms must ensure that they have the right people in place to carry out their vital tasks.

- iii. The right place at the right time and with the right procedure (process and location):
Process is significant because it affects production and efficiency, and location is more relevant to some types of organizations than others for a variety of reasons. Manual processes that might be automated or labor that is already being done by other departments can waste time and money for a business. A home-based consultant may merely require a workstation, but a pet groomer may require a parking space, and so on.
- iv. Internal resource coordination and cooperation:
Business operations also entail ensuring that specialization within a firm does not jeopardize the company's ability to function as a unit, as well as managing the company's resources, from capital to tangible products, and assigning roles and responsibilities in order to gain profit with specialized and unique, business operations. A corporation must have the appropriate amount and type of equipment and technology in order to function correctly and efficiently.
- v. Creating plans and implementing new technologies and equipment:
Designing strategies that aid the business in accomplishing a certain goal, as well as effective personnel usage, is critical. The proposal will benefit talented individuals, who will be able to find work depending on their qualifications. Individuals with technological knowledge have progressed over time and will be able to get work based on their qualifications. Implementing new technology and equipment, such as software or creating a company website, aids in achieving profit through appropriate plan execution.
Some functions of existing business operation:
 - a. Completing bank transactions: Completing fundamental financial transactions and maintaining records at an office which includes the collecting, counting, and disbursement of funds.
 - b. Establishing effective communication and offering upper-level management, as well as coaching and mentoring newcomers
 - c. Utilize a Technological Advantage Fresh product creation and user-friendly website advertising are critical in today's high-tech environment, where everyone is looking for the latest technology.
 - d. Complaints and praises must be analyzed, and improvements made as needed.
 - e. A long-term branding strategy should be incorporated, as well as monitoring and managing third-party collaboration.

2.2.1.1.1 Animal Life-Cycle (flowchart):

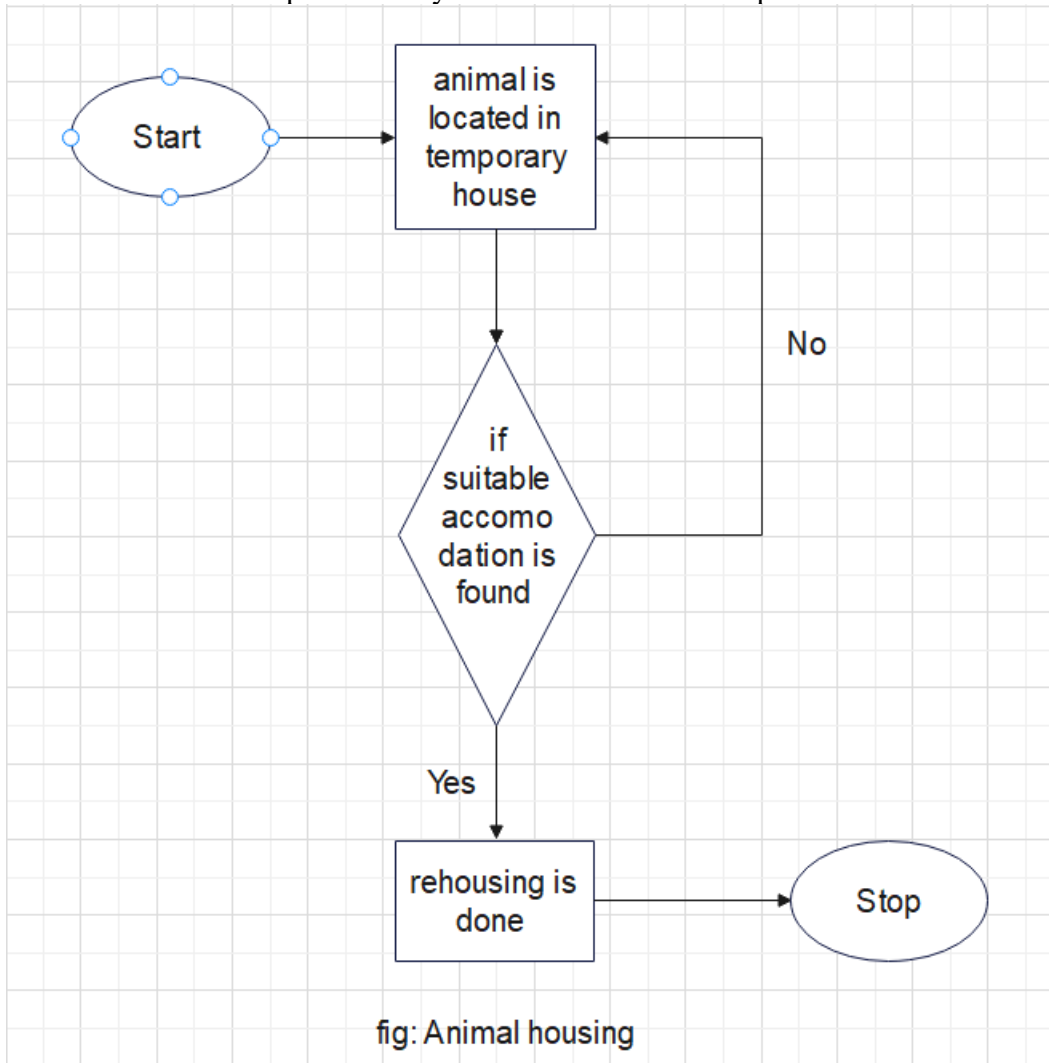
Animal arrival record:

If any animals arrive at the zoo recently, their records are handwritten manually. As a result, this type of data is difficult to handle; accessing it necessitates additional personnel and manual inquiry, while altering records may involve totally rewriting the facts, which may be lengthy and repetitive.



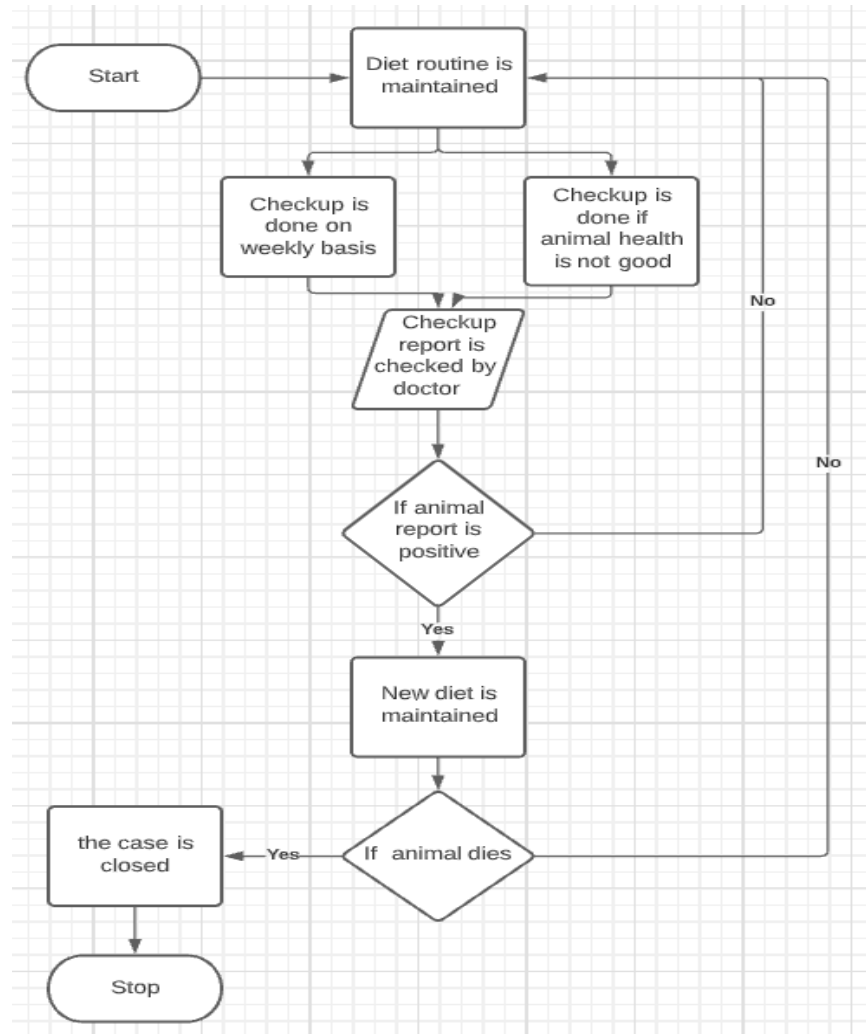
Animal housing record:

When the paperwork is finished, the animal is relocated to a temporary holding facility that is likewise closed to the public. They are re-housed after adequate accommodation is found.

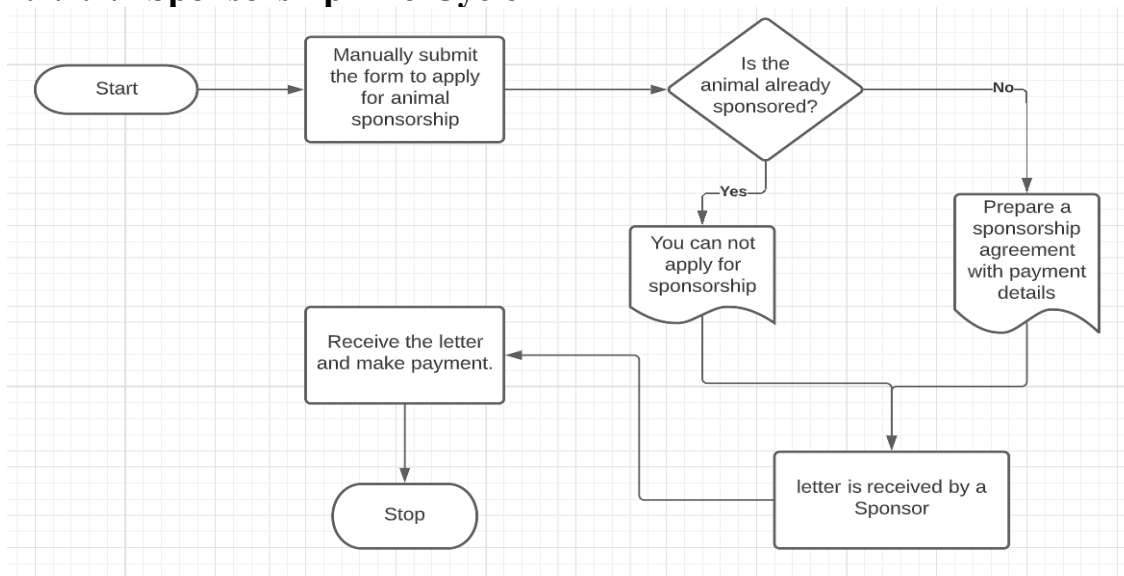


Animal health record:

Animals are monitored 24 hours a day, 7 days a week, so that if there is a problem with their health, the medical team can respond quickly. Dietary is tasked with caring for the animals and scheduling their diets. When an animal dies due to medical concerns or old age, the animal record is closed and the case-paper file is relocated.



2.2.1.1.2 Sponsorship Life-Cycle



A zoo takes the initiative and determines whether or not to give animal sponsorship; if yes, the zoo proceeds with the decision; if no, the zoo offers a project form for sponsors to manually fill out; and if yes, the zoo proceeds with the decision. The status of the animal's sponsorship is then decided. You can ask for sponsorship if the animal is already sponsored, but if it isn't, a sponsorship arrangement with payment is made, and the latter is delivered after the sponsor accepts. The transaction is concluded and payment is paid if the sponsor approves.

2.2.1.2 Summary of existing business limitations

Because the Claybrook zoo's existing business system is paper-based, the zoo has faced various issues as a result of its current operations' flaws, which were identified during the workshops and must be solved. This zoo is also working on building an online web site because the paper-based technique is outdated.

The following are the most significant drawbacks:

1. The maintaining of records and monitoring of animals and workers is a time-consuming and tedious
2. If the paper with the information is missing, lost, or an accident occurs during the procedure, it is incredibly simple to lose data on animals and other information.
3. While several of its rivals have transitioned to an electronic system, it continues to use a paper-based method.
4. In the zoo, there was no online ticketing system, no website for consumers and sponsors to login to, and no virtual tour or audio tour.
5. The update system for newly imported animals was not synchronized, and most of the zoo's signage and displays were frozen, leading many visitors to escape to competitors who had already established a computerized system.
6. During events, sending out fliers to zoo members costs a lot of money, and there's a risk that the letters will get lost in the commotion.

7. During events, the communication system was essentially irresponsive, time-consuming, and expensive. Sending out information to zoo members costs a lot of money, and there's a danger that the letters may get lost in the hubbub or won't arrive on time under the current system.

Due to an issue with the present system (i.e., a paper-based system), the customer wants to make a change and switch to an electronic system.

2.2.2 Functional Requirements

Many functional characteristics of the proposed system, such as features, access permissions, and data quality, are included in the functional requirements. The functional requirement includes the database table and it contains information about the website as well as the tasks that must be completed by the linked worker, admin. Access information, such as who has and does not have access, is also included in the database. The features necessary for the tourist kiosk system are all part of the functional requirements, which are stated and some of which are given in the form of a table.

Features Required

After reading the brief, the essential features have been gathered, and the features will be implemented in order of importance in the papers supplied. A list of features has also been compiled below:

Staff Management:

- It should be possible for employees to utilize the system.
- The system should keep track of the employee's record, information, and accomplishments.
- There should be different levels of access for personnel based on their responsibilities, Admins, for example, should have access to all the system has to offer. Managers should be able to remove and manage animal records as well as employee data.

Sponsorship Management:

- Sponsors should be allowed to log in to the page and be given a form that designates a distinct place for them to apply for and submit sponsorship forms.
- If possible, sponsors should be able to change their password, rename their account, and watch or receive crucial messages from the zoo as they are posted.
- There should be a guide to explain the advertisement to the sponsor and the system's functionality, as well as a complete troubleshooting area.
- Sponsors should have the option of using an electronic payment mechanism.
- The sponsor should be able to manage their own marketing, but the zoo administration should be able to manage sponsorships from the backend, where they may engage with present and potential sponsors about sponsorship acceptance, rejection, renewal, and expiration.

Animal Management:

- The system should allow users to search for animals and show information about them.

- All animal records procedures, such as adding, updating, and removing, should be managed by the system.
- Animal data, as well as the animal itself, should be removed from the system and archived instead.

Some other features:

- The system should be easy to navigate, and all employees and visitors should be able to access it.
- When searching for animals on the zoo's website, the system should allow visitors to utilize complex search combinations, and the animal's related choice should be offered.
- The system should offer online booking and ticketing.
- A game section for especially kids, but also for individuals of all ages, should be included, and the game should be related to animals.
- The system should be extremely secure, and anyone with malevolent intent should be denied access to any administrative functions.
- A tutorial and example should be provided to demonstrate any user how to use the system, and the kiosk system should assist with animal information retrieval.
- Visitors should be able to browse the website to learn more about the event, and the system should provide a list of events and deliver notifications when they are planned.

2.2.2.1 Records Management System (database)

The database shown in red in the figure below represents no access. The functions of each field in the database are listed below:

The recording activities at the Zoo are displayed below.

Operation	Admin	Manager	Sponsors	Staff	Visitor
Add	Add new animal information to the data list	Make a list of animals to add	No access	No access	No access
Remove	Able to delete the list of dead animals and the list of animals like dead animals and moving animals	Make a list of animals to be delete and transfer like the list of dead animals	No access	No access	No access

create	Create a new record.	Create a new record.	No access	No access	No access
amend	Change and correct the record details	Change and correct the record details	No access	No access	No access
archive	Move an unnecessary record to archive.	Move an unnecessary record to archive.	No access	No access	No access
delete	Erase the data from zoo catalogue.	Erase the data from zoo catalogue.	No access	No access	No access
display	Display all the list of information about animals and zoo, when an item is clicked	Display all the list of information about zoo	Display the list of animals that have been sponsored.	Show Information about the personnel and customers	Show their personal data as well as animal data.
assign	Assign the information of newly added animals to the appropriate categories.	Assign the information of newly added animals to the appropriate categories.	No access	No access	No access
news	Add future events to the calendar	Take a look at upcoming events.	Able to see almost all of the animal information.	Having the ability to add and suggest information	Able to see the announcements for forthcoming events.
map	Capable to edit the map	Capable to edit the map	Visibility to the changes	Can manage the content	Use the map to see where the various animals in the zoo are located.
advertisement	Add adds	Able to view ads	Insert ads	Able to view ads	Able to view ads

ticket price	Add the ticket details	Able to extend or diminish the cost of ticket	Able to view ticket details	Manage the ticket sales system	The price of a ticket can be viewed and buy.
search	Add necessary search for animals, events and correct them	Find the necessary animals and events and make the necessary corrections.	Able to search animals and events information	Able to find about the animals.	Search the zoo catalogue using a search engine.

Entity list

Animal Record Management:

The animal records will provide information on the zoo's animals.

Attributes	Type /format	Constraints	Notes
animal_id	NUMBER (10)	PK	Primary Key for the animals
species_name	VARCHAR2 (24)	UPPER	The species' name found in the zoo.
animals_name	VARCHAR2(20)	UPPER	Title of the Creature.
date_of_birth	DATE	NOT NULL	The year the creature was born.
Gender	CHAR	CHECK IN ('M', 'F'), NOT NULL	The animals' gender.
animal_life_span	DATE	NOT_NULL	The projected life expectancy of the animal.
species_type	VARCHAR2(50)	NOT_NULL	Classification of animal species categories.
food_requirements	VARCHAR2(38)	NOT_NULL	The types of food that the animals require on a daily basis.
Environment	VARCHAR2(5)	NOT_NULL	a natural habitat in which a specific animal can survive.
worldwide_population_analysis	NUMBER (10)	NOT_NULL	The number of animal species.
date_joined	DATE	NOT_NULL	The year the animal became a member of the zoo.

animal_dimensions	VARCHAR2(5)	NOT NULL	Animals Height, weight, length, etc.
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Staff record management

The employee information is stored in the staff record, and the zoo employs three distinct sorts of personnel, as outlined below:

1). Administrator

Below shows the list of Admin records:

Attributes	Type/format	Constraints	Notes
admin_id	NUMBER (8)	PK	Primary key for the manager.
admin_first_name	VARCHAR2(13)	UPPER	Admin first name.
admin_last_name	VARCHAR2(13)	UPPER	Admin Last name.
contact_number	NUMBER (30)	NOT NULL	Admin Contact number.
house number	VARCHAR2(5)	NOT NULL	Admin House number.
City	VARCHAR2(10)	NOT NULL	The city where the admin resides.
country	VARCHAR2(5)	NOT NULL	The country in which the admin resides.
date of birth	DATE	NOT NULL	Admins date of birth.

2). Manager

The following is a list of Admin records:

Attributes	Type/format	Constraints	Notes
manager_id	NUMBER (4)	PK	Primary key for the manager.
manager_first_name	VARCHAR2(5)	UPPER	First name of the manager.
manager_last_name	VARCHAR2(5)	UPPER	Last name of the manager.
contact_number	NUMBER (40)	NOT NULL	Contact number of manager.
gender	CHAR	CHECK IN ('M', 'F'), NOT NULL	The manager gender.
city	VARCHAR2(15)	UPPER	The city where the management resides.
country	VARCHAR2(6)	UPPER	The country in which the manager resides.
date of birth	DATE	NOT NULL	Manager's date of birth.

3). Staff

The following is a list of staff records:

Attributes	Type/format	Constraints	Notes
staff_id	NUMBER (10)	PK	Primary key for the staff.
staff_first_name	VARCHAR2(30)	UPPER	Staff first name.
staff_last_name	VARCHAR2(30)	UPPER	Staff Last name.
staff_address	VARCHAR2(40)	UPPER	Staff address.
town	VARCHAR2(20)	UPPER	The city where the Sponsors resides.
county	VARCHAR2(20)	UPPER	The country in which the Sponsors resides.
contact_number	NUMBER (25)	NOT_NULL	Staff Contact number.
gender	CHAR	CHECK IN ('M' , 'F') , NOT_NULL	The Staff gender.
email	VARCHAR2(30)	UNIQUE	Staff email address.
date_joined	DATE	NOT_NULL	The year the staff became a member of the zoo.

Sponsor record management

The following is a list of sponsor records:

Attributes	Type/format	Constraints	Notes
sponsor_id	NUMBER (10)	PK	Primary key for the sponsor.
Sponsor_first_name	VARCHAR2(20)	UPPER	Sponsors first name.
sponsor_last_name	VARCHAR2(20)	UPPER	Sponsors Last name.
contact_number	NUMBER (50)	NOT_NULL	Sponsors Contact number.
gender	CHAR	CHECK IN ('M', 'F'), NOT_NULL	The Sponsor gender.
city	VARCHAR2(30)	NOT_NULL	The city where the Sponsors resides.
country	VARCHAR2(8)	NOT_NULL	The country in which the Sponsors resides.
date of birth	DATE	NOT_NULL	Sponsors date of birth.

Visitor Record Management

The following is a list of visitor records:

Attributes	Type/format	Constraints	Notes
visitor_id	NUMBER (40)	PK	Primary key for the visitor.
visitor_first_name	VARCHAR2(50)	UPPER	Visitor first name.
visitor_last_name	VARCHAR2(50)	UPPER	Visitor Last name.
city	VARCHAR2(60)	UPPER	The city where the Visitor resides.
country	VARCHAR2(20)	UPPER	The country in which the Visitor resides.
house_number	VARCHAR2(50)	NOT_NULL	Visitor House number.
contact_number	NUMBER (80)	NOT_NULL	Visitor Contact number.
gender	CHAR	CHECK IN ('M', 'F'), NOT_NULL	The Visitor gender.
date_of_birth	DATE	NOT_NULL	Visitor date of birth.

Specialist information for specific types of animals:

Animals	Information				
Mammals	Gestational Period	Mammal Category	Average body temperature		
Birds	Nest Construction Method	Clutch size	Wing span	Ability to fly (yes/no)	Plumage color variants (possibly by gender)
Fish	Average body temperature	Water type	Color variants (possibly by gender)		
Reptile and amphibians	Reproduction type	Average Clutch Size	Average number of offspring		

All animals within the zoo are currently located in several specific areas, these are:-

Specific areas	Notes
The aviary	Where all the birds are housed.
The hothouse	Where all reptiles and amphibians reside.
The aquarium	For all fish within the zoo.
The cage/compounds	Housing all mammals.

2.2.2.2 Zoo Website

Some of the zoo's website's essential functional needs have been defined and described in brief in the table:

Features	Explanation
Home page	This page is the major attraction, and it includes a navigation bar with connections to various areas of the website. It's also possible to update it on a daily basis.
Contact page	This page allows users to contact the site's owner, and it includes an area where visitors may fill in their name, address, leave a comment, and so on, giving users a quick and easy way to ask the company questions.
Search bar	A search term is a graphical feature on a website that allows a user to seek and obtain data from a database.
Photo Gallery	This page has a collection of animal and bird images with little details such as what the species is, where it originated from, and when it arrived at the zoo, which might assist the user in recognizing the animal.
Ticketing system	This is an easy-to-use system that allows users to book tickets in advance or on the spot.
Events and program	This page contains information on the zoo's calendar and timetable, allowing users or visitors to organize their visit to the zoo.
About us	This page provides basic information on the zoo's history, as well as a place to highlight the founder.

2.2.3.3 Visitor Kiosk information System

A kiosk system is a user interface that is intended to be used interactively. It is made up of several sections that are categorized into three categories: "body," "brain," and "heart." The heart contains various components such as card readers, printers, bill, and coin collection, among others. The body is the enclosure that holds the display and protects and secures the inner workings, the brain is the smartest part that makes the entire system run, and the brain is the smartest part that makes the entire system run. The following are some of the user-friendly aspects of the kiosk information system:

1. It displays information about the animals and birds, such as what sort of behavior they have, what their names are, where they come from, and so on, in the form of a picture or a short video clip.
2. It aids in navigating, whether for the location of animals and birds, programs and activities, gift and book stores, or food and beverage stalls.
3. It's a fantastic approach to give consistent message, sales, and service while still being able to advertise, upsell, or present the customer with alternatives as needed.
4. It is fast, reliable and highly secure system.
5. Visitors are not disrupted by a language barrier between personnel and themselves because the technology allows them to speak in their own language.
6. It asks visitors questions about their experience at the zoo and collects input to help the zoo improve.

2.2.3 Performance Requirements (Records Management System)

Performance requirement is a specification that outlines the system's operation capabilities as well as limitations that help it work better. It specifies how a system must behave and the functional restrictions that must be met. Scalability, maintainability, speed, portability, security, capacity, reliability, usability and many other problems are addressed by performance Requirements. Performance Requirements are used to solve critical quality issues in software systems.

2.2.3.1 Speed

It specifies how quickly a software system or a specific piece of software responds to store data and records of animals, staff, members and sponsors. The speed of an application determines how quickly it maintains its data and records. Given the current number of users, this measure describes how long a user must wait before passing data and it refers to processes that are carried out in the background and are not visible to users. It refers to the system's ability to respond to keep record of animals, staffs, members, sponsors and visitors. User can keep and retrieve records of Animal Medical Record, Date and Destination of animal transfer, Date and Cause of Animal Death quickly and effectively.

2.2.3.2 Capacity

This feature reveals the storage capacity of your system, which is determined by its kind and attributes. A system's capacity refers to the quantity of storage it provides. It indicates the maximum amount of data that can be stored in the system and the maximum number of actions that can be executed at the same time. This implies more data of animal, member, staff and sponsors will be processed, and more transactions would be completed. Information referring animal's health, staff work time, sponsors and members details can be easily stored in the system.

2.2.3.3 Reliability

It estimates the probability that the system or its component will operate without failure for a specified amount of time under preset conditions. Defining critical failure, time, and regular usage conditions can be difficult and more straightforward technique is to count the number of serious bugs discovered in production over a period of time or calculate the mean time to failure. Probability percentage, the number of critical failures and the mean time between failures are three methods for calculating it. Small populations of held wildlife require intensive management that incorporates information from a variety of scientific fields, including genetics, nutrition, ethology, and veterinary medicine, to ensure their long-term survival. So, the system is designed to operate without failure with minimum amount of regular maintenance.

2.2.3.4 Usability

The usability of a system refers to how difficult it will be for a user to learn and operate it. It reveals how well people can learn and use a system. It refers to the average time it takes for a user to achieve their objectives, the number of data a user can store without assistance, the number of records stored without errors, how easy the interface, buttons, and headings are to understand and so on. In order to achieve effective animal management and disease control programs, modern zoos require excellent systems for acquiring important information from the field that are simple for personnel to use in executing necessary measures. So, Users can operate the system quickly and efficiently in the very short period of time.

2.2.4 Performance Requirements (Zoo Website)

2.2.4.1 Speed

Some of the features that characterize the system's rate and momentum:

Responding time:

The software's online search response time is quick, but it is also dependent on the number of simulators being used at the same time.

Throughput:

The number of data transfers per second for processing determines the software's speed; for example, this software can handle 100 transitions per second.

2.2.4.2 Capacity

The capacity of the system and the number of users it can manage at any given time determine the amount of data that can be stored. The system must be able to handle a large number of users at once.

- a. At any particular time, 1000 persons can join the online system.
- b. The system has such a storage capacity of 120 GB.

2.2.4.3 Reliability

The term "reliability" refers to how safe and accurate a system is to use. To complete the objective, the system must be dependable. If this is not done, the system may not provide correct results. This online system will be open for 165 hours each week, out of a total of 168 hours, and will be a reliable system to complete the assignment. However, because the existing system is paper-based, we can only access information about the zoo when it is open, and data can be lost or misplaced due to the paper-based method.

2.2.4.4 Usability

The concept "usability" refers to the ease with which a system can be operated. Anyone with a basic understanding of how to use a computer and navigate a website can run this system. However, when it comes to the current paper-based system, only zoo workers can understand and use it.

2.2.5 Performance Requirements (Visitor Information System)

2.2.5.1 Speed

Some of the characteristics that determine the system's speed and momentum

OPPORTUNITIES AT THE POINT OF SALE:

Nothing is more frustrating than attempting to make it to a particular event or performance, arriving just in time, and discovering a massive wait for tickets. However, with a self-service option that includes a credit card reader, you can keep lines moving and ensure that your guests don't miss a single second of the performance. Zoo kiosks are an excellent way to provide consistent message, sales, and service, as well as the possibility to promote, upsell, or provide the visitor with many alternatives.

2.2.5.2 Capacity

DEMAND OF STAFF AS REDUCED:

There are several potential applications for zoo kiosks in streamlining operations and better serving the public while optimizing zoo resources. The zoo kiosks at the Detroit Zoological Society are seen here. Users may discover more about the current attractions and sign up for the zoo's newsletter by visiting these wall-mounted kiosks.

Patron:

To be functional and capable of meeting stakeholders' requests, the suggested kiosks would require certain technology, installations, staff employees, and resources. RFID technology will be used to support bar codes on all publications including library cards (Raj and stha, 3). A book drop counter, tagging station, self-check in and out system, material shelves, and an anti-theft detection system will all be available (Surname, 4)

2.2.5.3 Reliability

If you are a zoo or aquarium with a problem to solve or a major change looming on the horizon, the first step is determining what data you need and how to get it. You can to customize and deliver a kiosk-based Visitor Survey Program that will help you get the right answers and ultimately achieve your goals.

You can use your survey in all kinds of different ways. In addition to capturing visitor demographics and experiences, the Detroit Zoo includes questions about their food and beverage service vendor. The vendor uses the metrics the zoo kiosks collect for an incentive program, motivating employees to provide superior service each time, every time.

2.2.5.4 Usability

It's best to go into any event as knowledgeable as possible in most parts of life (but especially in a zoo). A kiosk may be a fantastic method for visitors to learn more about the various creatures in your displays, avoiding the type of surprise that our long-necked buddy had when he encountered his first peacock.

Self-service information kiosks are fun and informative, and they're a terrific way to draw attention to upcoming events and attractions. They provide visitors with specific information that they may otherwise overlook, and they're also a great way to promote unique attractions, promotions, or subscriptions.

2.2.6 Design Constraints (All software components)

a) Target operation system: The purpose of this project is to provide a user-friendly online system for the Zoo to manage its information where they can store the zoo daily records, zoo information and online ticket system.

b) Distributed or local architecture: A distributed architecture is one in which components are presented on different platforms and several components can collaborate with one another over a communication network to achieve a specific objective or goal. It is centrally located in the system and maintains or supports the many components of a distributed system.

c) Hardware requirements: The hardware requirements for operating this system are mentioned below:

- Operating system: windows 8 or later
- Processor: Minimum 1GHz
- Ethernet connection (LAN) OR a wireless adapter (Wi-Fi)
- Minimum Graphics processing unit (GPU) or video memory.
- 1280x1024 or Larger Screen resolution
- Minimum 2GB system memory RAM is required
- Hard Drive: Minimum 32 GB is required
- Audio hardware such as sound cards and speakers is required.

d) Front-end graphic styles: The system is built using a consistent company format.

e) Programming languages to be used: HTML, CSS, JavaScript and Java are the programming language that has been used to create the system.

f) Application packages: MS-Word, database, graphics and presentation software are the application software that has been used to develop this system.

g) Development standards: IOS, IEC, and IEEE are the standards that were used to create this system.

h) Design approach: For the mock-up layout, we used a software program referred to as JUSTINMIND and for wireframe layout of the system we used diagrammatic approach.

Company Details:

As a software and mobile application development company, the zoo was formed in 2012 by a group of enthusiastic zoologists (Claybrook zoo) with the goal of safeguarding endangered species in the environment. The team has a great passion for animals and birds. They have been doing this project for some years and look forward to better protection as a software development. Claybrook Zoo provides the customized arrangement to clients focusing on the animal conversation and protection.

Contact Details – Claybrook Zoo

Manager: Matthew Jones Address: 45 Zoo Lane, Eastlands, North Yorkshire, YR12 3TH, UK

Purchased URL/Web-address: www.Claybrook-Zoo.com

Opening Times (BST) 10-8pm all other times of the year 12-6pm

Typical Staffing Levels (20-120 staff dependent on session - many are casual positions for the summer months).

Estimated total visitors per year: 1.4 million (Approx. average of 27,000 per day)

Structure of Software:

We put a lot of effort into each Project we take on. Our team follows a very systematic and coordinated approach to ensure that you are constantly informed and in the loop throughout the process. The wireframe makes a difference the client in understanding the total stream of the application, for occurrence what each portable screen will incorporate, what will be the

substance, the situating of the substance on the screen etc. The email includes various wireframe forms that can be provided to clients based on their hiring needs, ranging from basic to advanced, with ticket prices and payment systems, operating systems, and web browsers. Along with user-friendly information, the webpage has an intuitive search option with efficient components. Animal categories are categorized using a textual and graphical range, resulting in a user experience that is intuitive and complete. We offer an ease to our clients where a program can be altered in future and in case client needs to add modern highlights within the application, we make it simpler so that they can effectively update the application by including unused highlights as per the necessity of the environment.

2.2.7 Commercial Constraints (Total Project)

At this level, we have estimated the time scale and cost for this project based on factors such as:

- I. Estimated staff costs and the total hours of work required to complete the task.
- II. Project costs/resources.
- III. Required Profit Margin.

The following is a breakdown of the cost of the zoo project:

Project information			
Total length of project	77 days		
Hourly wage	£50 per person		
Number of workers	10		
Project breakdown			
Section	Weeks	Hours per week per person	Total wage
Requirement specification	4	4	£3796.55
Design and analysis	7	8	£7593.09
Project built breakdown	12		
Console	30%	13	£40000
Website	25%	11	£35000
Mobile application	10%	13	£35000
Testing	30%	15	£15000
Evaluation	5%	14	£10000
Additional cost			

Reason	Cost		
Software/hardware	£248400.40		
Office cost (£100 per week)	£45000.25		
Total built cost	£ 1,050,000		
Other expenses (10%)	£1526.77		
Profit margin (30%)	£4580.32		
Profit rights	Negotiable		
Total project cost	£ 2,500,000		

3 System Interface Designs:

Planning the System Interface incorporates two sections. One is to plan the User Interfaces and the other is to plan the connection points that communicate with different frameworks, which are indispensable to the working of the framework that is being planned. There are two parts in designing the system interface. One is to design user interfaces, and the secondly is to design interfaces that connect with other systems, which are both essential to the system's functioning.

Designing the interface involves the steps given below:

- Identify the Related Users - Determine the characteristics of the users who will interact with the system in order to fulfill the criteria under consideration in this iteration. Focus on characterizing the primary users because they are involved in the majority of the interactions.
- Identify the System's Key User Interface Elements - Identify the system's primary user interface windows. The user will interact with primary windows the most (those user-interface components that are fundamental to the user's mental image of the system).
- Define the Navigation Bar - The Navigation Map establishes the valid navigation pathways for the user to perform system functionalities through the user-interface components. It is not required to include all conceivable routes via the user-interface components. Its purpose is to act as a user interface road plan for the system.

Designing the System-to-System Interfaces involves:

- Identify and analyze the dependencies - the necessity for interaction with other systems and inside the subsystem must be recognized and assessed.
- Determine the API to be exposed - take into account the other systems' or subsystems' dependencies and determine the API to be exposed that will meet the demands of the dependent systems/subsystems.

3.1 Draft Interface Designs for RMS

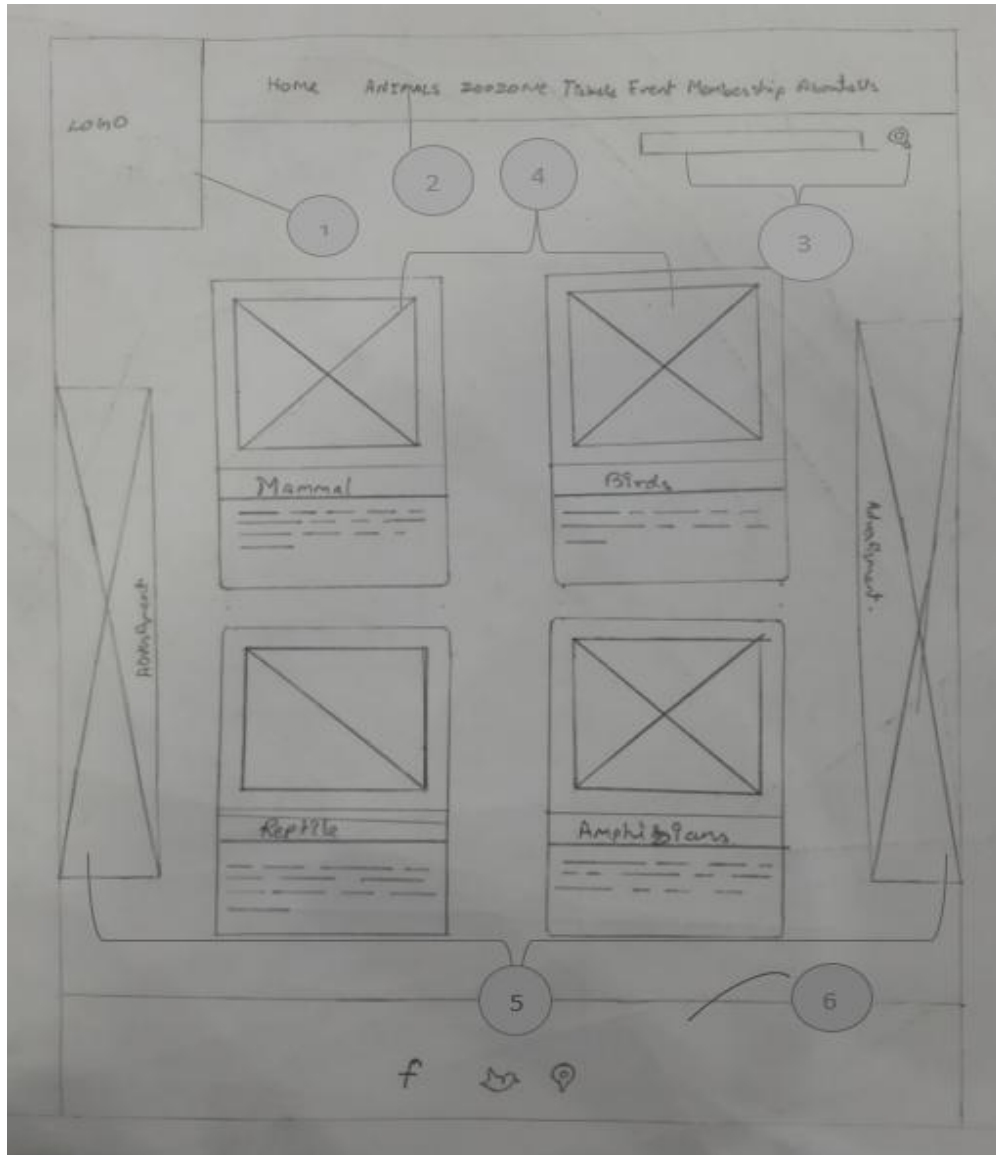
3.1.1 Wireframes

A wireframe is a physical expression of a website service. A wireframe is a method to sketching project sites and functionalities on a page while taking into consideration user needs and expeditions. Early in the development process, wireframes are used to establish the basic structure of a website before visual content and design are added.

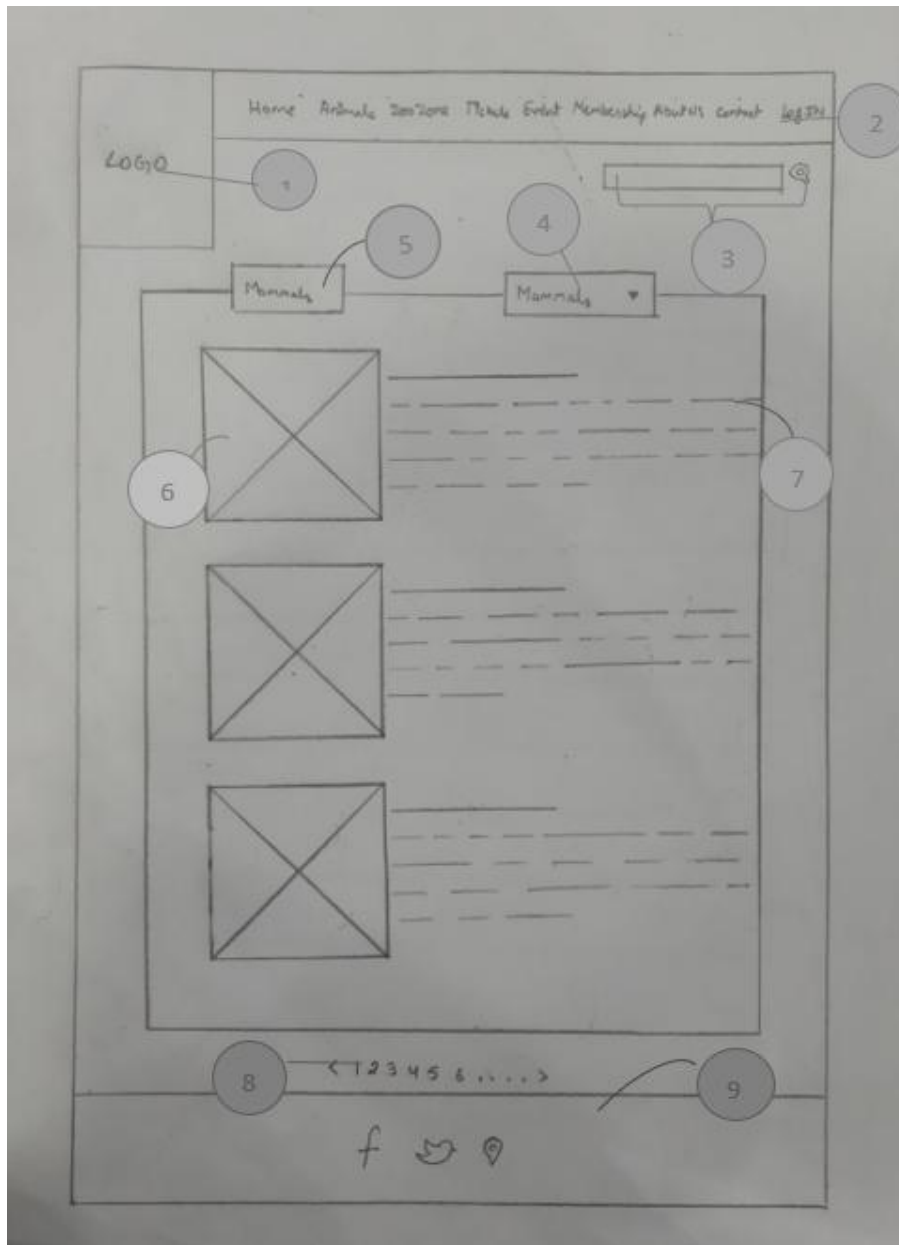
A wireframe is a web page layout that determines how interface elements will appear on key pages. It's an essential step in the interaction design phase.

A wireframe's purpose is to provide a visual representation of a page early in the project to achieve stakeholder and project team approval before the creative process starts. Wireframes also are helpful for prototyping.

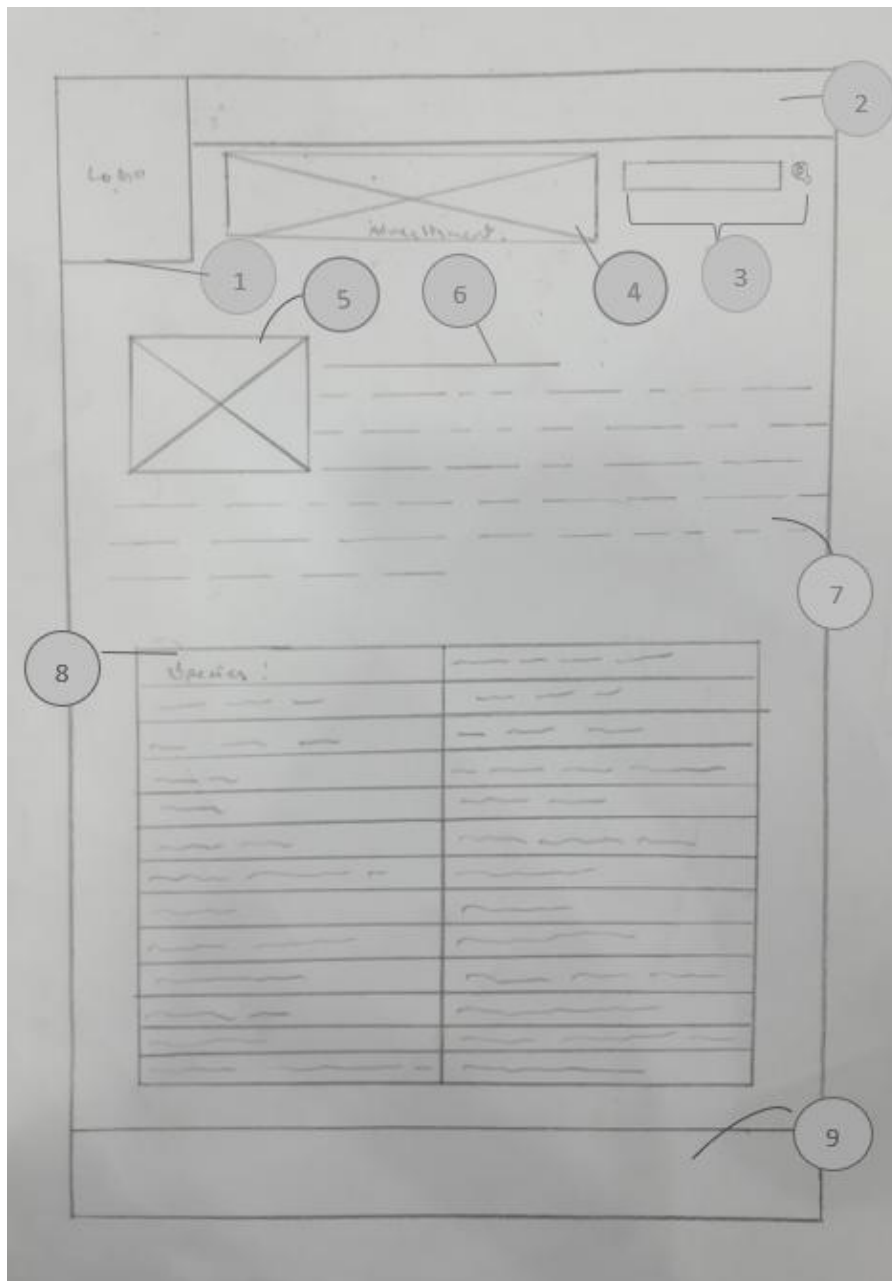
Animal RMS wireframe:



1. This is where the zoo logo kept.
2. This is the Navigation Menu
3. Search bar for searching animals RMS.
4. Pannels of animals based on species
5. This is the place for Advertisement
6. This is the Footer with contact info.

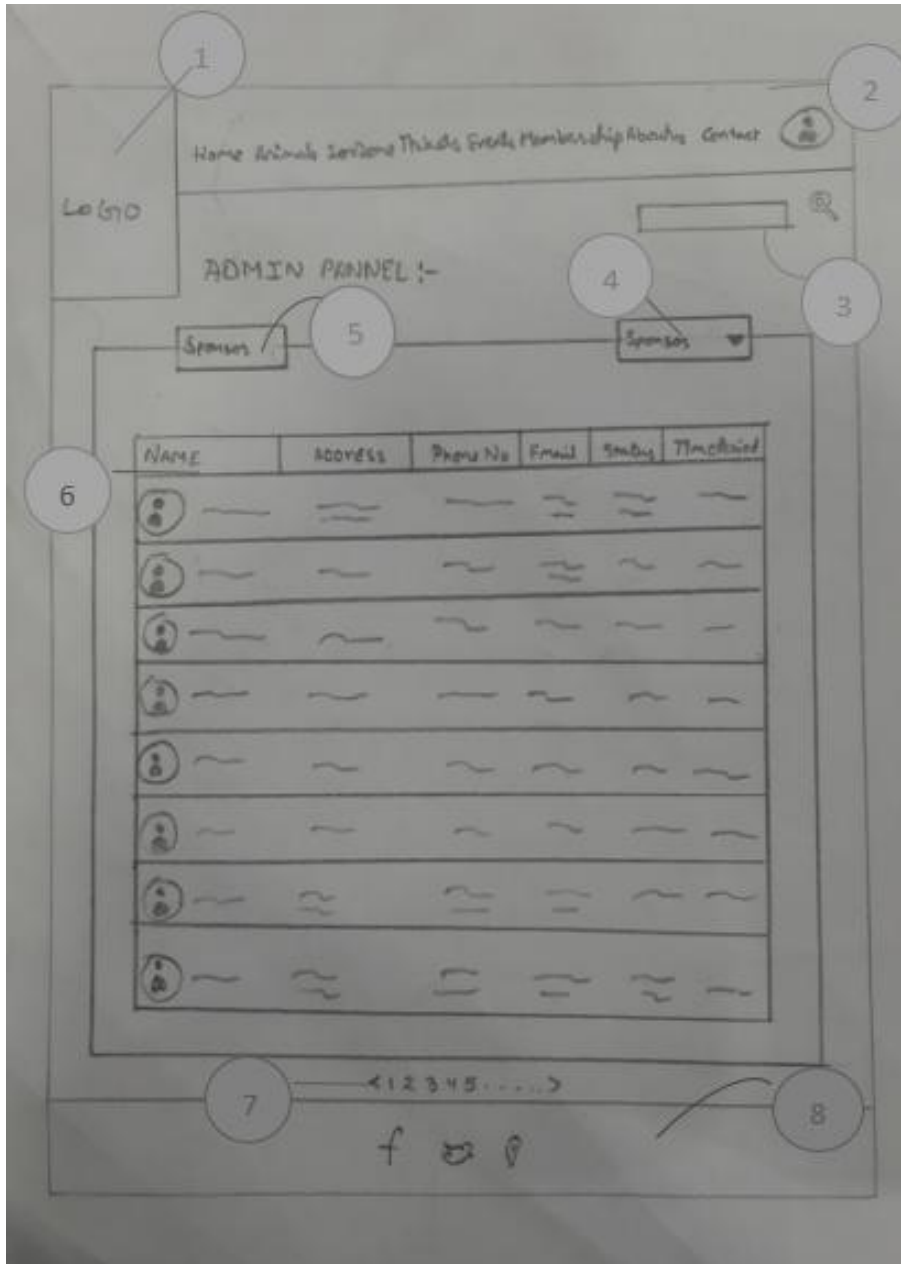


1. This is where the zoo logo kept.
2. This is the Navigation Menu
3. Search bar for searching animals RMS.
4. Drop down menu for animal species.
5. This is the animal Species table.
6. Images of animal.
7. Brief about the animal.
8. This is where the page is indication.
9. This is the Footer with contact info.



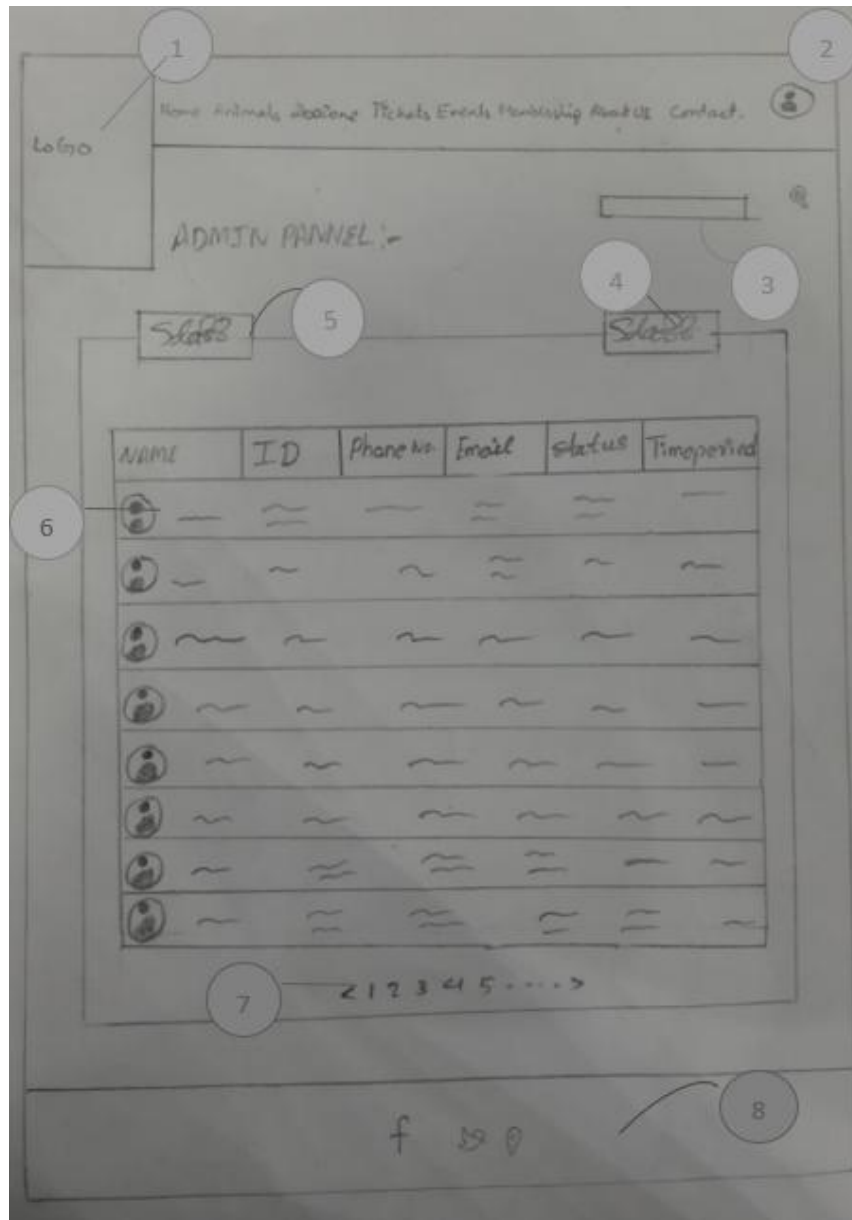
1. This is where the zoo logo kept.
2. This is the Naviagtion Menu
3. Search bar for searching animals RMS.
4. This ia the place for Advertisment
5. Images of the animal.
6. Name of the animal.
7. Brief about the animal.
8. Record of tabe about animal.
9. This is the Footer with contact info.

Sponsor RMS Wireframe:



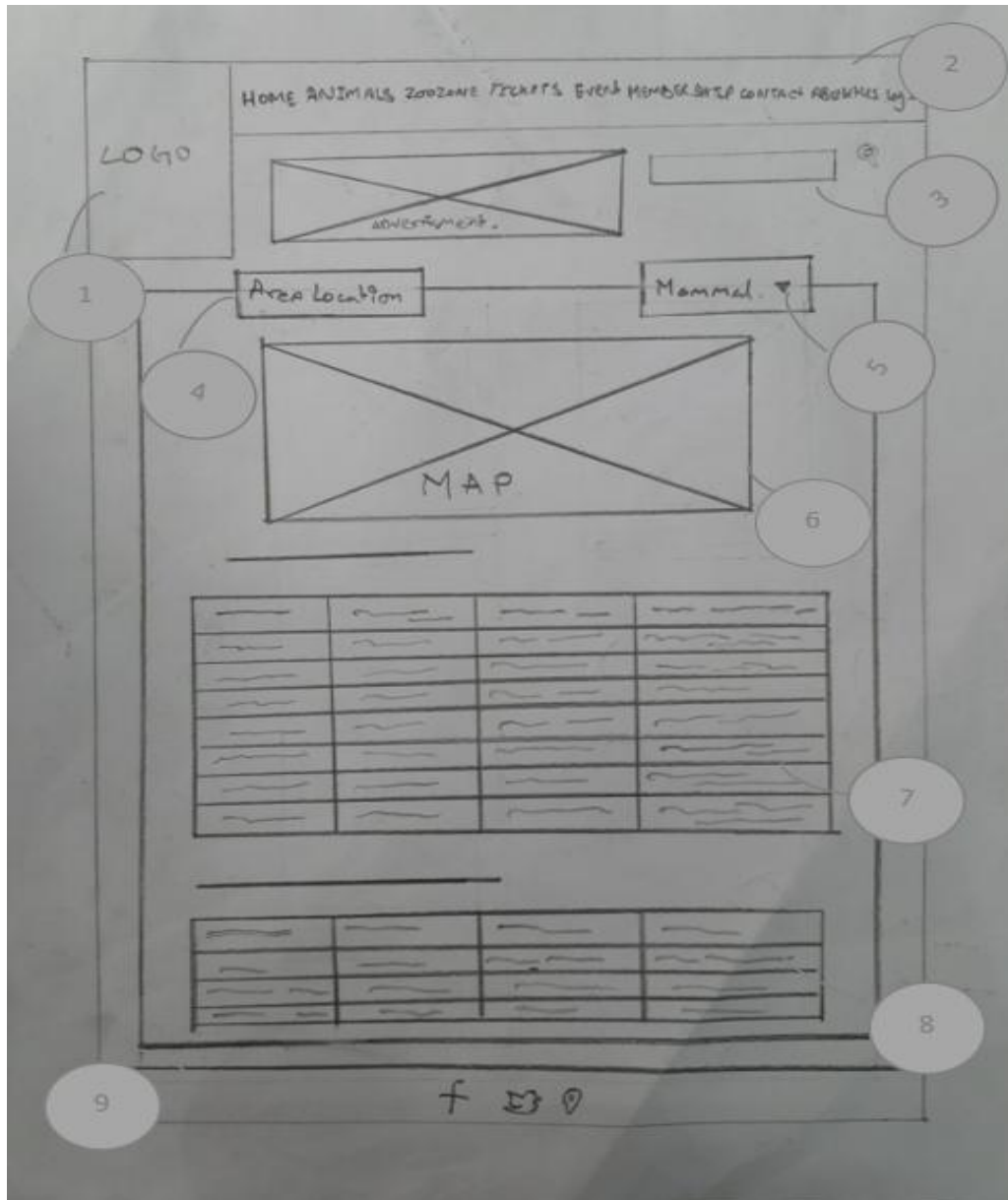
1. This is where the zoo logo kept.
2. This is the Naviagation Menu
3. Search bar for searching animals RMS.
4. Drop down menu for animal species.
5. This is the Sponsor Table.
6. Record table for Sponsor with photo, name and id etc.
7. This is where the page is indication.
8. This is the Footer with contact info.

Staff RMS Wireframe:



1. This is where the zoo logo kept.
2. This is the Naviagtion Menu
3. Search bar for searching animals RMS.
4. Drop down menu for staff.
5. This is the Staff Table.
6. Record table for Staff.
7. This is where the page is indication.
8. This is the Footer with contact info.

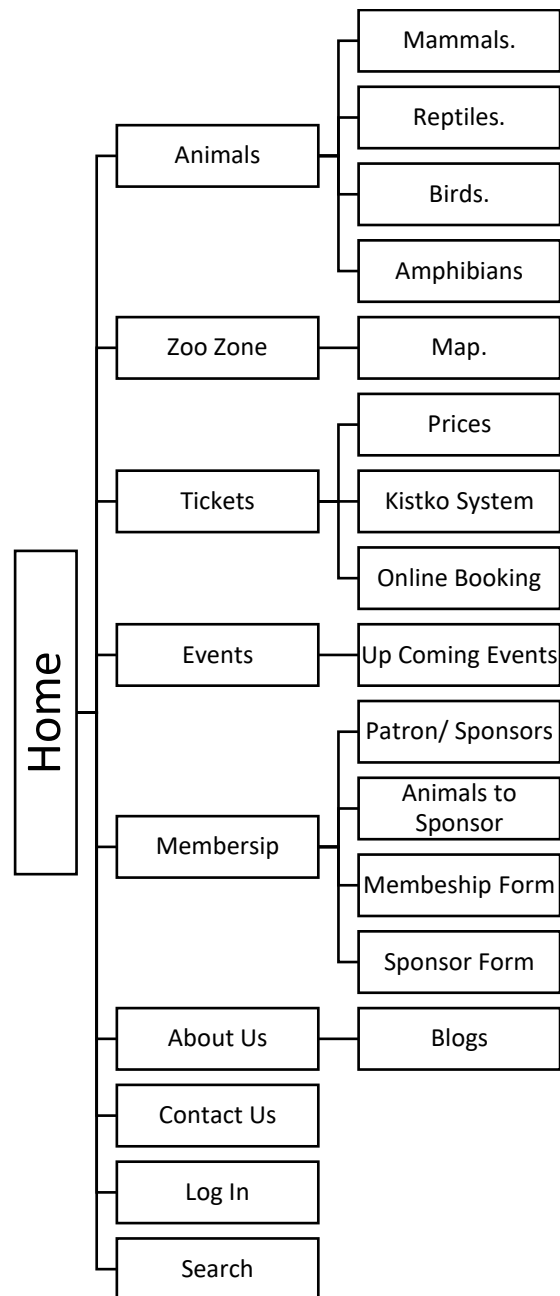
Animal Location RMS Wireframe:



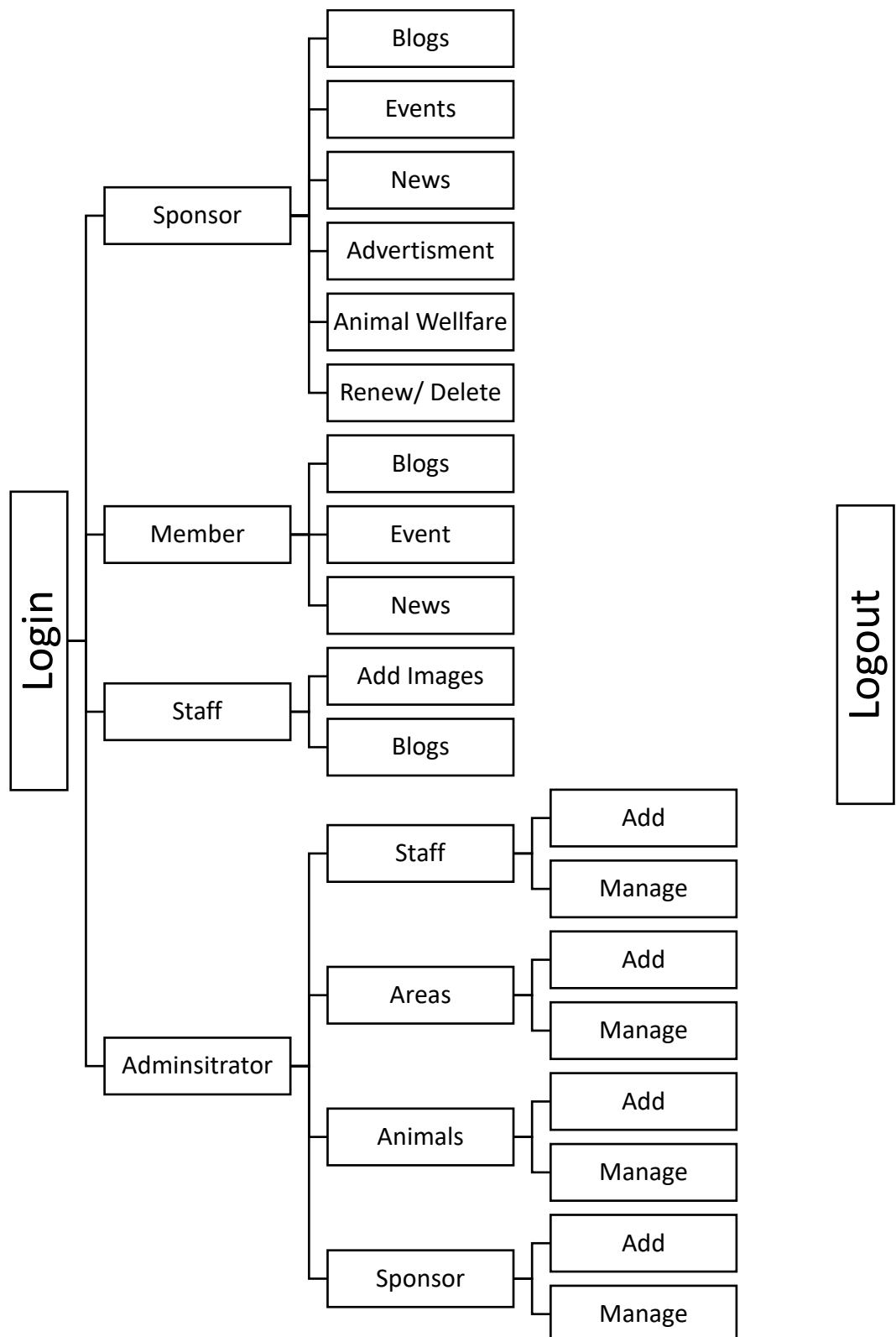
1. This is where the zoo logo kept.
2. This is the Naviagtion Menu
3. Search bar for searching animals RMS.
4. This is the Aminal Location Table.
5. Drop down menu for animal location.
6. This is palce is for map.
7. Area location table for Animals.
8. This is where Animal feeding list goes.
9. This is the Footer with contact info.

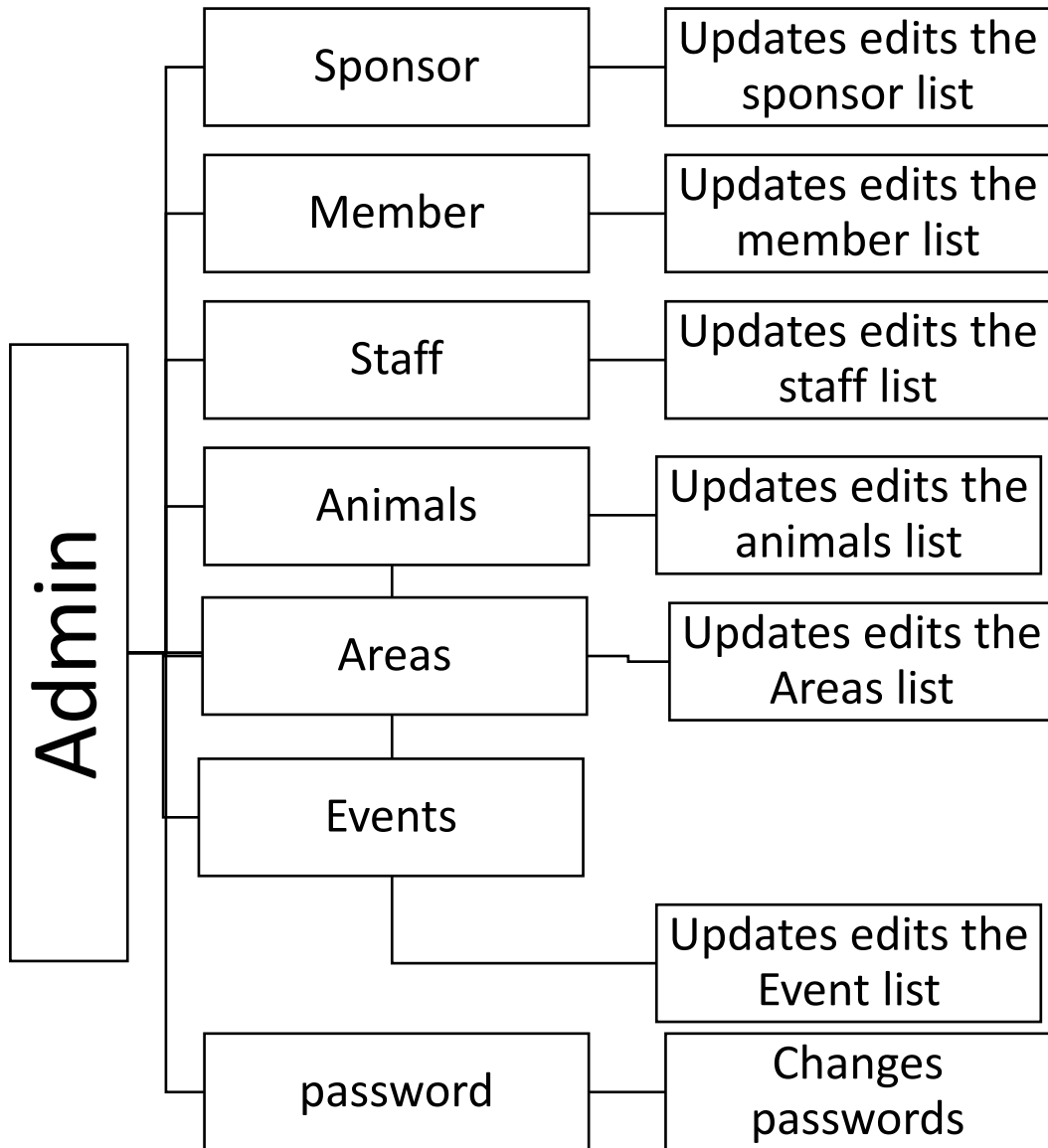
3.1.2 System Navigation Diagram

Front End:



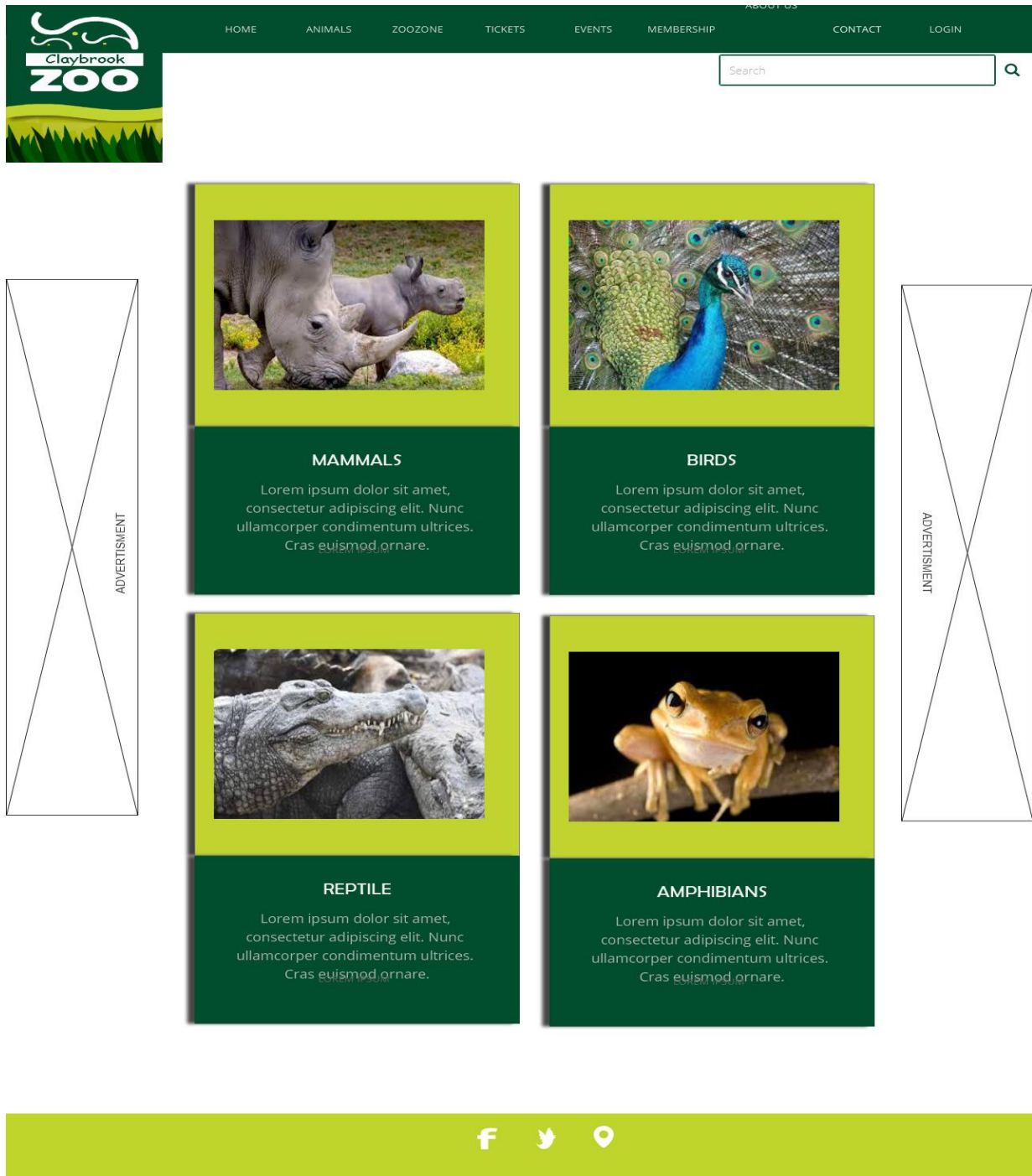
System Login:

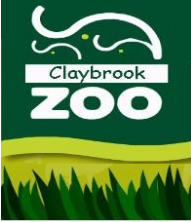




3.1.3 System Screen mock-ups:

Animal RMS Mockup:





[HOME](#)[ANIMALS](#)[ZOOZONE](#)[TICKETS](#)[EVENTS](#)[MEMBERSHIP](#)[ABOUT US](#)[CONTACT](#)[LOGIN](#)

Search

Mammals

Mammals



CRAS EUISMOD ORNADE

Rhinoceros

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc ullamcorper condimentum ultrices. Cras euismod ornare laoreet. Quisque vel efficitur quam, eu molestie odio. In hac habitasse platea dictumst.



CRAS EUISMOD ORNADE

Lowland Gorilla

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc ullamcorper condimentum ultrices. Cras euismod ornare laoreet. Quisque vel efficitur quam, eu molestie odio.



CRAS EUISMOD ORNADE

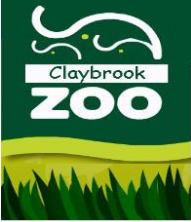
Elephant

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc ullamcorper condimentum ultrices. Cras euismod ornare laoreet. Quisque vel efficitur quam, eu molestie odio. In hac habitasse platea dictumst.

< 1 2 3 4 5 6 >

f

83



[HOME](#)[ANIMALS](#)[ZOOZONE](#)[TICKETS](#)[EVENTS](#)[MEMBERSHIP](#)[ABOUT US](#)[CONTACT](#)[LOGIN](#)

ADVERTISEMENT

Mammals

Mammals



Lowland Gorilla

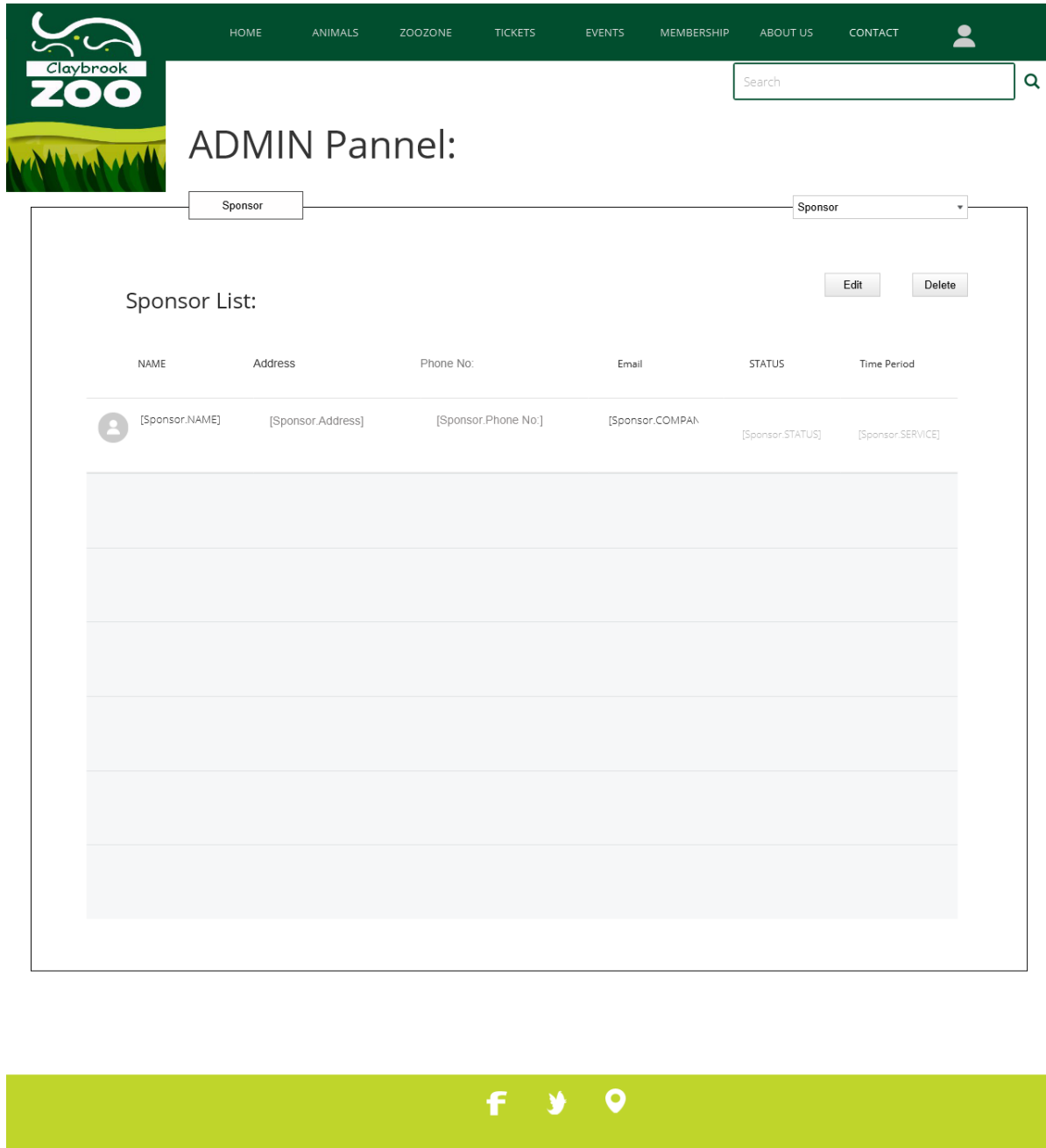
Lorem ipsum dolor sit amet, sapien etiam, nunc amet dolor ac odio mauris justo. Luctus arcu, urna praesent at id quisque ac. Arcu es massa vestibulum malesuada, integer vivamus elit eu mauris eus, cum eros quis aliquam wisi. Nulla wisi laoreet suspendisse integer vivamus elit eu mauris hendrerit facilisi, mi mattis pariatur aliquam pharetra eget.

Classification. :	Mammal - Primate
Animal Location Code :	MC1-A1
Average Life Span :	30–50 years
Average Animal dimension:	height 1.65–1.75 metres (5 ft 5 in–5 ft 9 in), and in weight 140–200 kg (310–440 lb).
Dietary Requirements (Daily)	2.2 Kg Green Leaf Matter 1.5 Kg Assorted Fruit
Natural Habitat Description	West and Central Africa

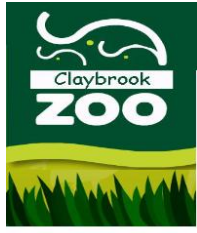
< 1 2 3 4 5 6 >

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Sponsor RMS Mockup:



Staff RMS Mockup:



HOMEANIMALSZOOZONETICKETSEVENTSMEMBERSHIPABOUT USCONTACT

Search

ADMIN Pannel:

Staff

Staff

Staff List:

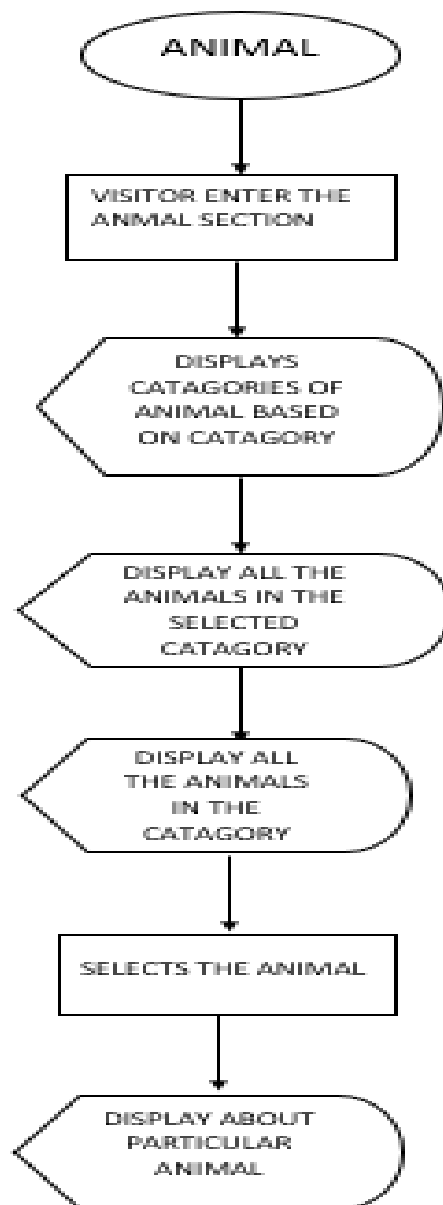
EditDelete

NAME	ID No:	Email	Phone No:	STATUS	SERVICE
 [staff.NAME]	[staff.ID No.]	[staff.Email]	[staff.Phone No.]	[staff.STATUS]	[staff.SERVICE]



3.1.4 System Activity Event Diagrams

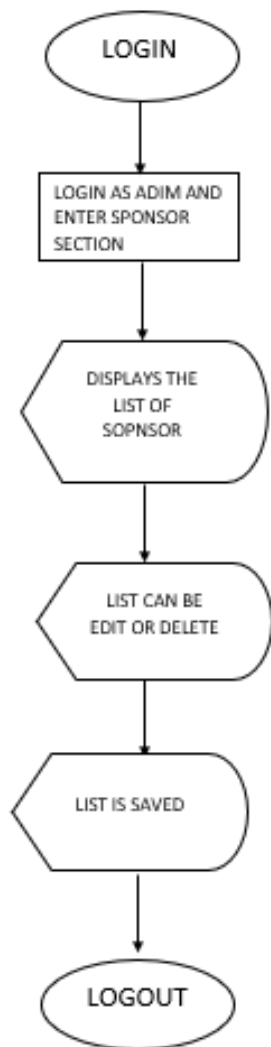
1)Animal RMS:



2) Staff RMS:



3) Sponsor RMS



3.2 Design Revisions

We tried to keep the system as simple and user-friendly as possible while designing it. We tried to maintain our system as clear as possible. The system was developed in accord with the user's needs. We've incorporated a few additional features in response to client requests.

We sought to make our system stand out from the competition. As according to our clients, we wanted to add some unique features to our system. Like adding sponsor name on the sign boards on the zoo premises.

3.3 Draft Interface Designs for Zoo Website

Draft interface design is a diagrammatic representation that depicts the operation of all system parts. It aids in the creation of system wireframes.

A wireframe is a physical representation of an online service. A wireframe is a method of laying up material and functionality on a page while taking into account user requirements and desires. Early in the development process, wireframes are used to build the fundamental structure of a page before visual design and content are added.

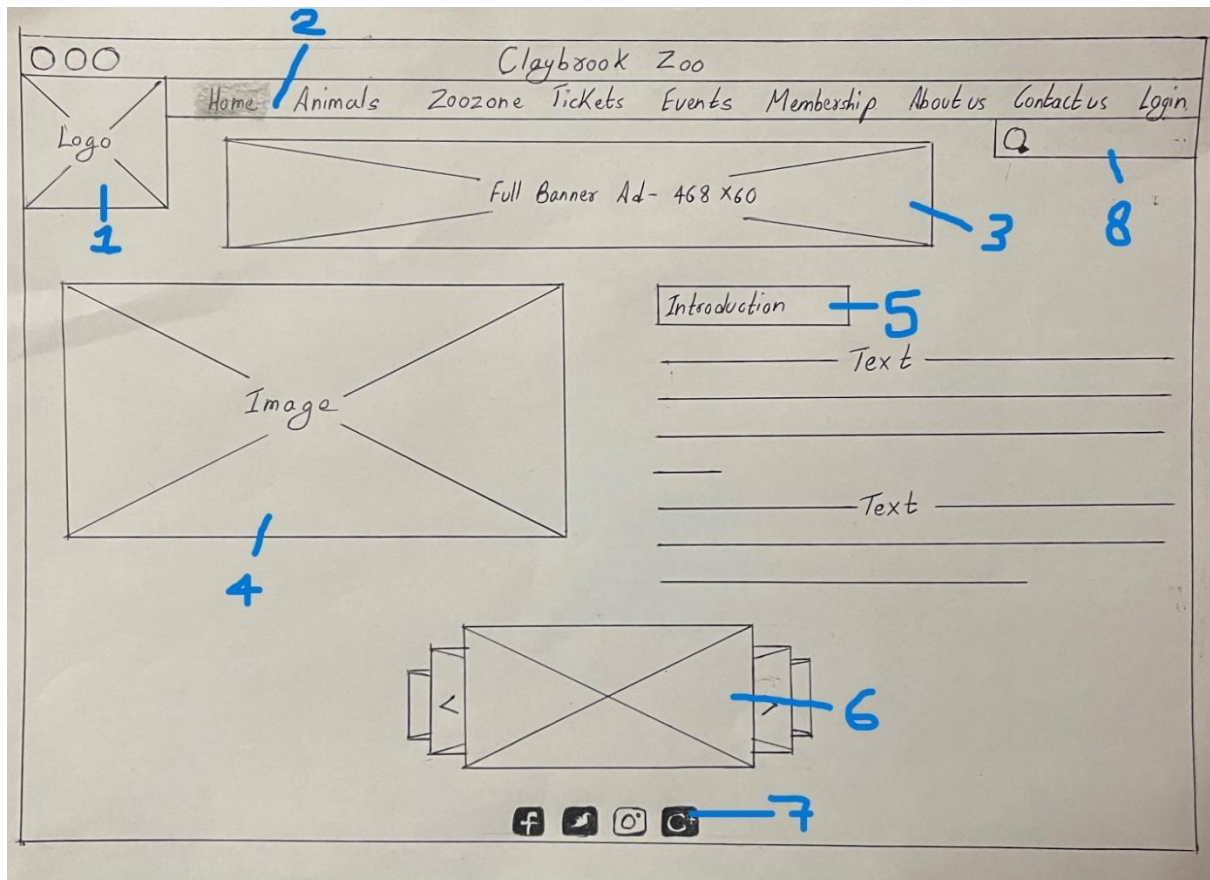
A wireframe is a web page layout that specifies which interface components will appear on important pages. It's an important step in the interface design process.

A wireframe's purpose is to offer a visual overview of a page early in a project to gain stakeholder and project team approval before moving forward with the creative phase. Wireframes may also be utilized to build the global and secondary navigation, ensuring that the site's vocabulary and structure fit the needs of the users.

In practice, the wireframes guarantee that page information and functionality are properly positioned depending on user and business requirements. They may also be utilized as an excellent way for project team members to agree on the project's vision and scope as the project progresses.

3.1.1 Wireframes

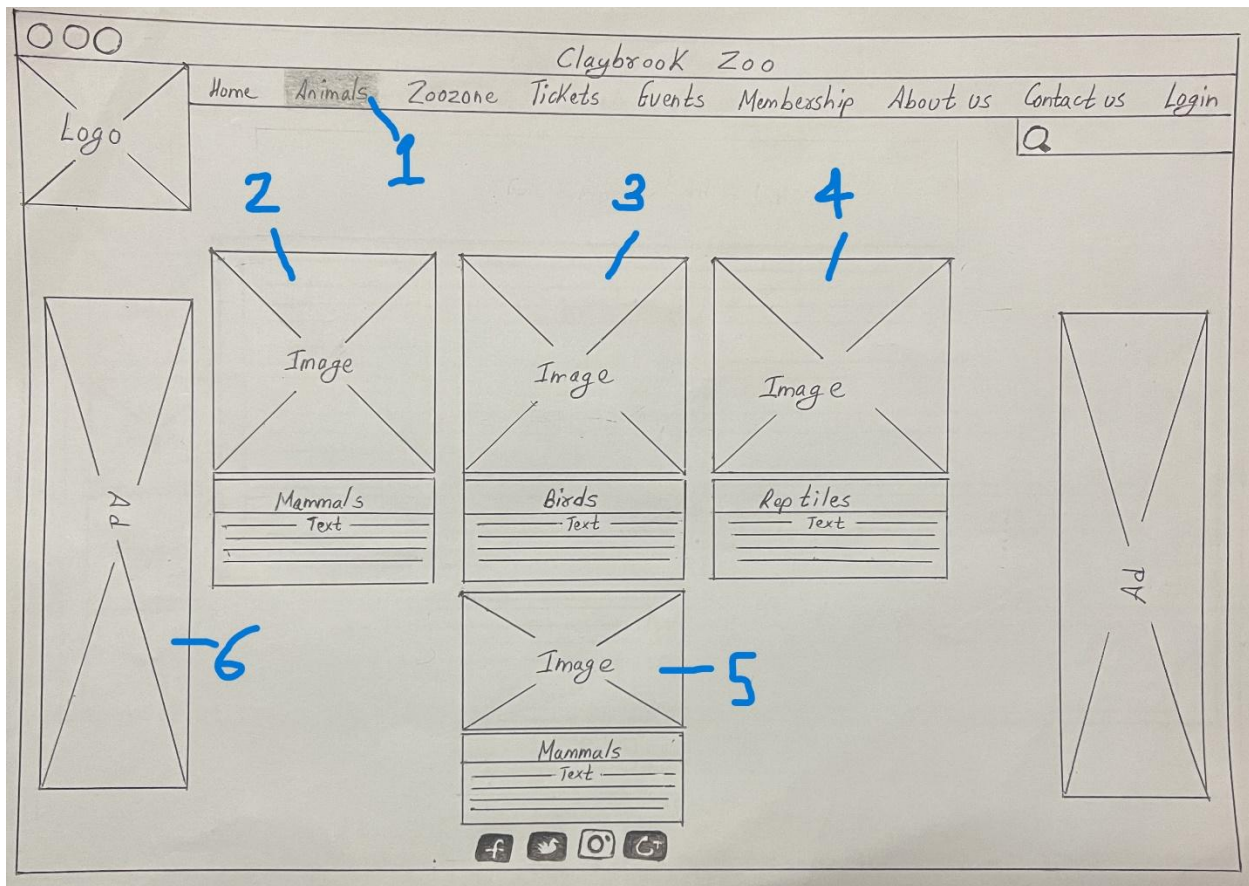
Home:



Description:

1. It is the logo of Claybrook Zoo.
2. This is home page.
3. Banner for displaying advertisements.
4. An Image which represents Claybrook Zoo.
5. Introduction to the Zoo.
6. Slide show Images of Zoo.
7. Social contacts of the zoo where users can contact the zoo.
8. Search bar to search data about zoo.

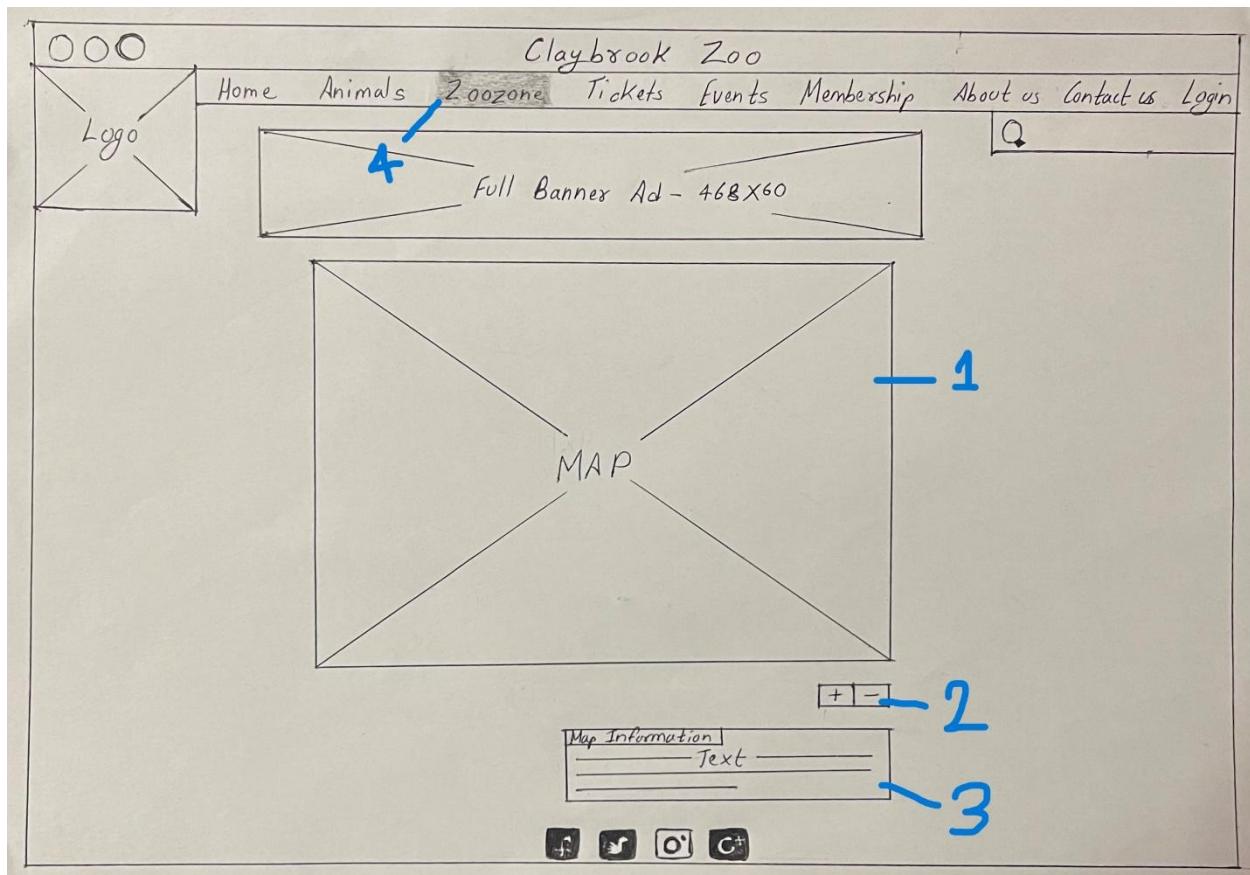
Animals



Description:

1. This is Animal page.
2. Tab about Mammals with an image.
3. Tab about Birds with an image.
4. Tab about Reptiles with an image.
5. Tab about Mammals with an image.
6. Banner of Advertisements.

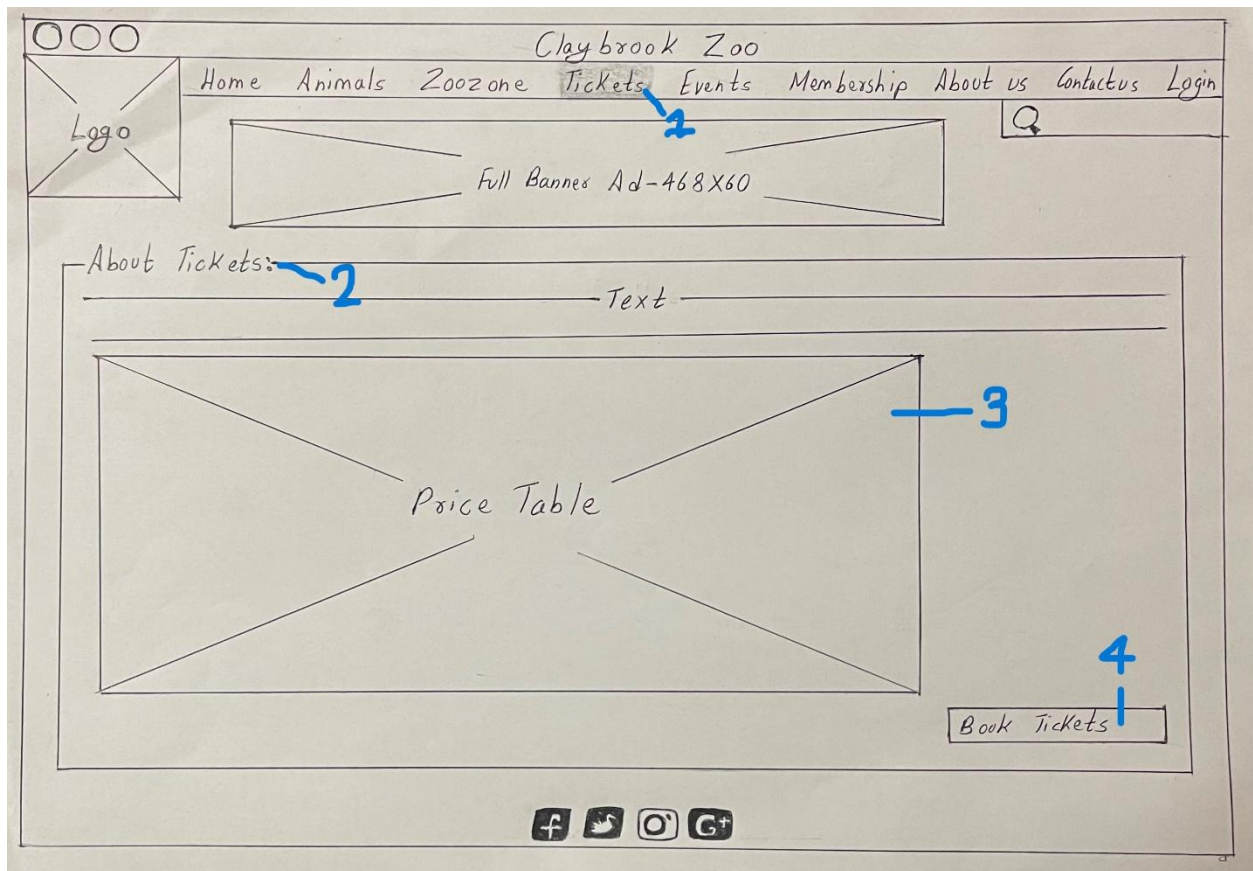
Zoo zone



Description:

1. Map area of the zoo.
2. Zoom in and zoom out button for map.
3. Information about the signs in map.
4. This is Zoo zone page.

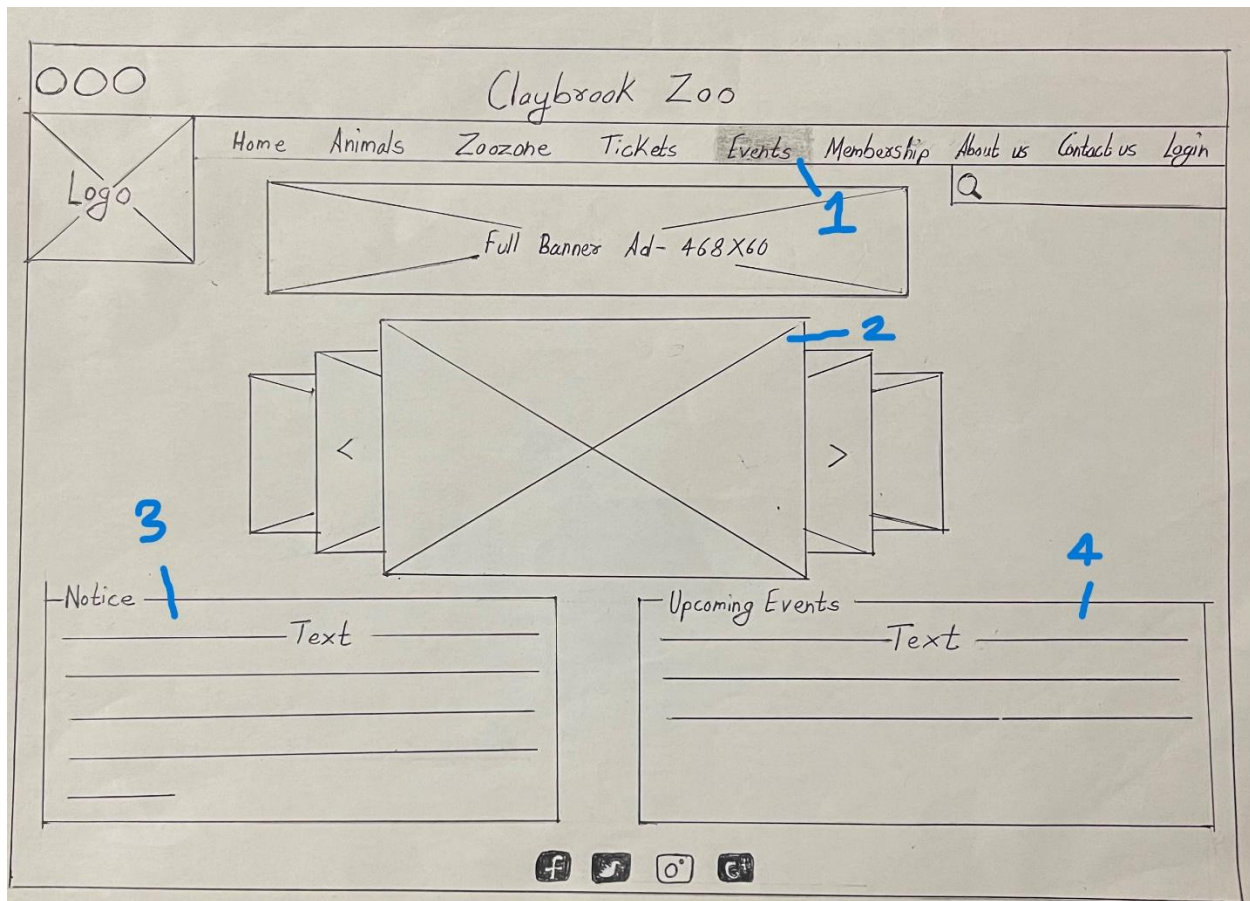
Tickets



Description:

1. This is Tickets page.
2. Information about tickets is here.
3. This is the price table of the tickets.
4. Button for booking tickets.

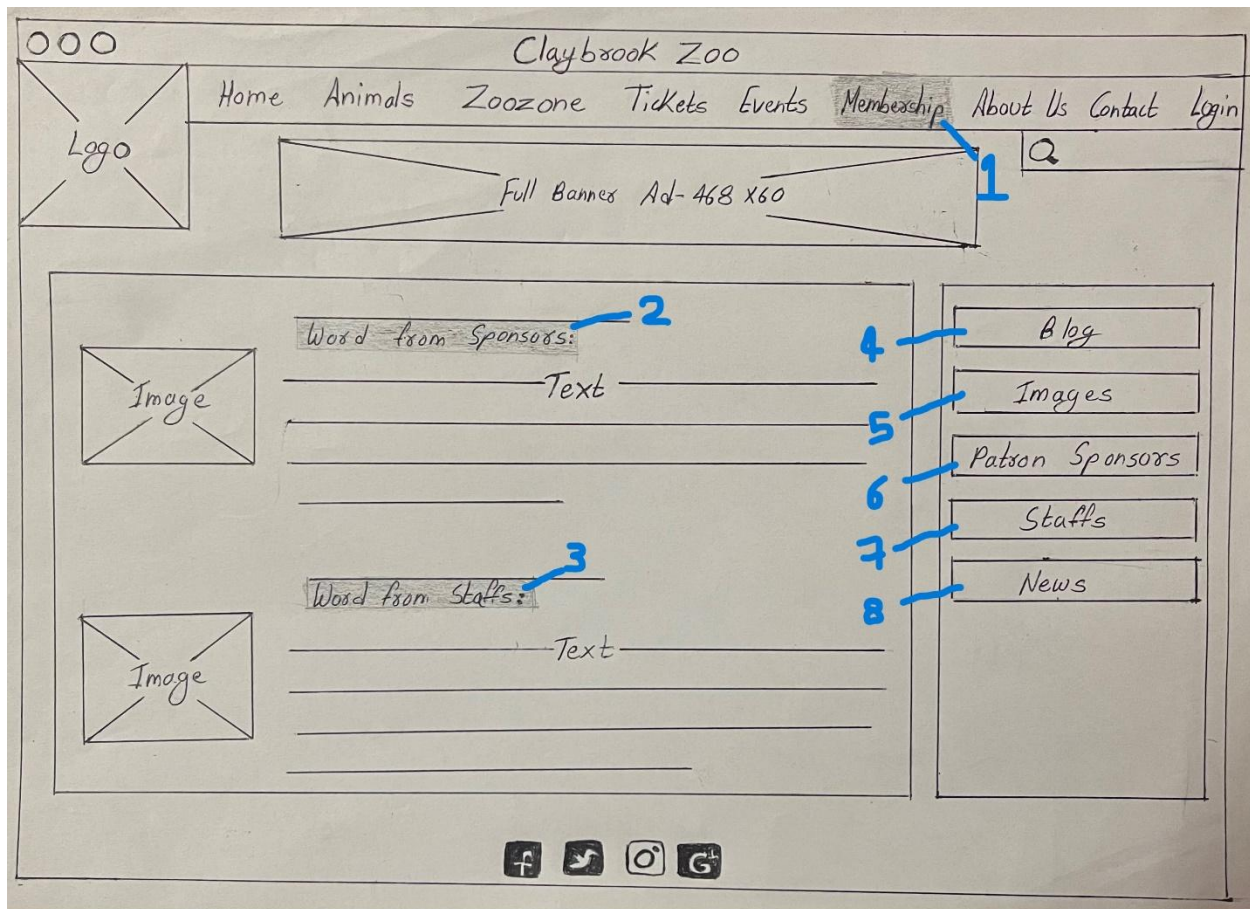
Events



Description:

1. This is Event page.
2. Slideshow images of events.
3. Notice about the Event.
4. Information about the upcoming events.

Membership



Description:

1. This is membership page.
2. Here are some words from Sponsors.
3. Here are some words form Staffs.
4. This is for blog tab.
5. This is for Images tab.
6. This is for Patron Sponsors tab.
7. This is for Staffs tab.
8. This is for News tab.

Sponsor

Record Management [sponsors]

Logo 1

2

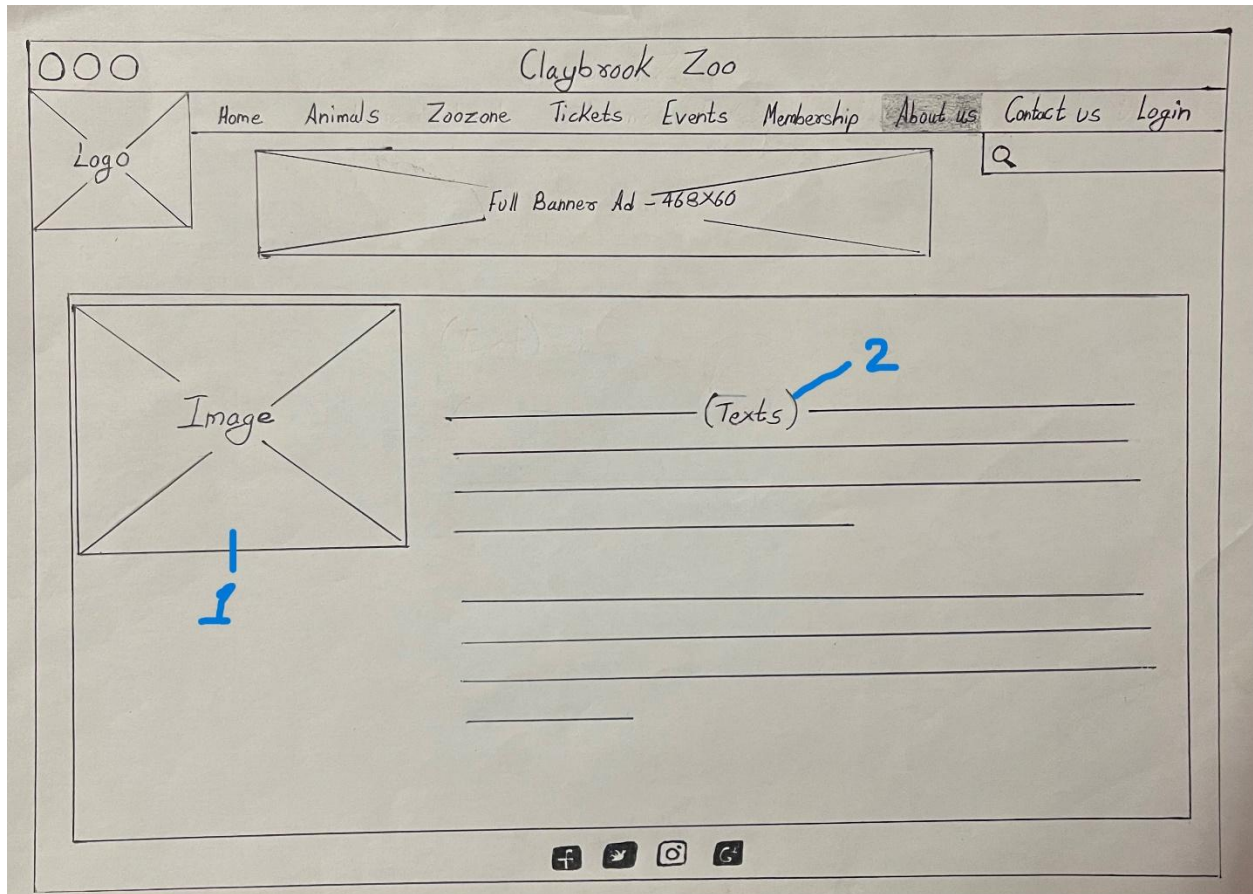
3

sponsor logo in each box

Description:

1. This is place for logo of Zoo.
2. There will be some motivation texts here for sponsors.
3. Logos of sponsors are here, by clicking on them users can view their informations.

About us



Description:

1. Image which represents the zoo.
2. Some information of the zoo.

Contact us

Hand-drawn wireframe of a 'Contact us' web page for Claybrook Zoo. The page layout includes:

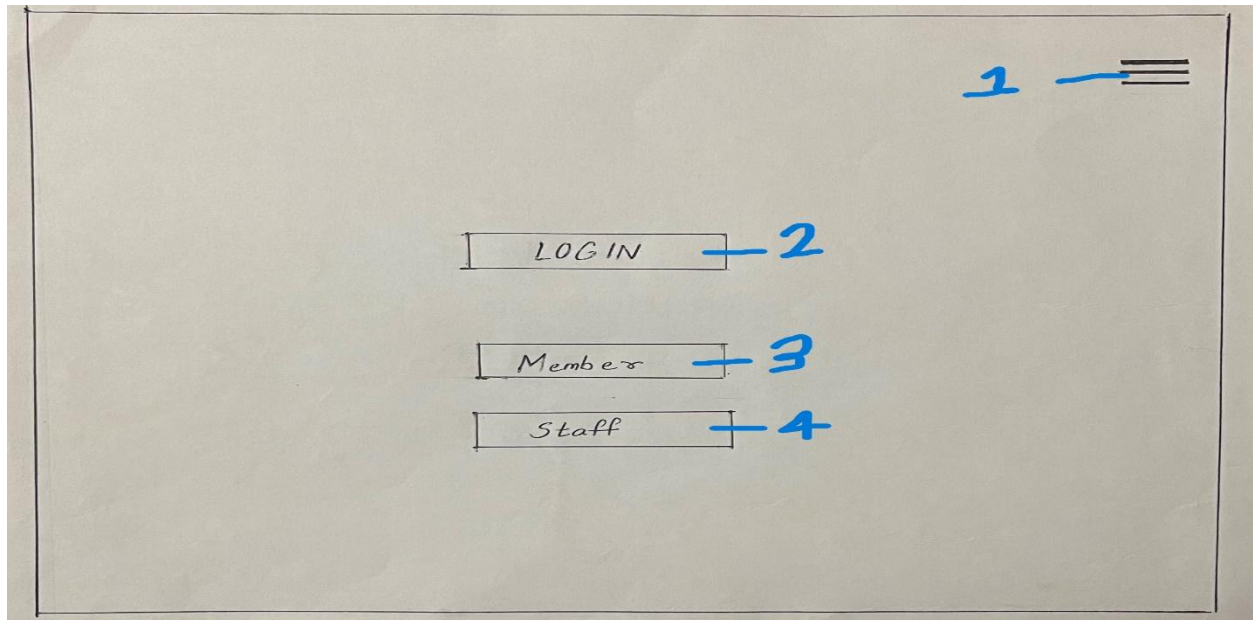
- Header:** Claybrook Zoo, Home, Animals, Zoozone, Tickets, Events, Membership, About us, Contact us, Login.
- Search Bar:** A search bar with a magnifying glass icon.
- Form:** A contact form with three input fields: Name, Email, and a text area for 'Write your message here...'. A 'Submit' button is located below the form.
- Map:** A placeholder for a map, labeled 'Map'.
- Footer:** Social media icons for Facebook, Twitter, Instagram, and YouTube.

Blue handwritten numbers 1, 2, and 3 are placed next to the form, map, and submit button respectively, corresponding to the description below.

Description:

1. This is a form for contacting the zoo.
2. A map which shows where zoo is located.
3. From submit button.

Login



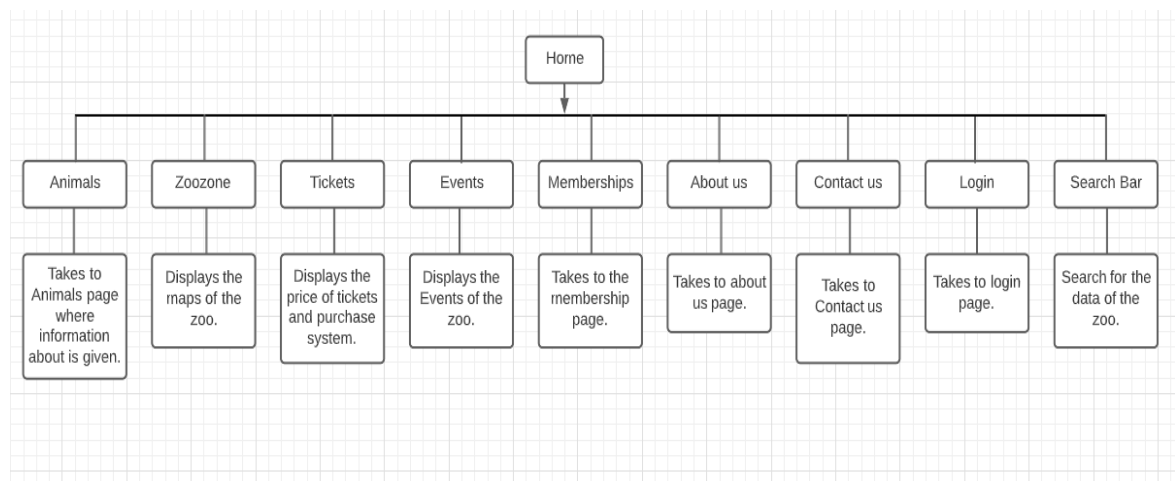
Description:

1. Hamburger icon for choosing other pages.
2. Login button.
3. Member login button.
4. Staff login button.

3.1.2 System Navigation Diagram

System navigation diagram represents an overview pathway of different systems of the website.

System Navigation Diagram for Home:



3.1.3 System Screen mock-ups

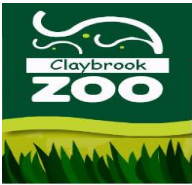
Home:



Animals:



Zoo zone:



HOMEANIMALSZOOZONETICKETS

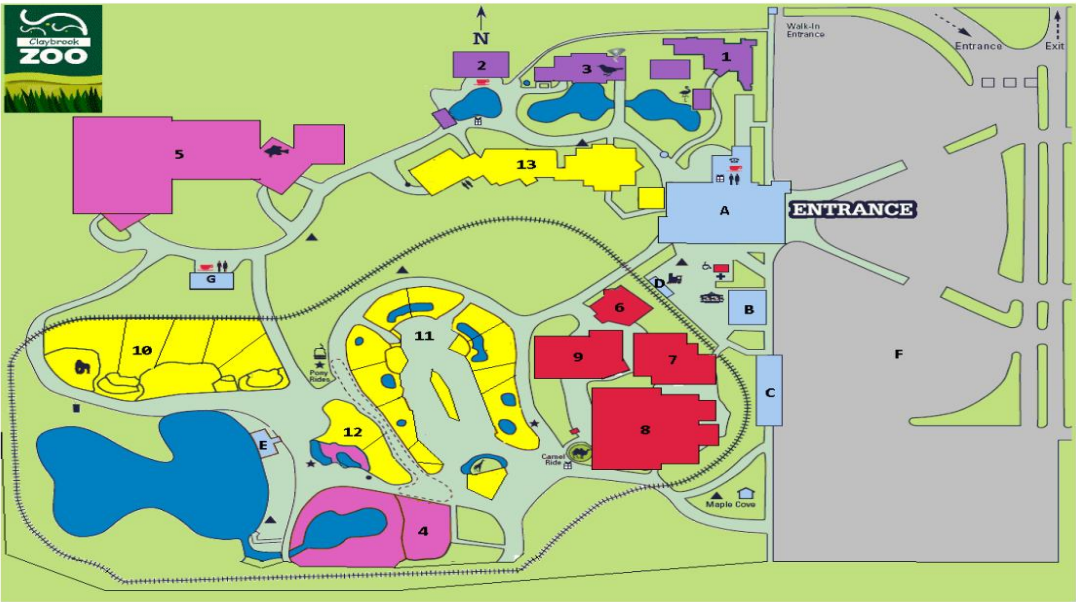
EVENTSMEMBERSHIP

CONTACTLOGIN

Search

Q

ADVERTISEMENT



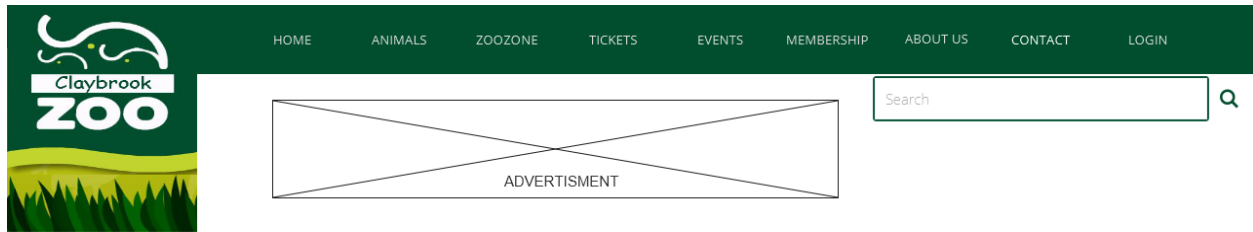
Key	Type of Area/Building	Description
A	Administration	Visitors Centre
B	Administration	Managers Office
C	Administration	Administration and File Storage
D	Administration	Zoo Train (Ticket Sales)
E	Administration	Boat House (Ticket Sales)
F	Administration	Car Park
G	Administration	Cafe Block
1	Aviary	Indoor and Outdoor Aviary Block 1
2	Aviary	Indoor Aviary Block 2
3	Aviary	Indoor Aviary Block 3
4	Aquarium	Outdoor Aquarium Block 1
5	Aquarium	Indoor Aquarium Block 2
6	Hothouse	Amphibian/Reptile Block 1
7	Hothouse	Amphibian/Reptile Block 2
8	Hothouse	Amphibian/Reptile Block 3
9	Hothouse	Amphibian/Reptile Block 4
10	Mammal Compound/Cage	Mammal Outdoor Compound 1
11	Mammal Compound/Cage	Mammal Outdoor Compound 2
12	Mammal Compound/Cage	Mammal Outdoor Compound 3
13	Mammal Compound/Cage	Mammal Indoor Caged Area 4

f

y

9

Tickets:



Tickets

Lorem ipsum dolor sit amet, sapien etiam, nunc amet dolor ac odio mauris justo. Luctus arcu, urna praesent at id quisque ac. Arcu es massa vestibulum malesuada, integer vivamus elit eu mauris eus, cum eros quis aliquam wisi. Nulla wisi laoreet suspendisse integer vivamus elit eu mauris hendrerit facilisi, mi mattis pariatur aliquam pharetra eget.

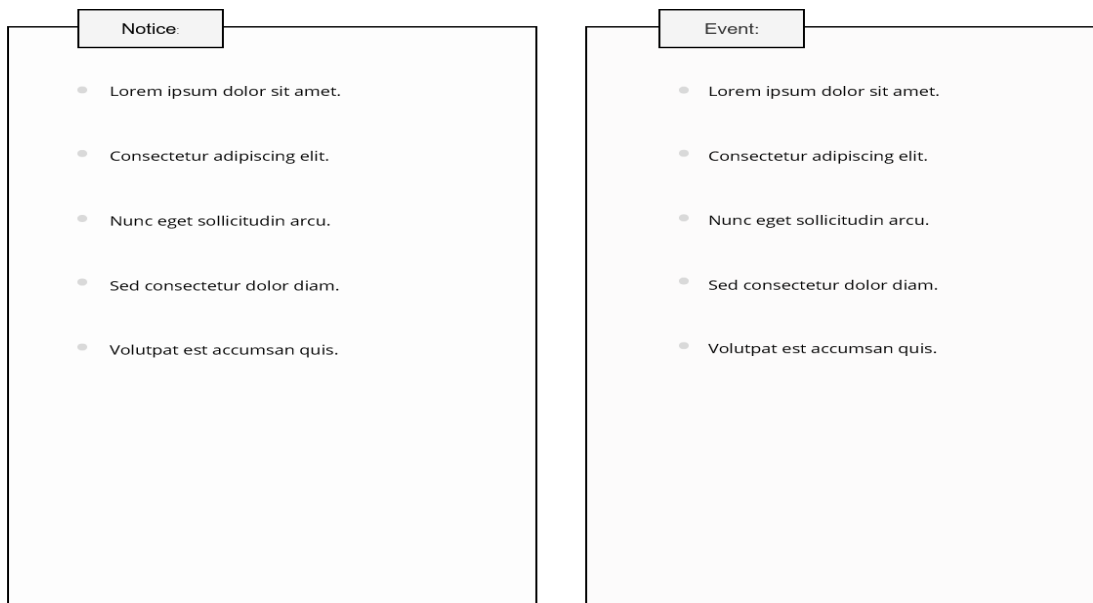
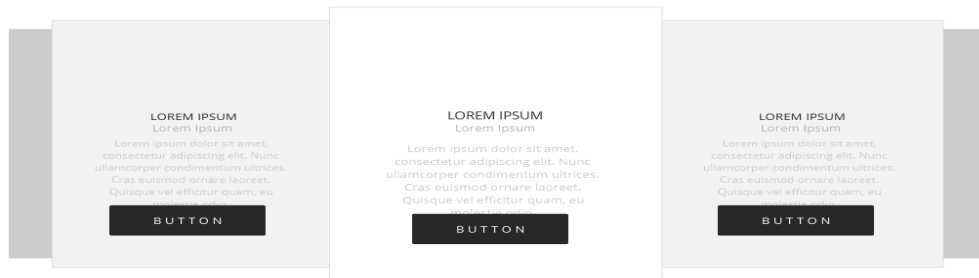
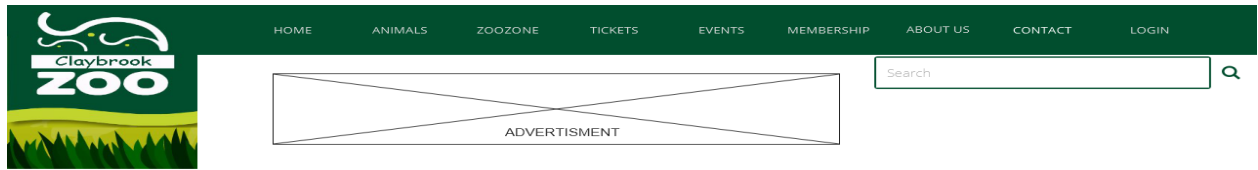
Lorem ipsum dolor sit amet, sapien etiam, nunc amet dolor ac odio mauris justo. Luctus arcu, urna praesent at id quisque ac. Arcu es massa vestibulum malesuada, integer vivamus elit eu mauris eus, cum eros quis aliquam wisi. Nulla wisi laoreet suspendisse integer vivamus elit eu mauris hendrerit facilisi, mi mattis pariatur aliquam pharetra eget.

Visitor	Price
Child	8 euro
Adult	10 euro
Family (2&2)	32 euro
Family (2&3)	40 euro

Book Tickets



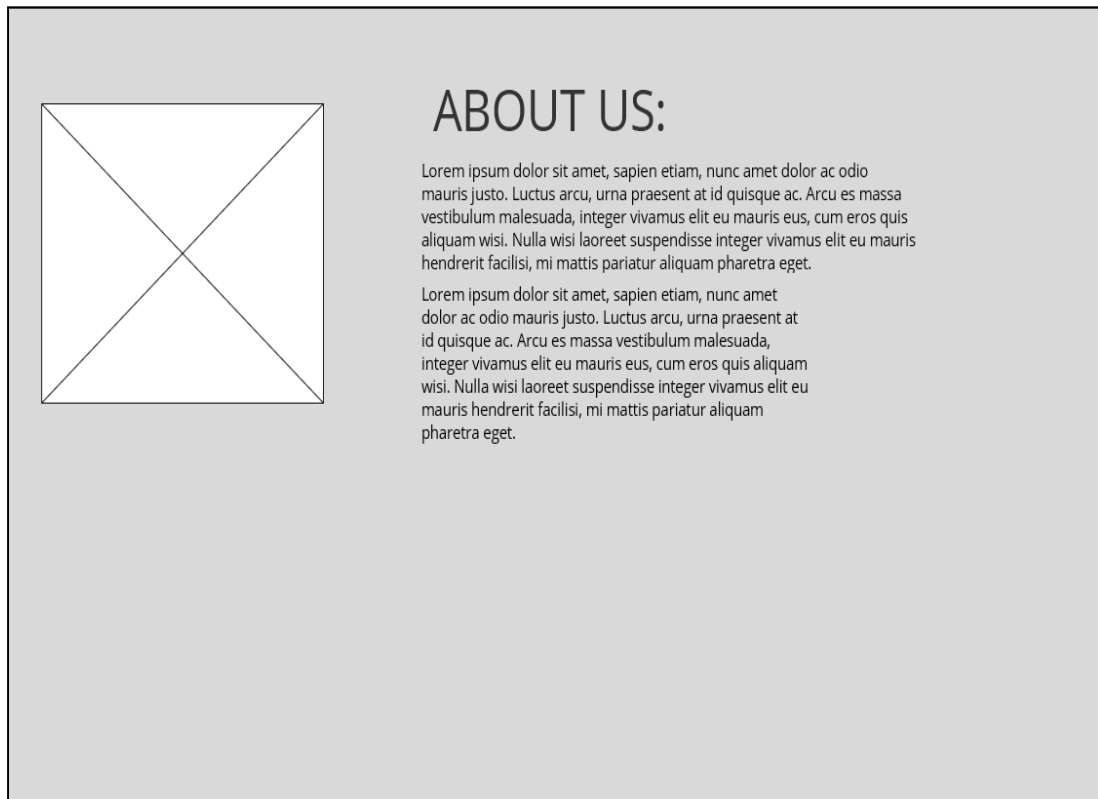
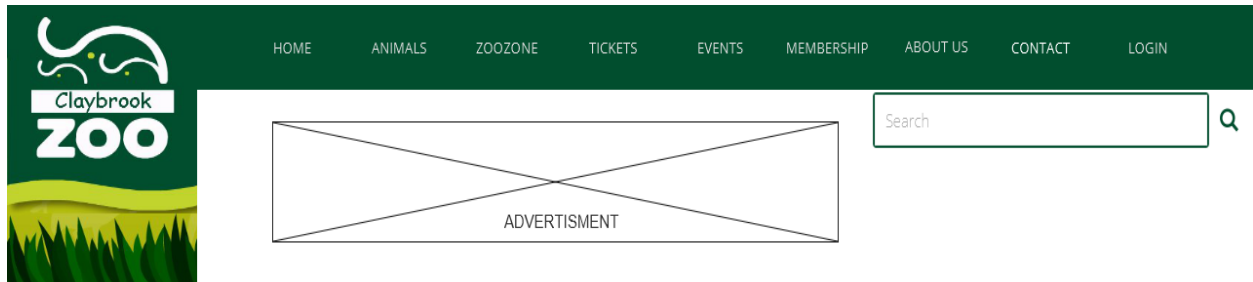
Events:



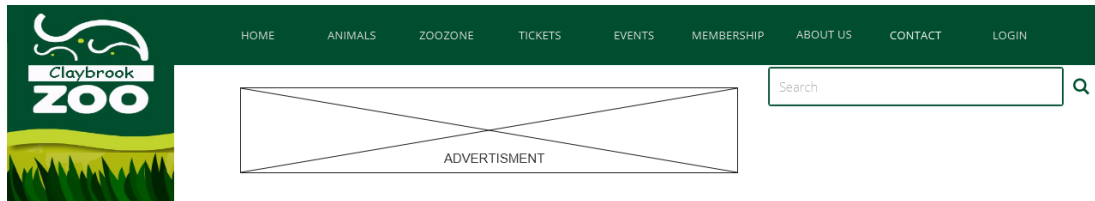
Membership:



About us:



Contact us:



Contact Us

First Name

Email

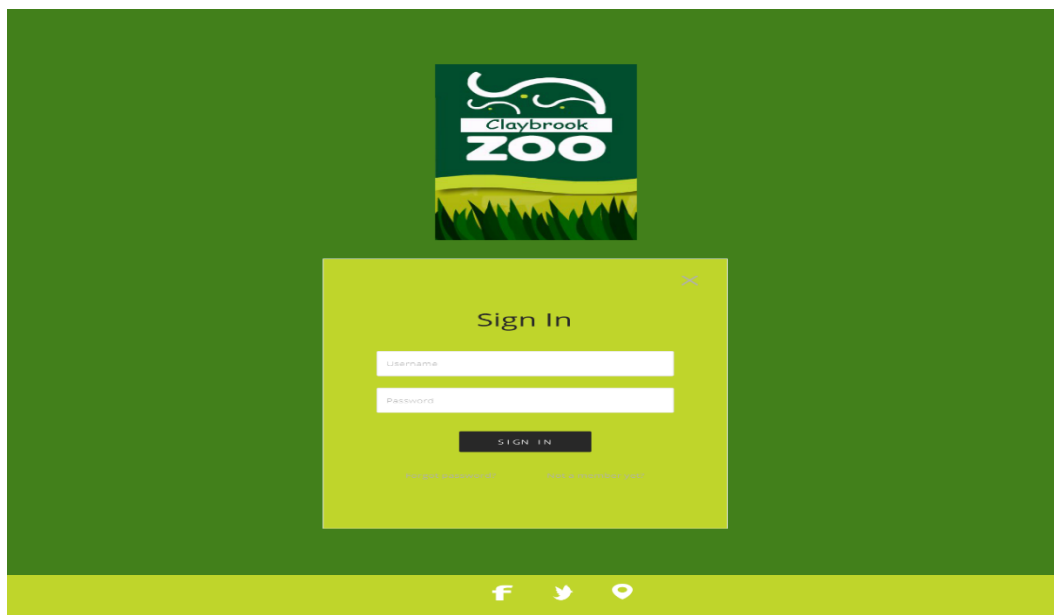
Write your message here...

240, Waterview Lane - San Francisco
CA 94104, USA

 acme.info@2mail.com

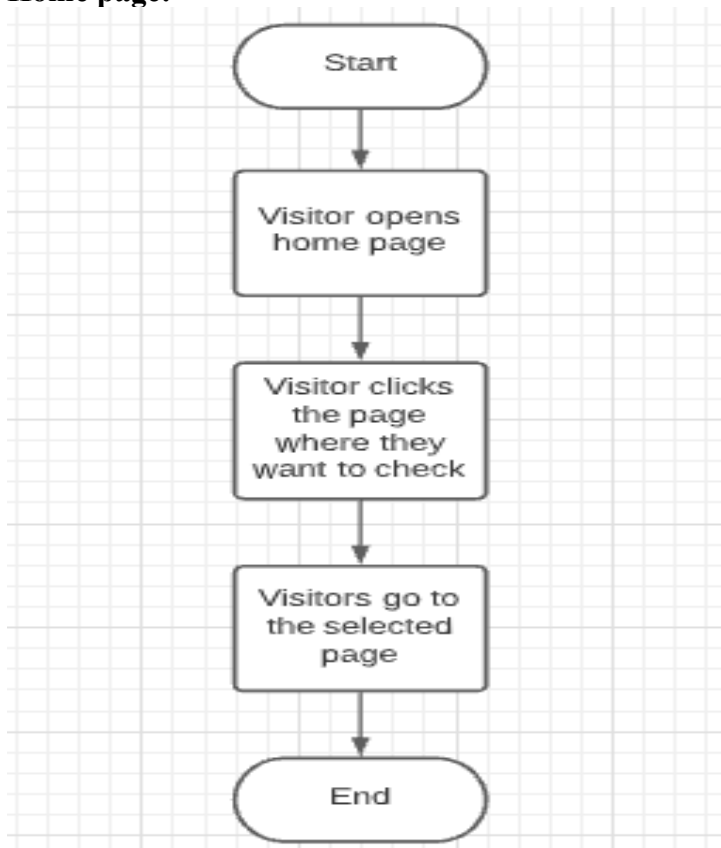


Login:

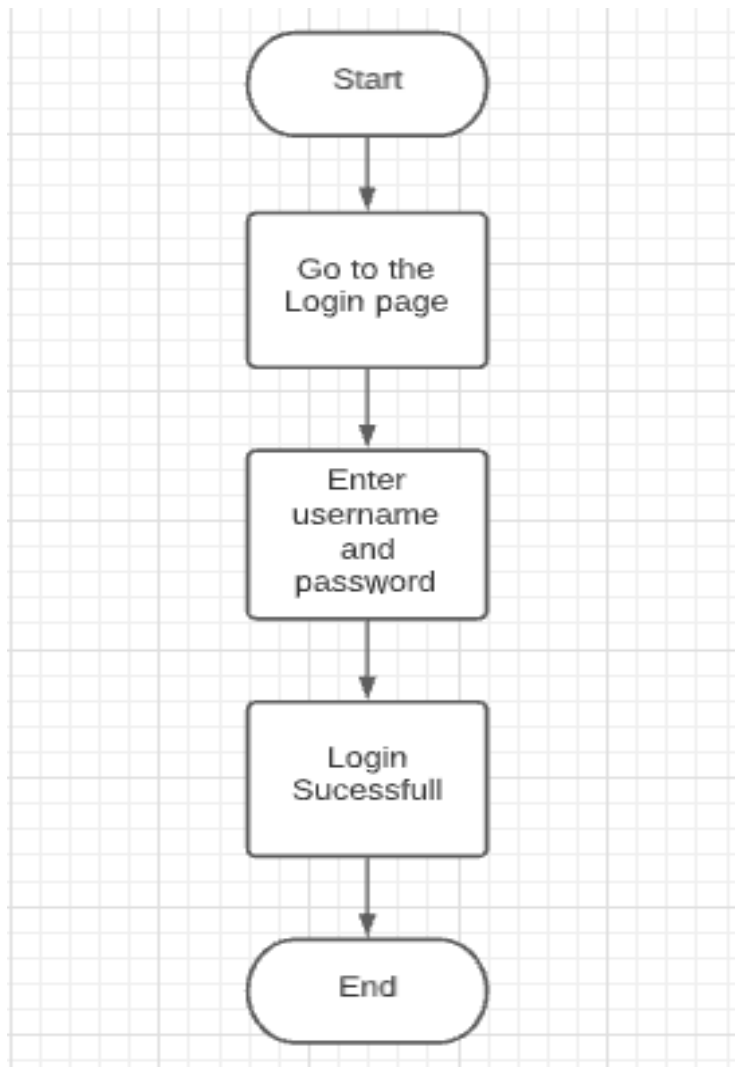


3.1.4 System Activity Event Diagrams

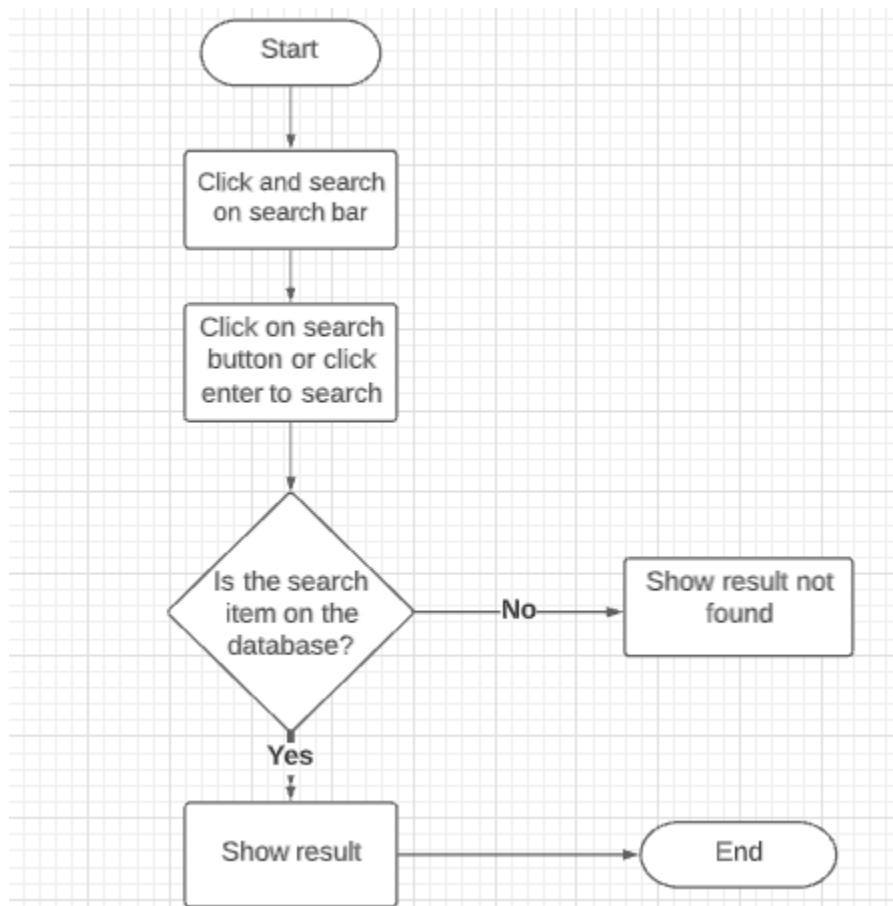
Home page:



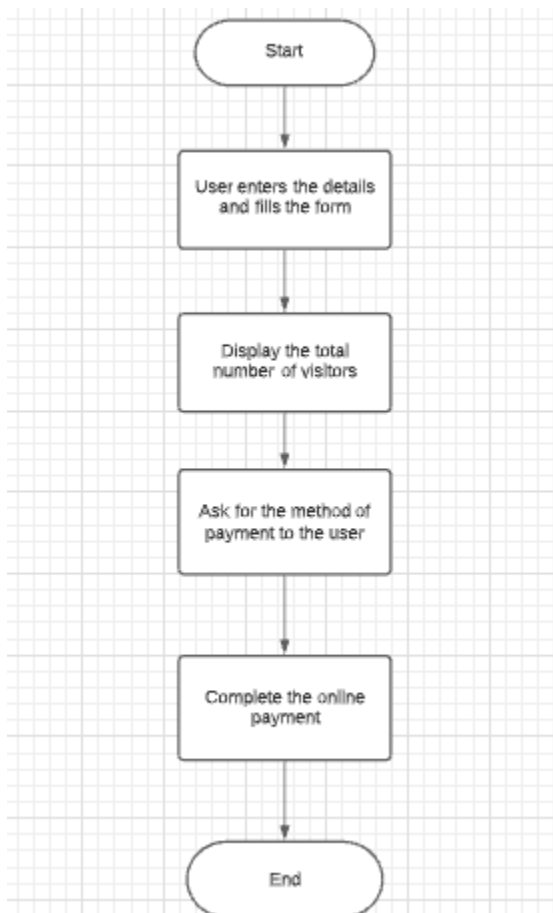
Search bar:



Ticket booking:



Login:



3.2 Design Revisions

In developing the system, we tried to make it as simple and easy to use as possible. We tried not to make the system too complicated. The system was designed according to the client's idea. We have added some functions according to the customer's request.

We added a logo of the zoo on the top left corner because most of the users directly see the web page from the top left to right and then downwards. We have put mainly green backgrounds according to our clients demands. We have included most of the features which are understandable by all people. We tried to distinguish our system from other existing systems. We wanted to create

some unique features in our system, following the contact from our customers. We have made it easier and simpler to use so that both adults and children can easily understand it. We also added more information and more data about the zoo.

3.1 Draft Interface Designs for Kiosk/Visitor Information

The operation of all system components is depicted in a diagrammatic representation called a draft interface design. It facilitates the development of system wireframes. Good user interface design facilitates finishing the task at hand without drawing unnecessary attention to itself.

A kiosk is a small, temporary, stand-alone booth that is used for marketing in high-traffic areas. Kiosks come in a variety of shapes and sizes, including job, foodservice, healthcare, and photo kiosks.

3.1.1 Wireframes

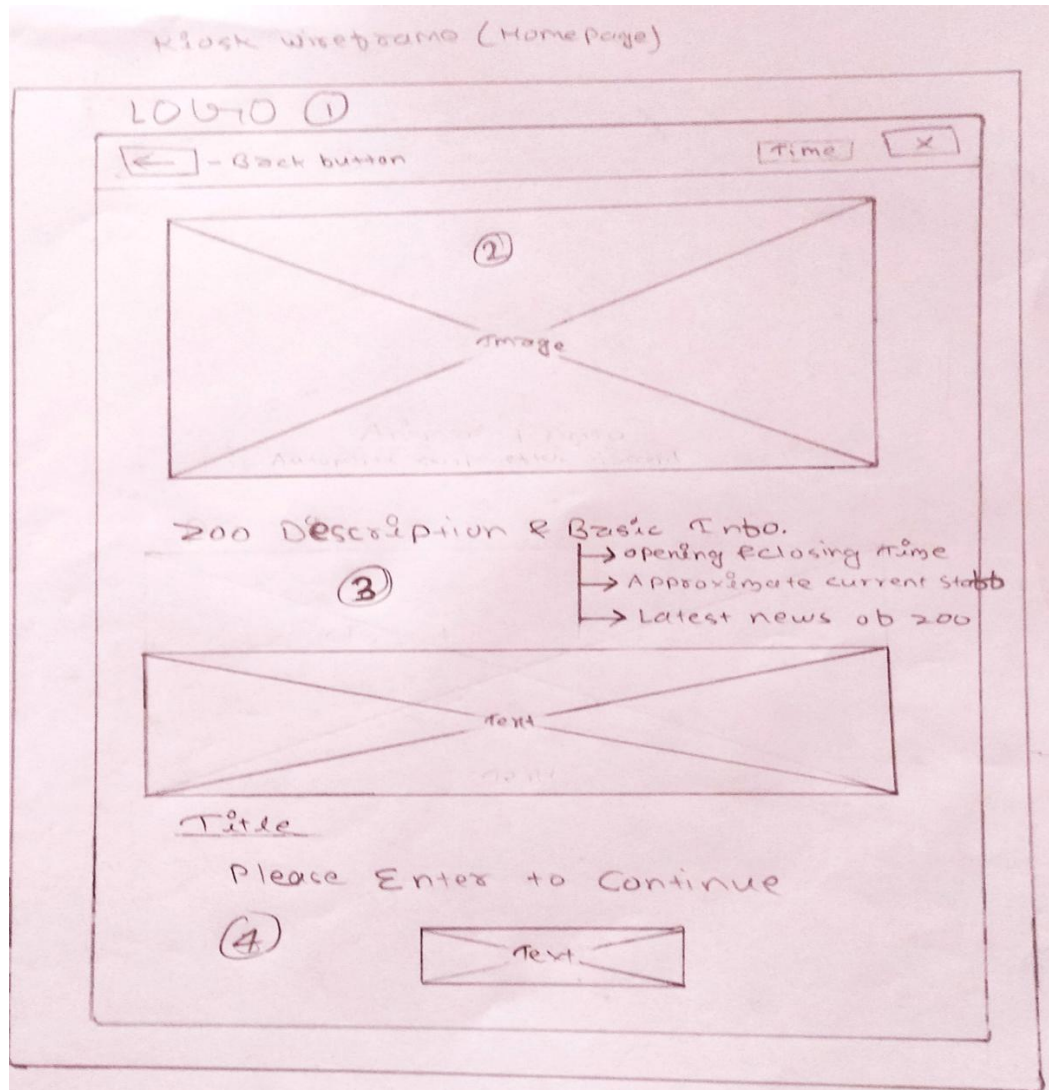
The screen can be made in a variety of ways, but only a handful of them will accurately convey your message and result in simple software or a website. The most crucial aspect of software design is determining a proper interface structure. Doing this effort before writing any code and before finalizing the visual design, i.e., wireframe, will save you a lot of time and grief later on. A wireframe is a schematic or blueprint that is useful for helping you, your programmers and designers think and communicate about the structure of the software or website that you have been building.

Simply, a wireframe is a basic visual representation of elements of any structure of software. Using pencil and paper (or some easy-to-use interface builder), the designer, with the aid of the client, attempts to draft the provisional structure and layout of the proposed system. In wireframe, key screen elements and their location are decided on and drafted. The wireframe should include specific details that are important to the intended user-group. The wireframe can be changed according to the client.

Wireframe typically have an intentional low-fidelity (simple and low-tech concept) look and feel to them for following reason.

- Wireframes make it clear that this is not the final design
- Wireframes convey that this is all up for discussion
- Wireframes make it clear that no code has been written yet

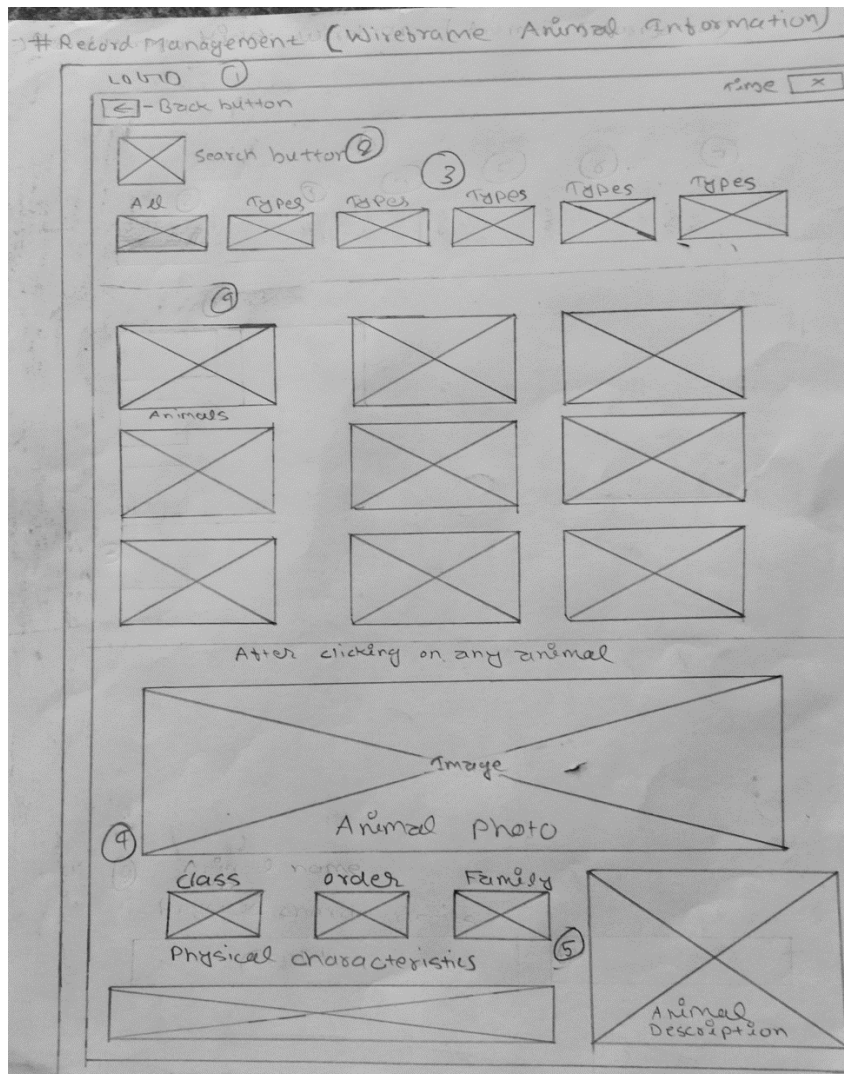
Wireframe of kiosk homepage:



Description:

1. This is where the logo of Claybrook zoo stands.
2. This is where the image of animal is shown. (Pictures changes randomly after 5 sec)
3. This is where the zoo's fundamental information is displayed, such as the zoo's opening and closing times, as well as the zoo's most recent news.
4. This is where you'll find the enter key, which will take you to the next page.

Animal Information section:

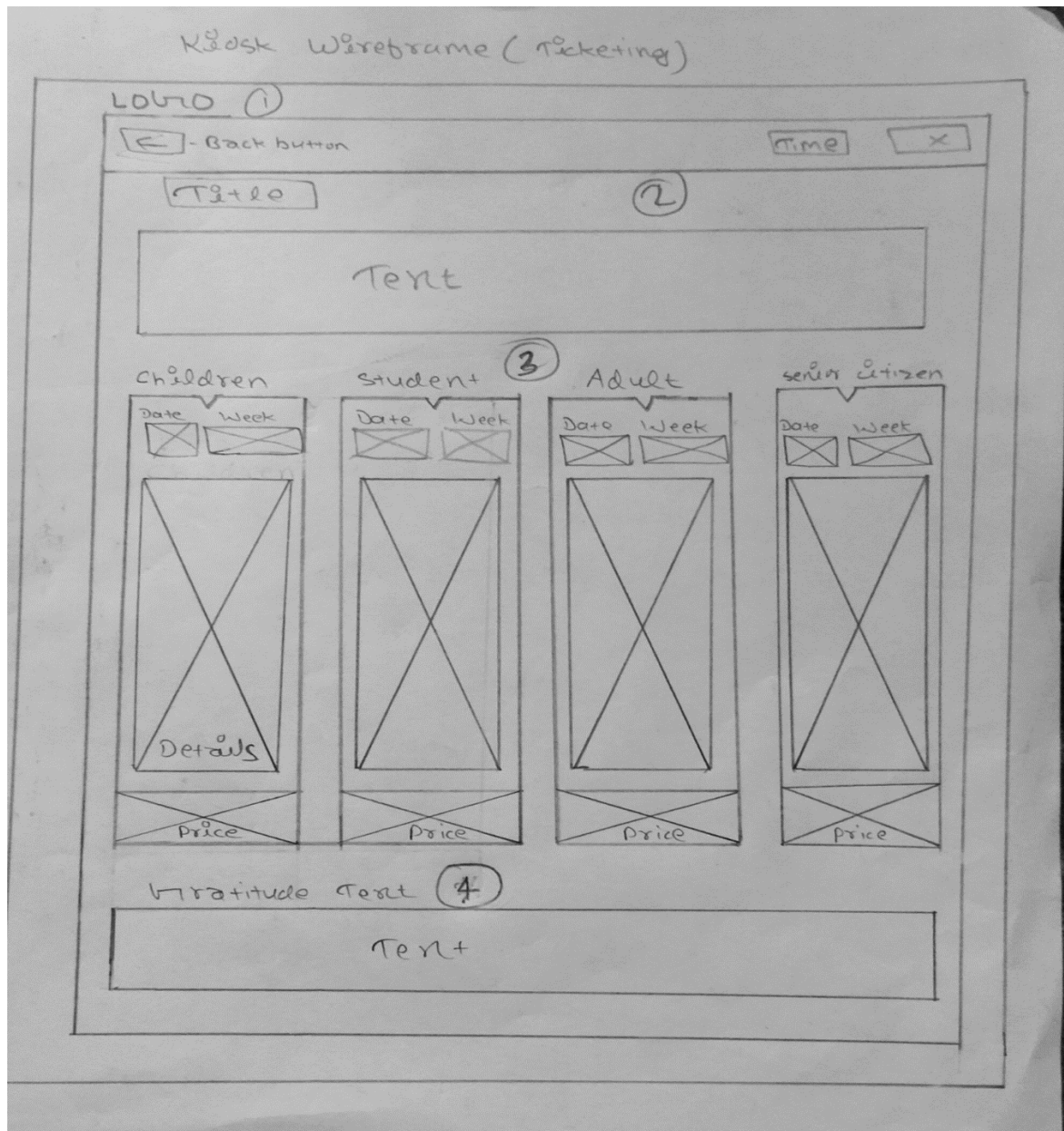


Description:

This section is for the animal information within the zoo. Here, the visitors can select the species they want to see. All the animal information will be displayed such as their physical traits, class etc.

1. This is where the logo of Claybrook zoo stands.
2. This part is a search bar. (Search any animal directly by its name).
3. This area allows visitors to select the species they want to see, such as mammals, amphibians, reptiles, fish, and birds.
4. After pressing on any animal, information about that animal will appear, including basic information, physical traits, class, order, and family.

Ticketing Section :

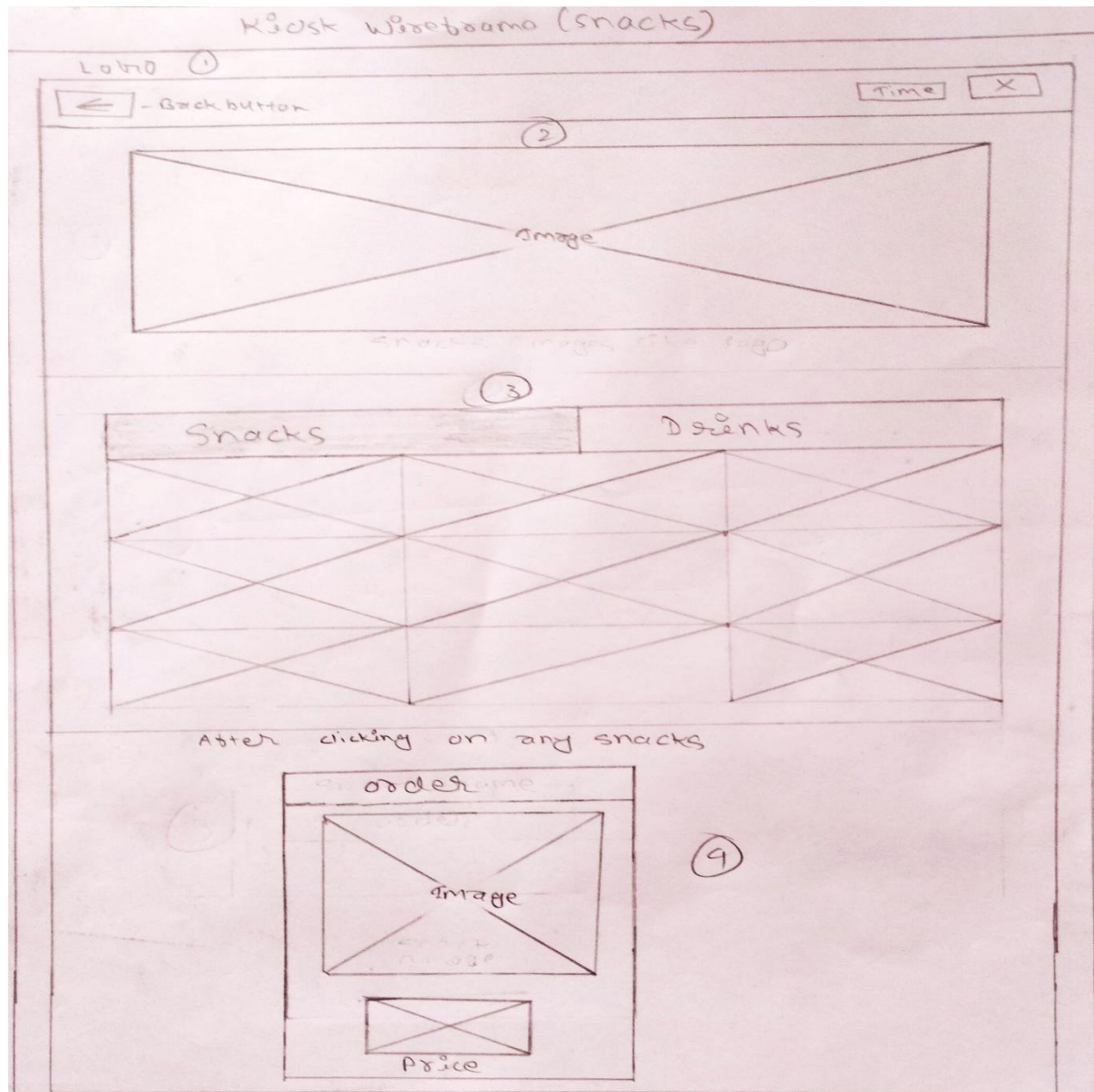


Description:

This is online ticketing system which is accessed when clicked upon the ticket button. Here the customer has to select the ticket according to their age and pay through online payment system in order to book their tickets.

1. This is where the logo of Claybrook zoo stands.
2. This part is for the basic information about the ticket.
3. This section allows them to select a ticket based on their age, with the price displayed alongside the ticket.
4. This is where the gratitude text will be displayed.

Snacks:

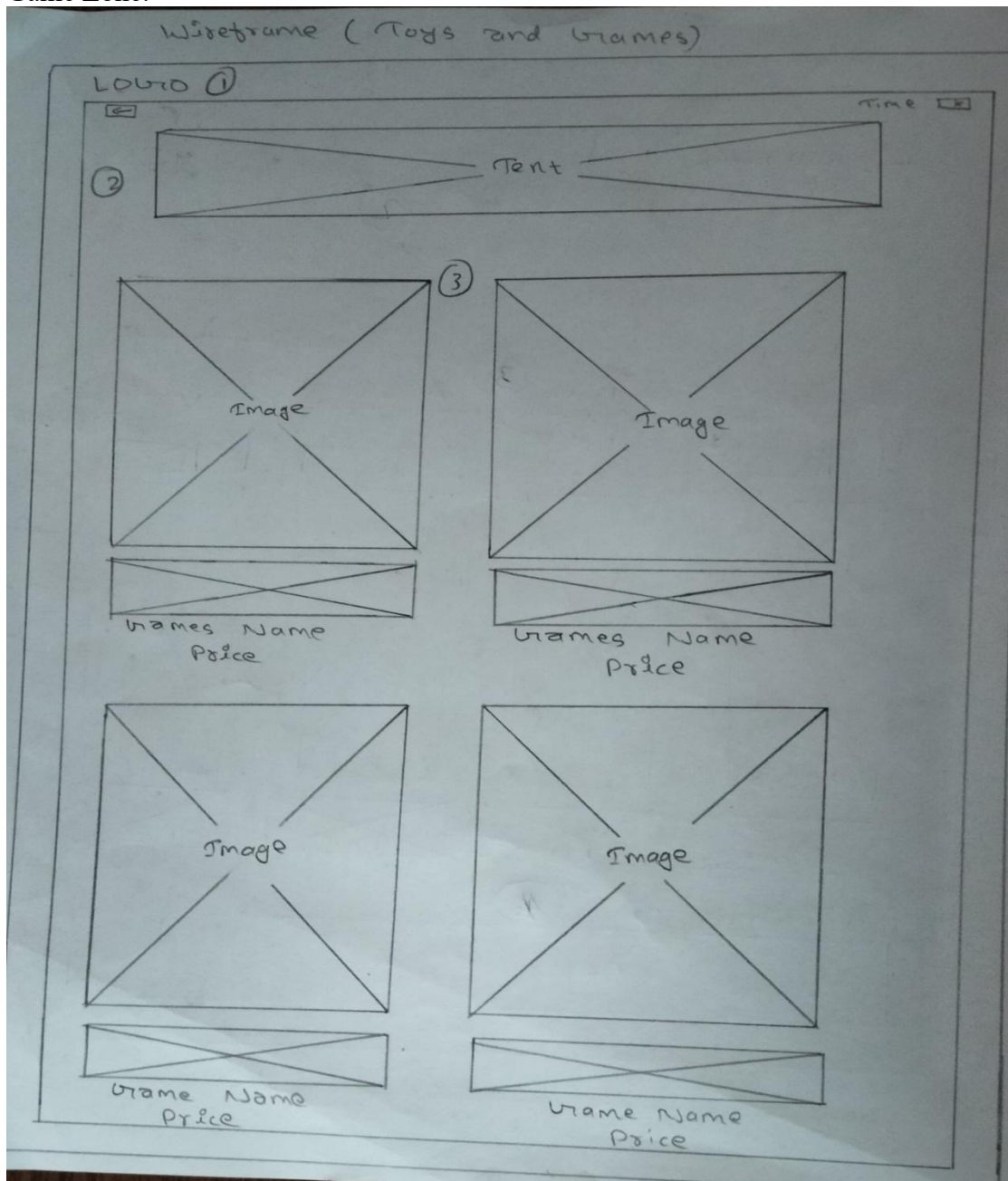


Description:

The use of self-service kiosks is becoming more common across the globe. Customers place and pay for their own orders at kiosks. This enables totally contactless and frictionless service. Customers enjoy the opportunity to buy with no waiting and no delay.

1. This is where the logo of Claybrook zoo stands.
2. This part is for some images of foods and drinks.
3. The visitor select their desire item with the price displayed alongside the item.

Game Zone:

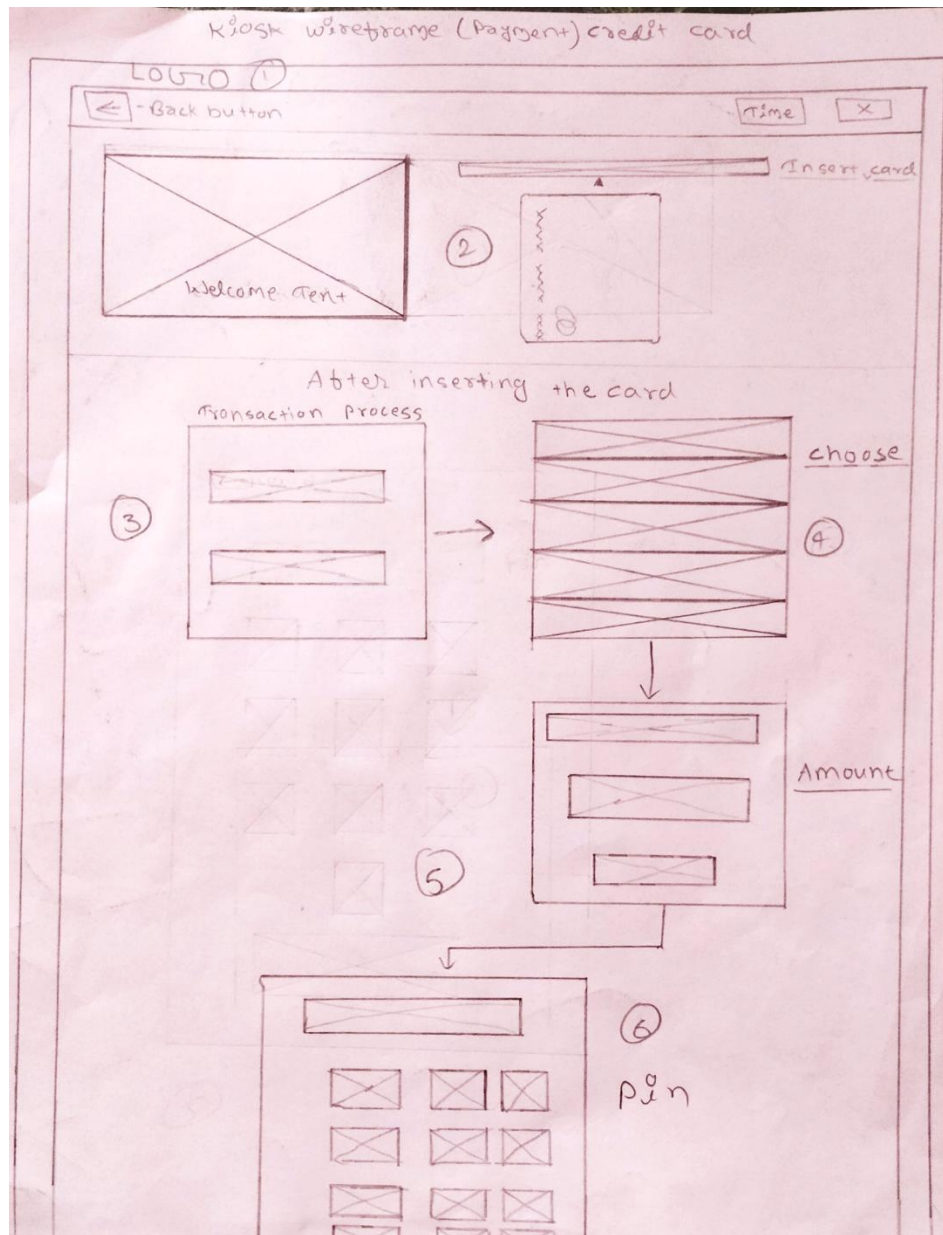


Description:

Visitors or children can play a variety of games in the gaming section and earn prizes. They can play the game by making a payment.

1. This is where the logo of Claybrook zoo stands.
2. This section contains a small amount of information regarding the game.
3. This is where the games are held, and visitors can play in them for a fee

Payment

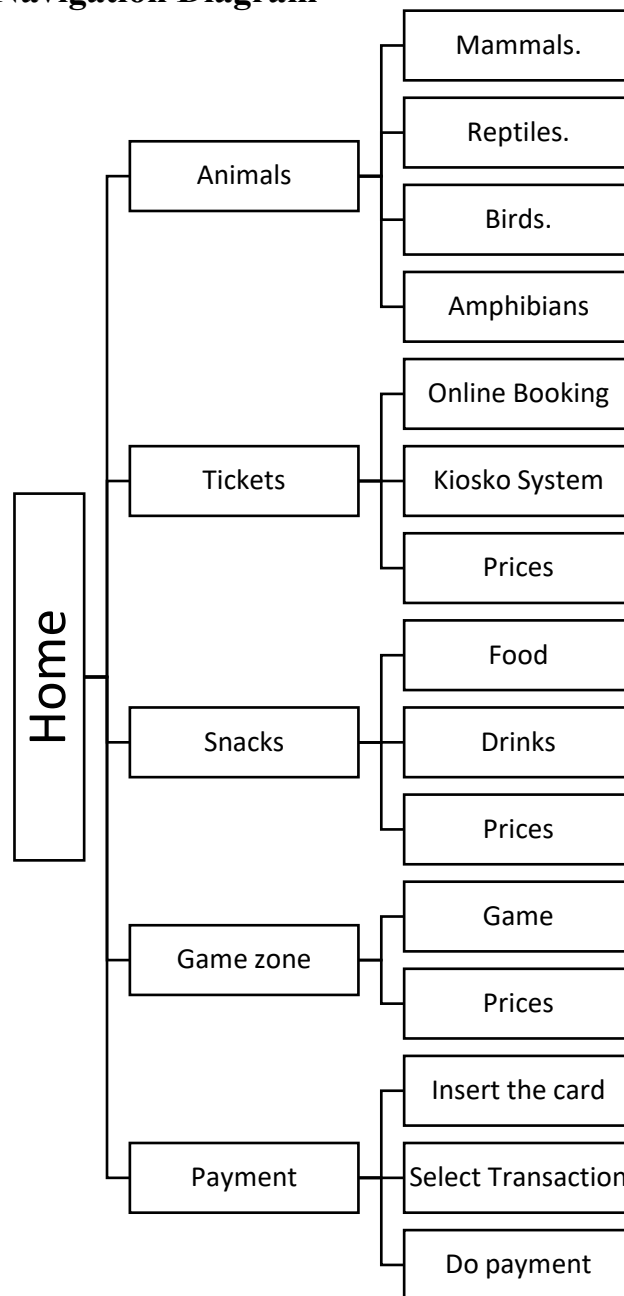


Description:

Visitors are given a variety of card systems to use in order to make payments more simply.

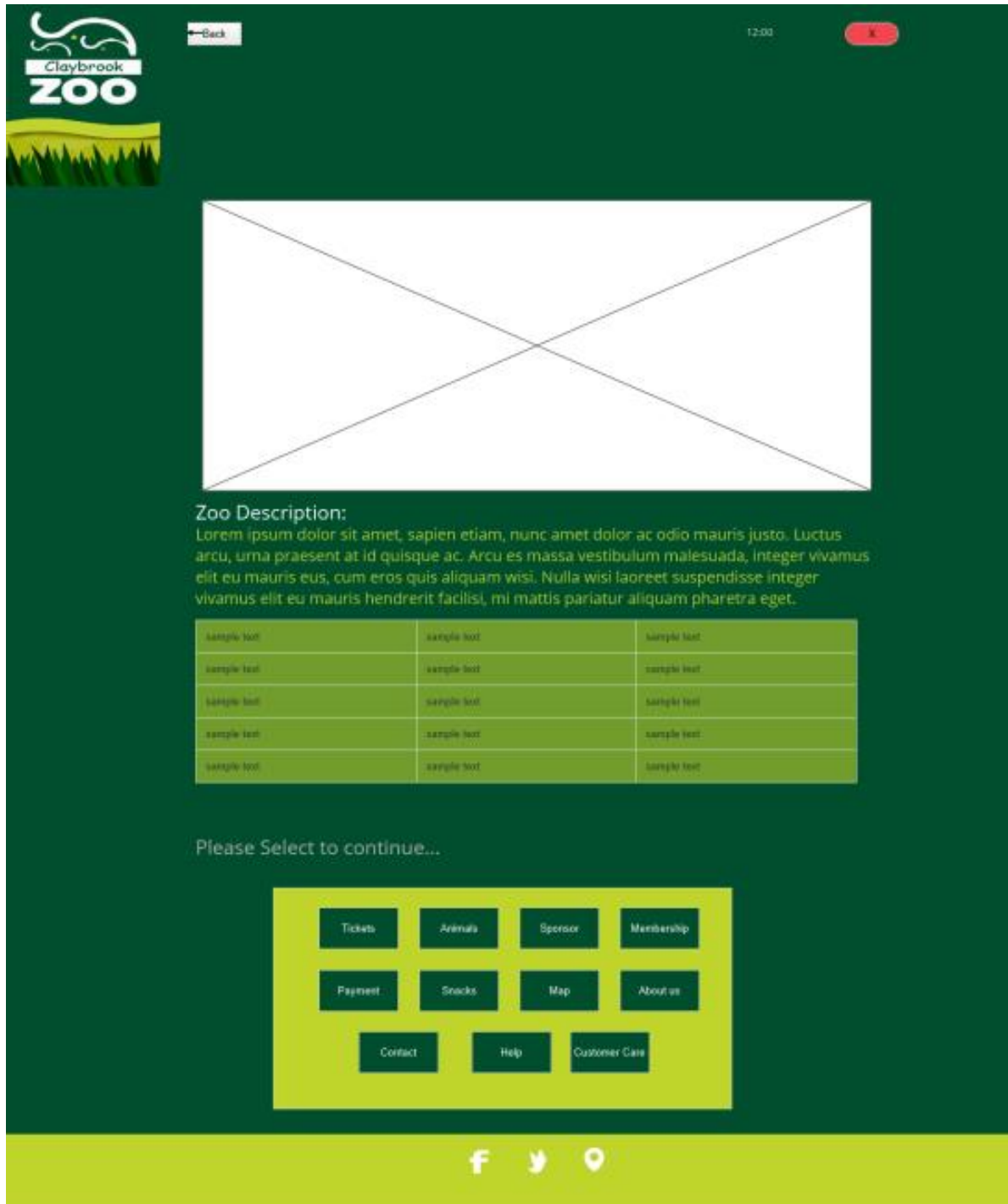
1. This is where the logo of Claybrook zoo stands.
2. This is where the visitor can insert the card.
3. This part is where the transaction process is displayed.
4. This section is where the visitor can select the transaction process.
5. This is the part where visitor can enter the amount.
6. This section is where the visitor can enter the pin.

3.1.2 System Navigation Diagram



3.1.3 System Screen mock-ups

Homepage



Animal information

← Back12:00X



MAMMALS

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc ullamcorper condimentum ultrices. Cras euismod ornare.



BIRDS

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc ullamcorper condimentum ultrices. Cras euismod ornare.



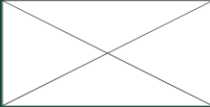
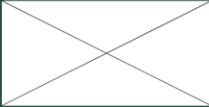
REPTILE

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc ullamcorper condimentum ultrices. Cras euismod ornare.



AMPHIBIANS

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc ullamcorper condimentum ultrices. Cras euismod ornare.





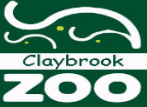
Lowland Gorilla

Lorem ipsum dolor sit amet, sapien etiam, nunc amet dolor ac odio mauris justo. Luctus arcu, urna praesent at id quisque ac. Arcu es massa vestibulum malesuada, integer vivamus elit eu mauris eus, cum eros quis aliquam wisi. Nulla wisi laoreet suspendisse integer vivamus elit eu mauris hendrerit facilisi, mi mattis pariatur aliquam pharetra eget.

Classification. :	Mammal - Primate
Animal Location Code :	MC1-A1
Average Life Span :	30–50 years
Average Animal dimension:	height 1.65–1.75 metres (5 ft 5 in–5 ft 9 in), and in weight 140–200 kg (310–440 lb).
Dietary Requirements (Daily)	2.2 Kg Green Leaf Matter 1.5 Kg Assorted Fruit
Natural Habitat Description	West and Central Africa



Ticketing



← Back

Search

×

Tickets Prices:

Lorem ipsum dolor sit amet, sapien etiam, nunc amet dolor ac odio mauris justo. Luctus arcu, urna praesent at id quisque ac. Arcu es massa vestibulum malesuada, integer vivamus elit eu mauris eus, cum eros quis aliquam wisi. Nulla wisi laoreet suspendisse integer vivamus elit eu mauris hendrerit facilisi, mi mattis pariatur aliquam pharetra eget.

Child

✓

 Lorem ipsum dolor sit amet.

✓

 Consectetur adipiscing elit.

✓

 Nunc eget sollicitudin arcu.

✓

 Sed consectetur dolor diam.

✓

 Volutpat est accumsan quis.

✓

8 euro

Buy

Adult

✓

 Lorem ipsum dolor sit amet.

✓

 Consectetur adipiscing elit.

✓

 Nunc eget sollicitudin arcu.

✓

 Sed consectetur dolor diam.

✓

 Volutpat est accumsan quis.

✓

10 euro

Buy

Family (2 & 2)

✓

 Lorem ipsum dolor sit amet.

✓

 Consectetur adipiscing elit.

✓

 Nunc eget sollicitudin arcu.

✓

 Sed consectetur dolor diam.

✓

 Volutpat est accumsan quis.

✓

32 euro

Buy

f

t

o

Snacks



Payment

[← Back](#)

Search



Payment:

Lorem ipsum dolor sit amet, sapien etiam, nunc amet dolor ac odio mauris justo. Luctus arcu, urna praesent at id quisque ac. Arcu es massa vestibulum malesuada, integer vivamus elit eu mauris eus, cum eros quis aliquam wisi. Nulla wisi laoreet suspendisse integer vivamus elit eu mauris hendrerit facilisi, mi mattis pariatur aliquam pharetra eget.

Customer information

FIRST NAME

LAST NAME

EMAIL

ADDRESS

COUNTRY

Payment details

CARD NUMBER

NAME ON CARD

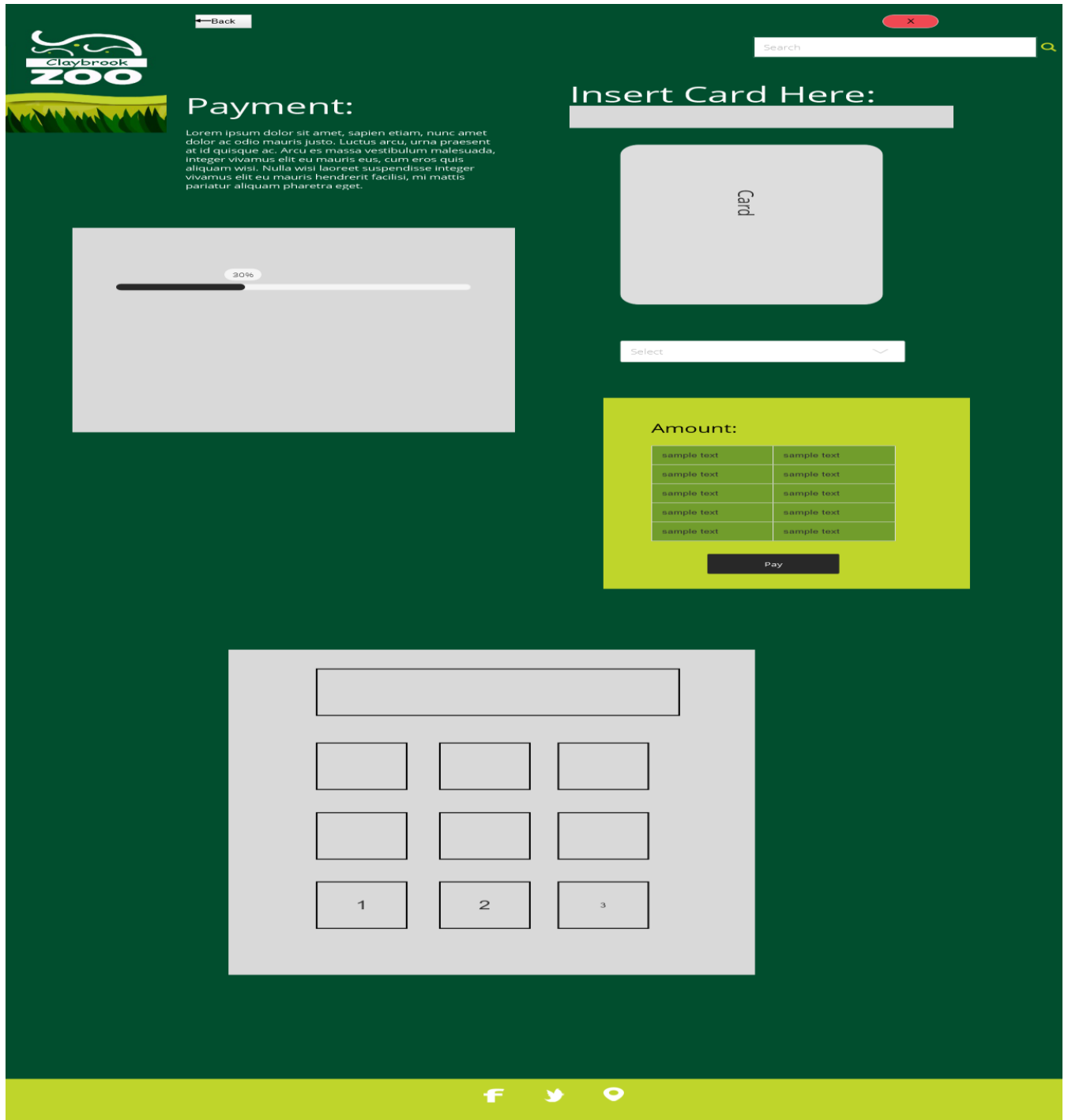
CARD TYPE

VALID THROUGH

CVC CODE

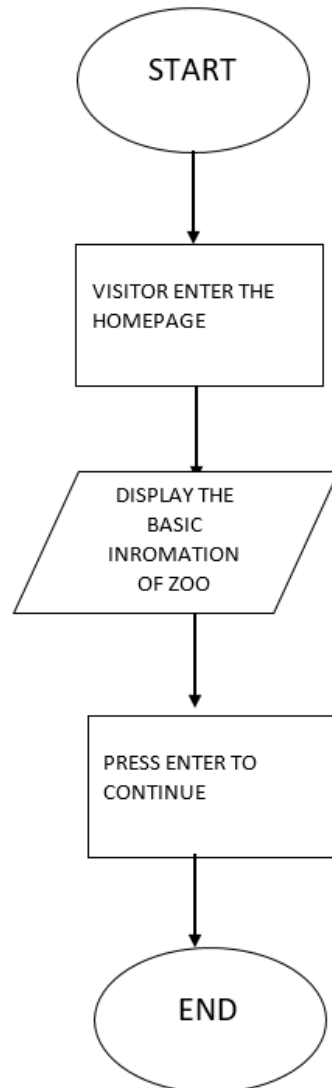
Pay



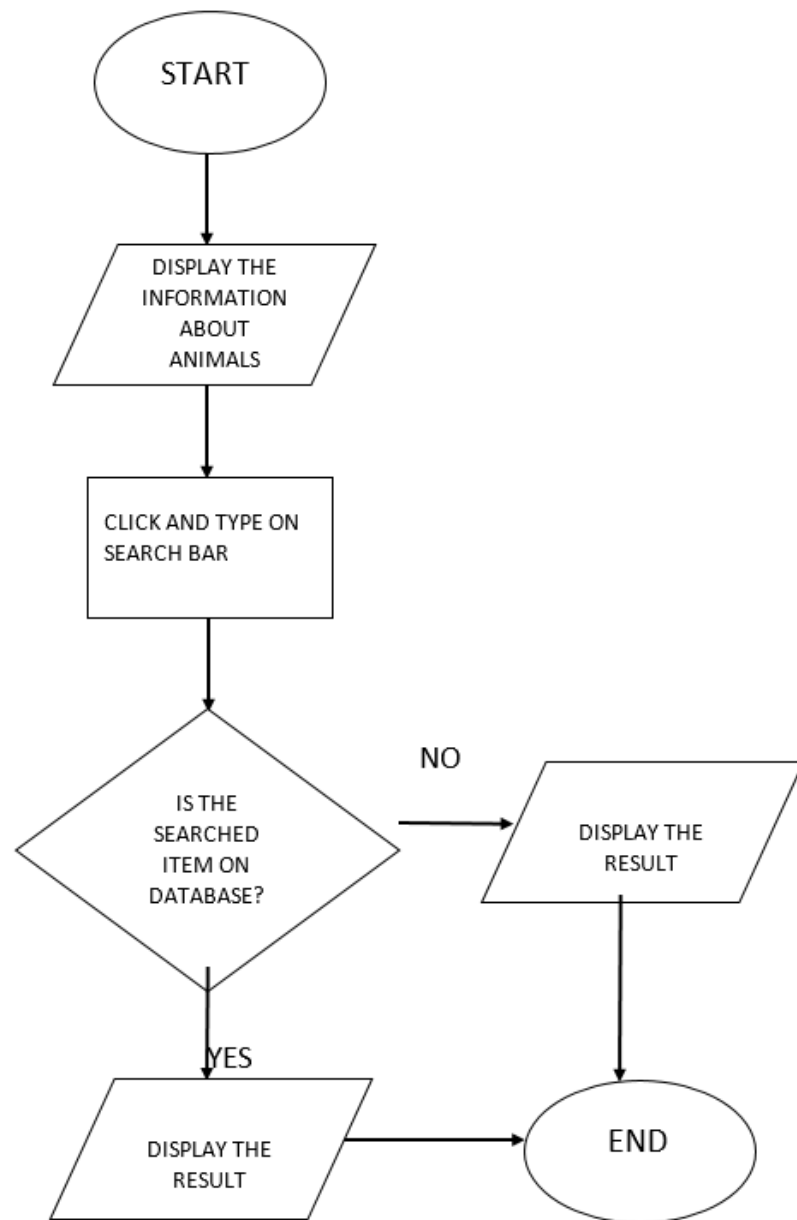


3.1.4 System Activity Event Diagrams

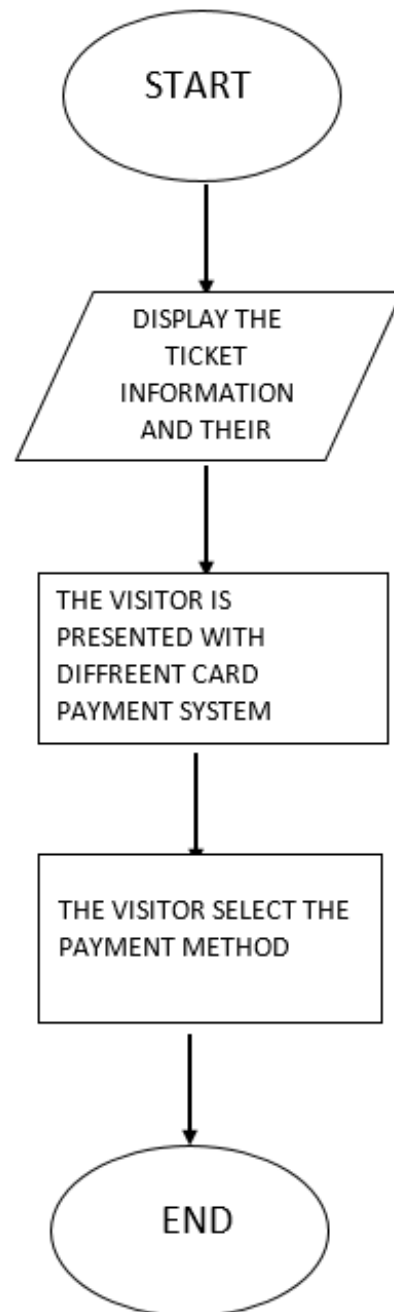
Homepage



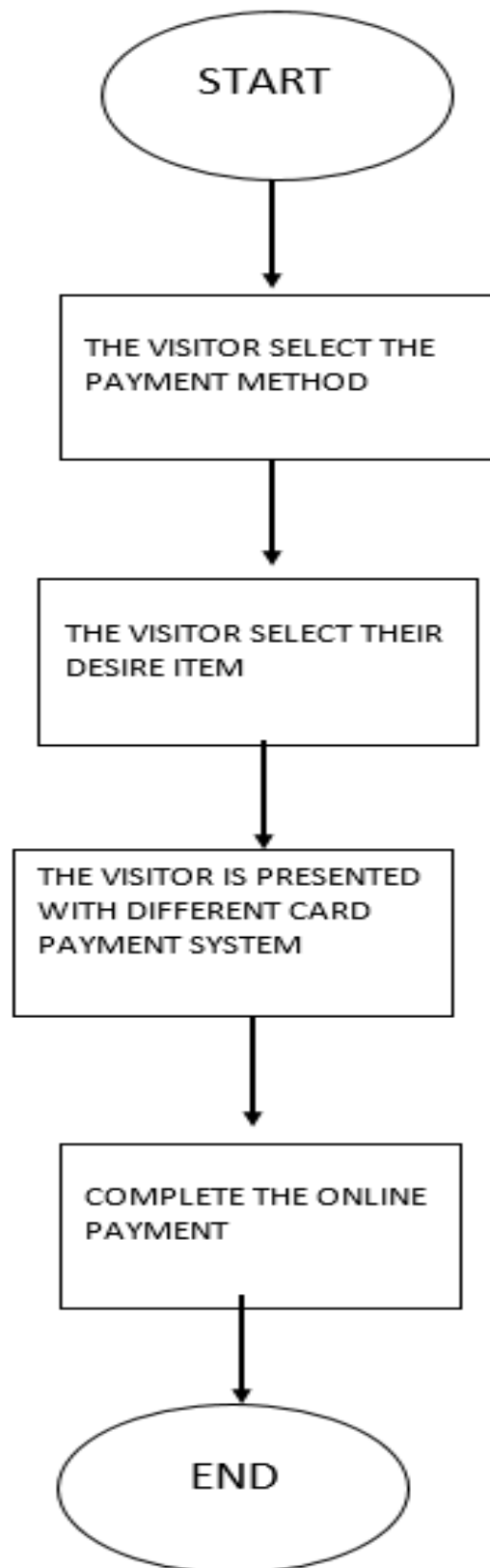
Animal Information



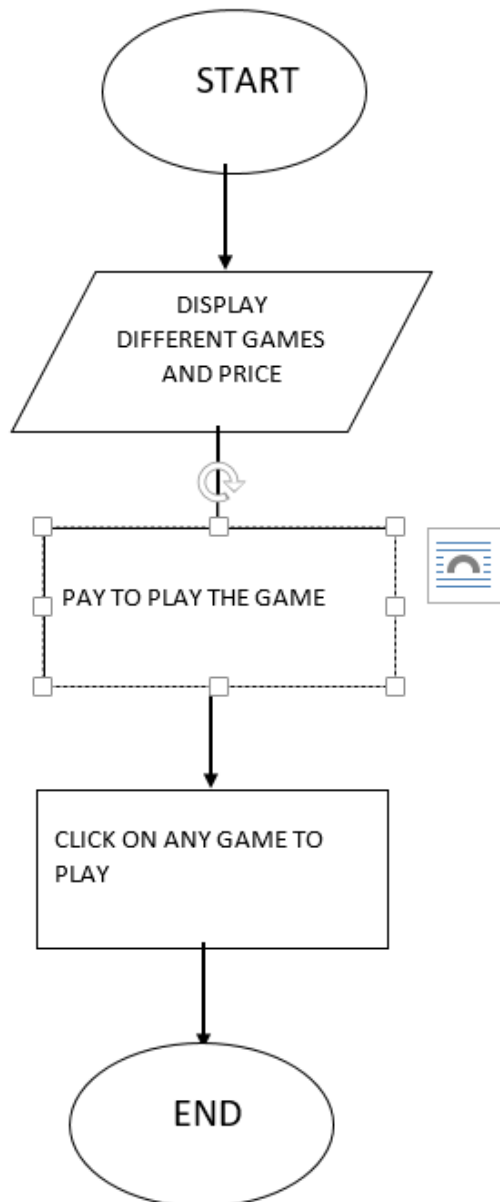
TICKETING



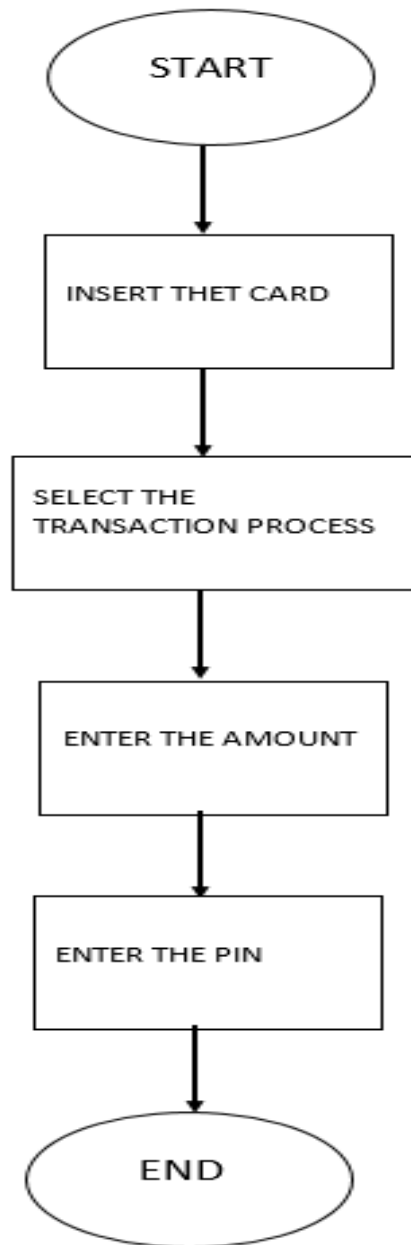
SNACKS



Game Zone



PAYMENT



3.2 Design Revisions

We created several designs for the kiosk system design modification, and this was the final drafted design. After examining many kiosks, we have decided that our kiosk will be a little different from the rest.

There are elements such as a ticketing system that allows visitors to buy tickets based on their age group.

The system was created in accordance with the client's wishes. Snacks were a client suggestion. He declared that the kiosk should also be self-serve. Fast meals such as chips, noodles, and drinks are available to visitors. This allows for completely frictionless and contactless service. Customers like the convenience of being able to purchase without having to wait.

Another wonderful aspect was that we created a game section where kids could pay to play games. There are games like general knowledge, shoot the animal, and zoo farm.

We tried to keep as much animal information as possible. More information and data about the zoo have also been included. We made every effort to keep our kiosk as simple and user-friendly as possible, so that both children and adults could comprehend and operate it with ease.

4 Analysis and Design Records Management System

4.1 Preliminary Analysis Stages

4.1.1. Textual Analysis

Candidate Class	Candidate Routine
Admin	Add, Remove, Create, Amend, Archive, Delete, Assign, Display, News, Map, Advertisement, Ticket, Search
Manager	Add, Remove, Create, Amend, Archive, Delete, Assign, Display, News, Advertisement, Ticket, Search
Sponsor	Display, News, Map, Advertisement, Ticket, Search
Staff	Display, News, Map, Advertisement, Ticket, Search
Visitor	Display, News, Map, Advertisement, Ticket, Search

4.1.2. Significant Event Analysis

Event	Performers	Attributes
Add	Admin, Manager	admin_id, admin_first_name, admin_last_name, manager_id, manager_first_name, manager_last_name contact_number, city, gender Date of birth
Remove	Admin, Manager	admin_id, admin_first_name, admin_last_name, manager_id, manager_first_name, manager_last_name contact_number, city, gender Date of birth
Create	Admin, Manager	admin_id, admin_first_name, admin_last_name, manager_id, manager_first_name, manager_last_name contact_number, city, gender Date of birth
Amend	Admin, Manager	admin_id, admin_first_name, admin_last_name, manager_id, manager_first_name, manager_last_name contact_number, city, gender Date of birth
Archive	Admin, Manager	admin_id, admin_first_name, admin_last_name, manager_id,

		manager_first_name, manager_last_name contact_number, city, gender Date of birth
Delete	Admin, Manager	admin_id, admin_first_name, admin_last_name, manager_id, manager_first_name, manager_last_name contact_number, city, gender Date of birth
Assign	Admin, Manager,	admin_id, admin_first_name, admin_last_name, manager_id, manager_first_name, manager_last_name contact_number, city, gender Date of birth
Display	Admin, Manager, Sponsor, Staff, Visitor	admin_id, admin_first_name, admin_last_name, manager_id, manager_first_name, manager_last_name staff_id, staff_first_name, staff_last_name, staff_address, visitor_id, visitor_first_name, visitor_last_name contact_number, city, gender Date of birth
News	Admin, Manager, Sponsor, Staff, Visitor	admin_id, admin_first_name, admin_last_name, manager_id,

		manager_first_name, manager_last_name staff_id, staff_first_name, staff_last_name, staff_address, staff_joined visitor_id, visitor_first_name, visitor_last_name contact_number, city, gender Date of birth
Map	Admin, Manager, Sponsor, Staff, Visitor	admin_id, admin_first_name, admin_last_name, manager_id, manager_first_name, manager_last_name staff_id, staff_first_name, staff_last_name, staff_address, staff_joined visitor_id, visitor_first_name, visitor_last_name contact_number, city, gender Date of birth
Advertisement	Admin, Manager, Sponsor, Staff, Visitor	admin_id, admin_first_name, admin_last_name, manager_id, manager_first_name, manager_last_name staff_id, staff_first_name, staff_last_name, staff_address, staff_joined visitor_id, visitor_first_name, visitor_last_name

		contact_number, sponsor_id, sponsor_first_name, sponsor_last_name city, gender Date of birth
Ticket	Admin, Manager, Sponsor, Staff, Visitor	admin_id, admin_first_name, admin_last_name, manager_id, manager_first_name, manager_last_name staff_id, staff_first_name, staff_last_name, staff_address, staff_joined, visitor_id, visitor_first_name, visitor_last_name contact_number, sponsor_id, sponsor_first_name, sponsor_last_name city, gender Date of birth
Search	Admin, Manager, Sponsor, Staff, Visitor	admin_id, admin_first_name, admin_last_name, manager_id, manager_first_name, manager_last_name sponsor_id, sponsor_first_name, sponsor_last_name staff_id, staff_first_name, staff_last_name, staff_address, staff_joined visitor_id, visitor_first_name, visitor_last_name contact_number, city, gender

		Date_of_birth
--	--	---------------

4.1.3. Class- Responsibility-Collaborator (CRC)

Class Name: -Admin
Responsibilities Add Remove Create Amend Archive Delete Assign Display News Map Advertisement Ticket Search
Collaborators Manager

Attributes admin_id, admin_first_name, admin_last_name, contact_number, house_number, city, country, date_of_birth
purpose To control all over the system all the time

Class Name: - Manager
Responsibilities Add Remove Create Amend Archive Delete Assign Display News Map Advertisement Ticket Search
Collaborator staff, sponsor

Attributes manager_id, manager_first_name, manager_last_name, contact_number, gender, city, country, date_of_birth
Purpose To control all the problem domains and control over all the system

Class Name: - Sponsor
Responsibilities Display News Map Advertisement Ticket Search
collaborator Manager

Attributes
sponsor_id, sponsor_first_name, sponsor_last_name, contact_number, gender, city, country, date_of_birth
Purpose
To sponsor the animals

Class Name:-staff
Responsibilities Display News Map Advertisement Ticket Search
collaborator
Visitors

Attributes
staff_id, staff_first_name, staff_last_name, contact_number, staff_address, gender, town, country, email, staff_joined
Purpose
To guide the visitors and work inside the zoo in different fields

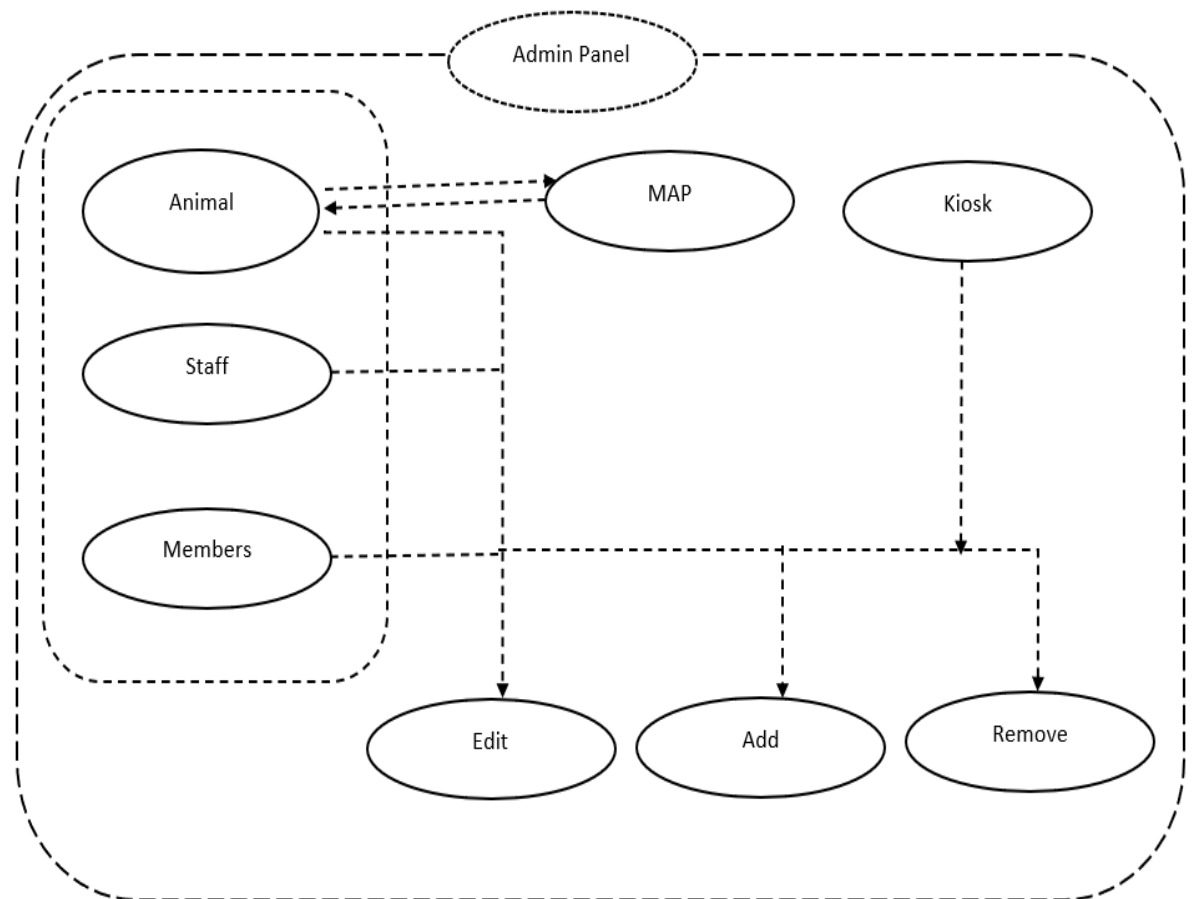
Class Name: -Visitor	
Responsibilities	
Display	
News	
Map	
Advertisement	
Ticket	
Search	
collaborator	
Staff	

Attributes	
visitor_id, visitor_first_name, visitor_last_name, city, country, house_number, contact_number, gender, date_of_birth	
Purpose	
To visit inside the zoo	

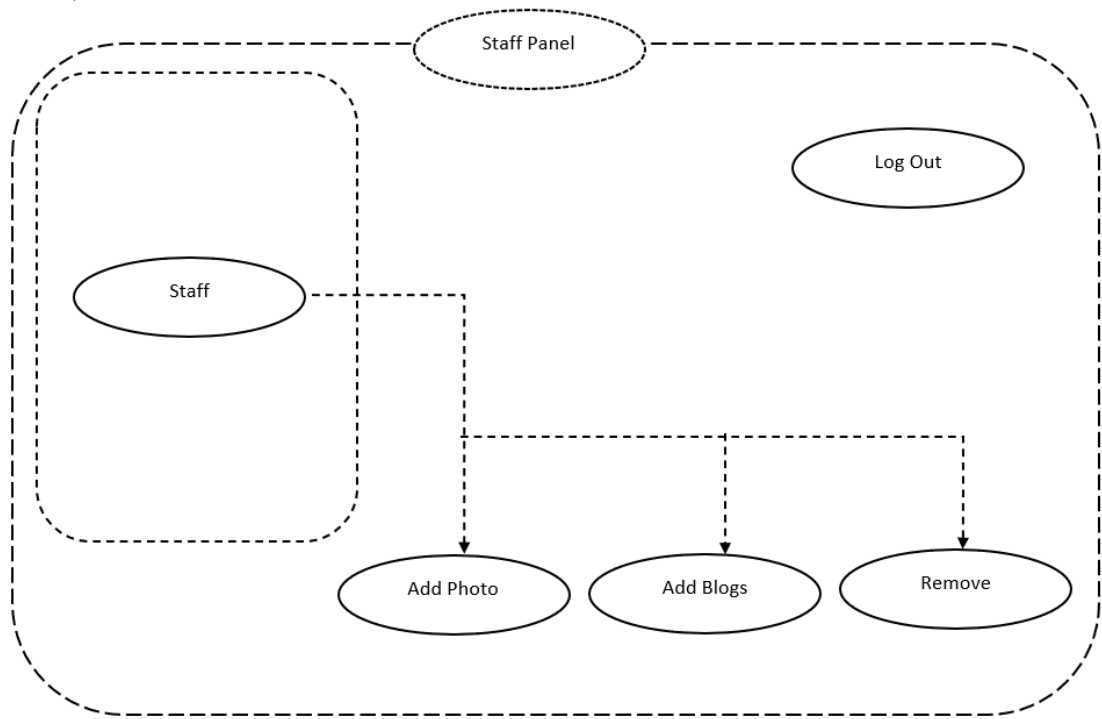
4.2. Detailed Static System Designs

4.2.1. First Draft BON System Architecture Diagram

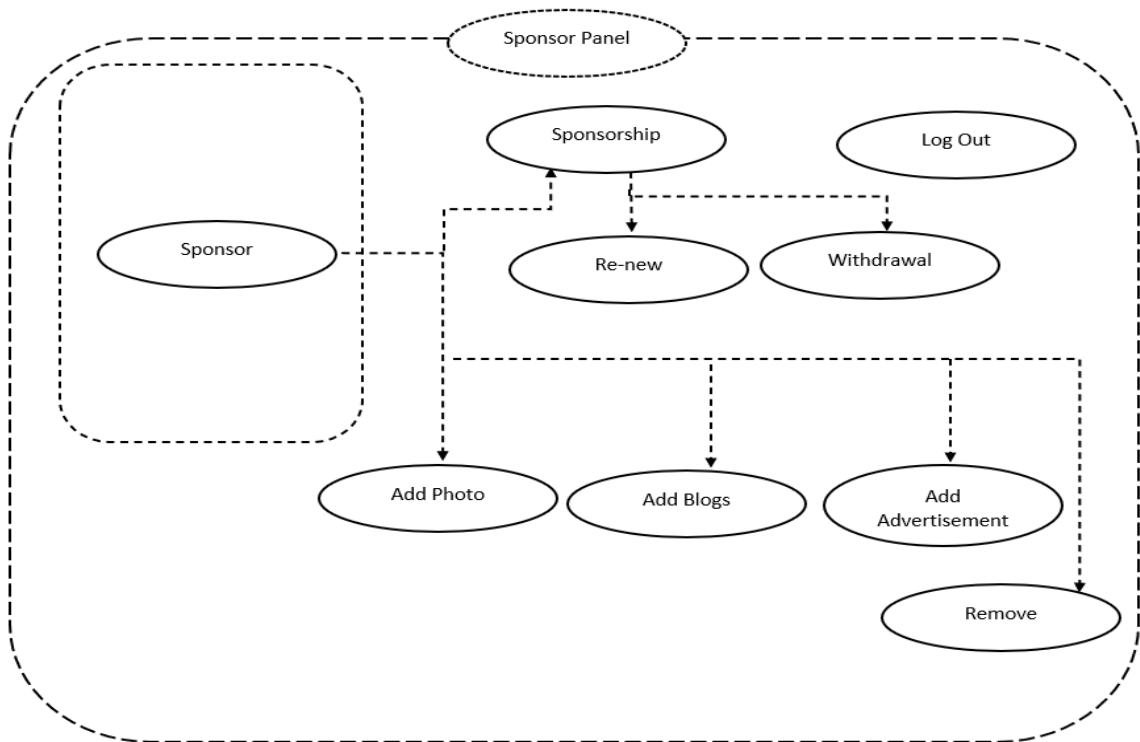
Admin:



Staff:



Sponsor:



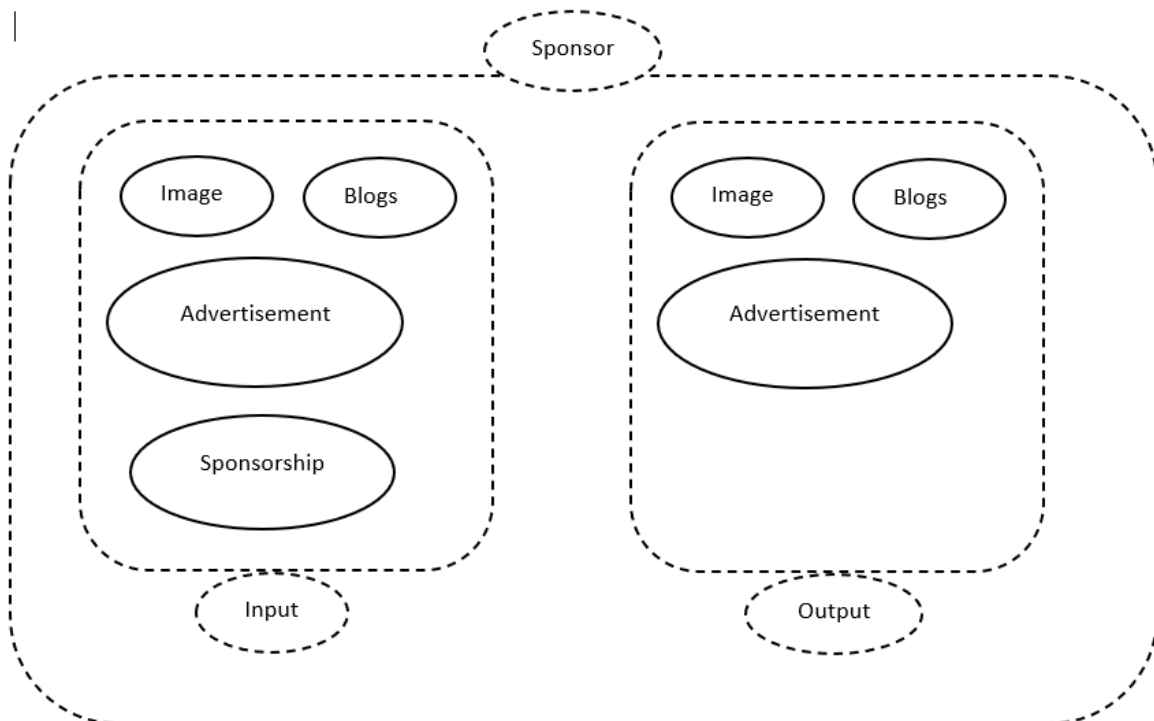
4.2.2. BON System Chart

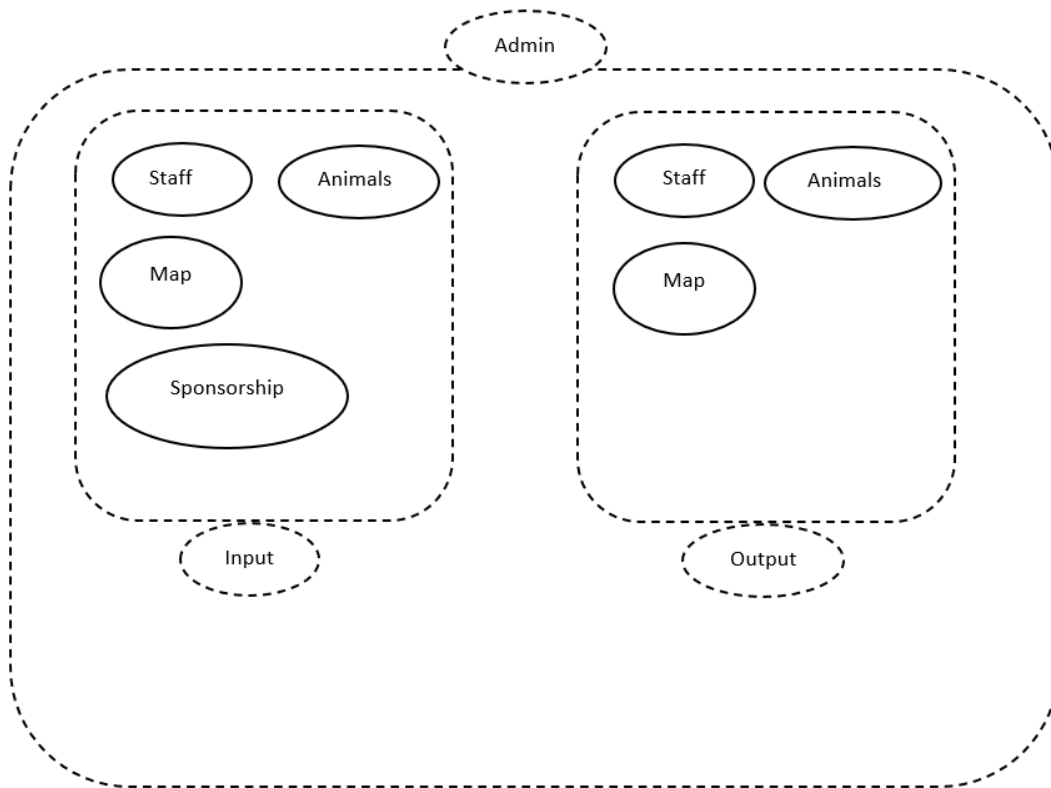
System	<i>Record Management System (RMS)</i>	Part:1/3
Purpose: System keeping record of all the staff, members, visitor's and animals of the zoo		INDEXING Keywords: Animal, staff, Sponsor
Cluster	Description	
Animal_Record	Store the data of all the animals present inside the zoo and the history of the animal	
Staff_Record	Store the data of all the staff of the zoo	
Visitor_Record	Store the data of all the visitors who visited the zoo	
Sponsor_Record	Store the data of all the sponsors of the animals	

System	<i>Zoo Website</i>	Part:2/3
Purpose: System to describe all the features and facilities of the zoo		INDEXING Keywords: About zoo, animal species, Membership
Cluster	Description	
Animals Page	Brief description about all the species of animals	
Events Page	Information about events conducted inside the zoo and upcoming events of the zoo	
Membership Area	Consists the form where members can apply for membership of the zoo	
Sponsorship Area	Consists the form where interested people can apply for the sponsorship	

System	<i>Kiosk System</i>	Part:3/3
Purpose: System to make a visiting experience more entertainment		INDEXING Keywords: digital catalogue, games, ticket
Cluster	Description	
Digital Animal Catalogue	To see the animal's location and animal species found in the zoo	
Quiz Games	For the visitor's entertainment Purpose	
Online Ticketing	To Book the ticket online and buy it through online platform	
Food and Drink	To buy food and drink	

4.2.3. BON Cluster Charts





4.2.4. BON Class Charts

CLASS	MANAGER		Part:1/1
TYPE OF OBJECT A Manager object		INDEXING Created:2021/12/19	
Queries	manager_id, manager_first_name, manager_last_name, contact_number, gender, city, country, date_of_birth		
Commands	Add, Remove, Create, Amend, Archive, Delete, Display, Assign, News, Map, Advertisement, Ticket price, Search		
Constraints	- Administrator has complete control over everything. - Active manager cannot be deleted.		

CLASS	<i>SPONSORS</i>	Part: 1/1
TYPE OF OBJECT Person who sponsors the animals	INDEXING Created: 2021/12/18	
Queries	sponsor_id, sponsor_first_name, sponsor_last_name, contact_number, gender, city, country, date_of_birth	
Commands	Display, News, Map, Advertisement, Ticket price, Search	
Constraints	• Sponsors with valid sponsorship cannot be deleted. • One cannot apply for more than sponsorship of animals. • Sponsorship of sponsors will automatically expire after certain time period.	

CLASS	<i>STAFF</i>	Part: 1/1
TYPE OF OBJECT Person who works inside the zoo	INDEXING Created: 2021/12/15	
Queries	staff_id, staff_first_name, staff_last_name, staff_address, town, county, contact_number, gender, email, date_joined	
Commands	Display, News, Map, Advertisement, Ticket price, Search	
Constraints	• Active staff data cannot be deleted. • Only the zookeeper has access to the watchlist.	

CLASS	VISITORS		Part:1/1
TYPE OF OBJECT Person who visits the zoo		INDEXING Created: 2021/12/20	
Queries	visitor_id, visitor_first_name, visitor_last_name, city, country, house_number, contact_number, gender, date_of_birth		
Commands	Display, News, Map, Advertisement, Ticket price, Search		
Constraints	- Daily visit data of visitors cannot be deleted. - Have access to the location of animals		

5 Report Conclusion

Since the Claybrook Zoo's existing Clerical-based system was causing a lot of problems. Having a paper-based management system in this era, where everything is online and digitalized, would undoubtedly generate a lot of problems, especially when we have to deal with large amounts of data. Our group has attempted their best to develop a system that meets all of the zoo's requirements. We have attempted to incorporate all of the features requested by our clients for their newly based system.

Before beginning the project, we did a lot of research about the background of the zoo and the main goal of the zoo for this project. Since we chose the Project Methodology for our project to complete the project in a sequential way. We interviewed with the key stakeholders of the zoo to find out the problem domain for the system. According to the research with the different key stakeholders of the zoo the main requirement of the zoo was to develop a system that can handle many paperwork electronically, saving both time and money. The Claybrook Zoo does not consist of a website. So, the Zoo wants to develop a zoo website for the visitors and sponsors. The clients wish to use the web system to attract as many visitors and sponsors as possible. As a result, they want the system to be simple to use and informative. We also did a lot of research on the existing Zoo whose marketing was very high. We examined the features and functions of those websites to guarantee that we don't miss out on anything. The client requirements were our first priority and research on other existing websites made us to add more functions and features

for better management system. We did not only comprehend the design of a management system, but we also comprehended its working mechanism.

The disadvantage of paper-based system is that data about animals can be lost or misplaced, and the system is also difficult to manage. The logo and the information to be kept in the Zoo was provided by our clients. In terms of design constraints, the client requested a website with minimal information, a simple homepage, information about events, an online ticketing system, sponsorship, and a registration form. Our clients also suggested that we include a game section for entertainment. The overall estimated budget of this project was 2.5 million pound and the client was also satisfied with cost. We've also defined different laws related to the system itself to prevent any illegal activities. The laws are in place to protect personal information.

The system is built in such a way that it works on a variety of devices. The website is user-friendly, with features such as a search box that allows users to find detailed information about various species of animals existing at the zoo. Next to the search box is a very informative message area. The website is suitable for people of all ages. For the admin, staff and the sponsor the log in accessibility is given. The admin panel can only be accessed by using the correct username and password. There is a rating area where users can rate the zoo and get feedback on how to improve or maintain the zoo facility.

We have attempted our exceptional to consist of all of the capabilities that become required with the aid of using our clients, we're inclined to feature greater capabilities on the future at the gadget if necessary.

6 References

Chester Zoo:

Anon., n.d. *Chester Zoo*. [Online]
Available at: <https://www.chesterzoo.org/>
[Accessed 5 December 2021].

Colchester Zoo:

Anon., n.d. *Colchester Zoo*. [Online]
Available at: <https://www.colchester-zoo.com/>
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Available at: <https://www.gov.uk/data-protection>
[Accessed 3 January 2022].

BON Diagram:

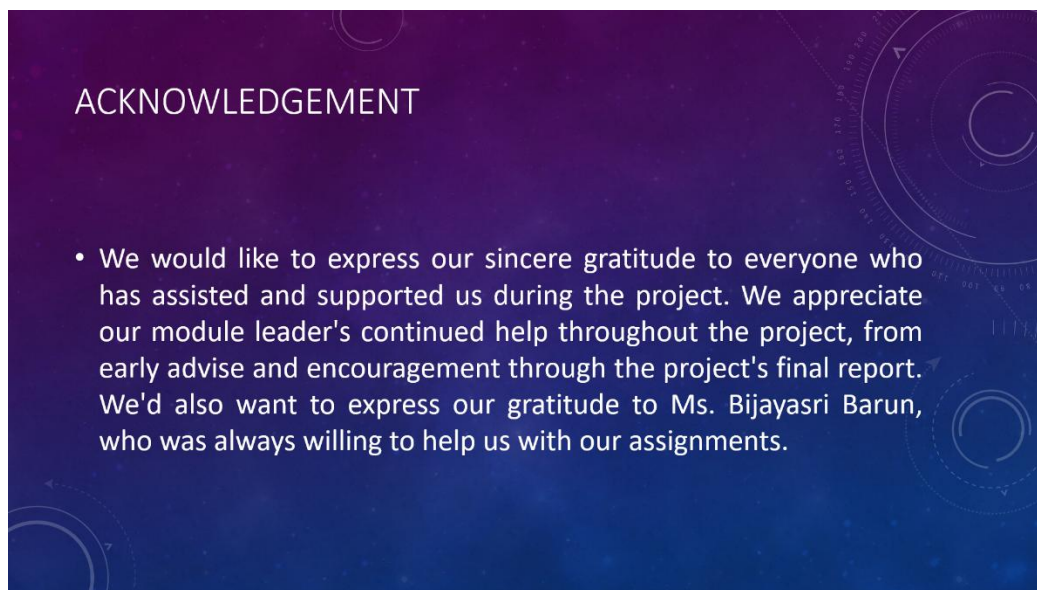
Anon., n.d. [Online]
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Software to develop Mock-up:

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Available at:
https://uizard.io/wireframing/?utm_source=google_traffic_lo-fi_IN&utm_medium=cpc&utm_campaign={Search_traffic_lo-fi}&utm_content=120693266325&utm_term=wireframing%20tools&gclid=CjwKCAiA24SPBhB0EiwAjBgkhkau5TZyeu3EQG0Iy2xQBw16xNCLtv6QdMZpbDEI4O7GRJeR1KsbPBo
[Accessed 27 December 2021]

7 Appendix

This section contains screenshots from our presentation.



INTRODUCTION

- Developing or in need of a new electronic management system for Claybrook zoo
- The current paper-based system has been the obstacle that is preventing the zoo from performing its best.
- The four main stakeholder:
 - Matthew Jones
 - Thomas Smith
 - Jonathan Rodgers
 - Phillip Brown

PROJECT AIMS AND OBJECTIVE

- Provide detail info on animals
- To make user-interactive website
- Attract more visitors and sponsor
- Effective navigation and search bar
- Responsive site
- Highly secured kiosk system

ELICITATION ACTIVITIES



Gather as much data and information about zoo while interviewing them



Collect all the features and functionalities required by them



Compare similar website to not to miss out the important features like

Record management system
Zoo website
Visitor Kiosk Information System



Focus on client requirements

PROBLEM DOMAIN



Maintaining of records and monitoring of animals and workers is a time-consuming and tedious



It is incredibly simple to lose data on animals and other information.



Ticketing is time consuming with no online ticketing facilities



Sharing important document with the fliers is difficult, could be risky.



Updating the record of animal is unmanaged

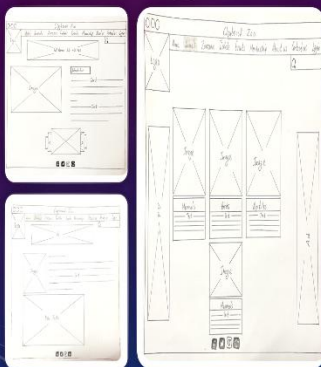
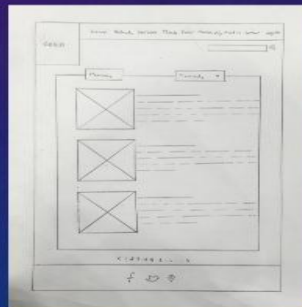
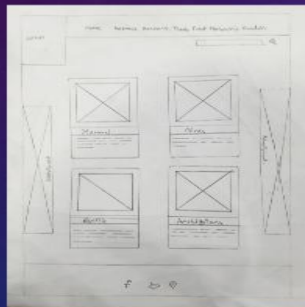
SOLUTION OF PROBLEM DOMAIN

- Easy database management system
- Admin panels for authorized person of the zoo
- Electronized record management system
- Online ticketing and booking system
- Online game for all age group visitors.
- Digital advertisement for sponsors
- User friendly and highly secured website
- Updating information, changing facts and feature ability according to the zoo's personal

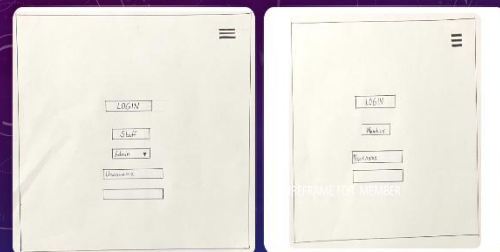
SYSTEM INTERFACE DESIGN

WIREFRAME

- It keeps the concept of focused on user , which clarify and define website features
- It is a screen blueprint, is a visual guide for understanding the layout which is represented by the skeletal framework of a website .



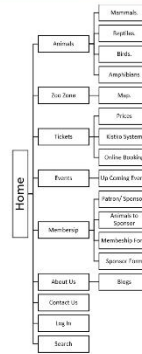
WIREFRAME OF WEBSITE



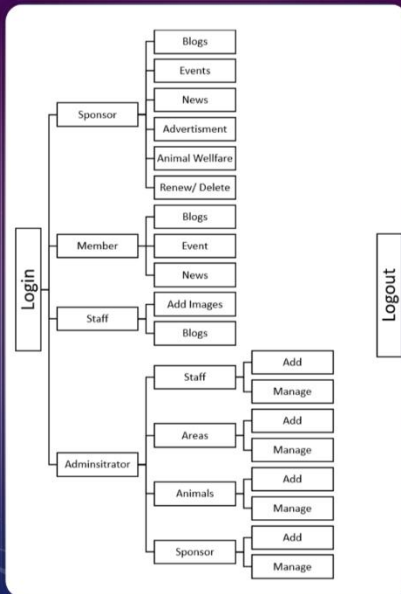
WIREFRAME FOR ADMIN WIREFRAME FOR MEMBER

B). SYSTEM NAVIGATION DIAGRAM

- A system diagram is a visualization of a system as a flow-chart-like diagram.
- With supporting documentation, it captures all the essential information of a system's design.



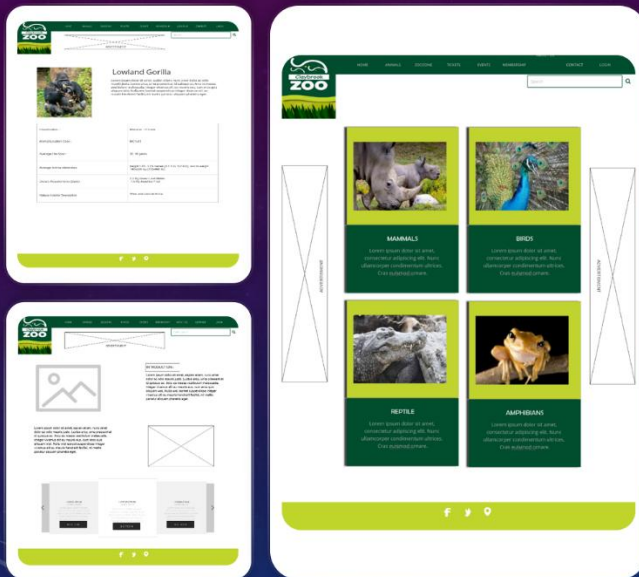
FONT END DIAGRAM (SYSTEM NAVIGATION DIAGRAM FOR VISITORS)



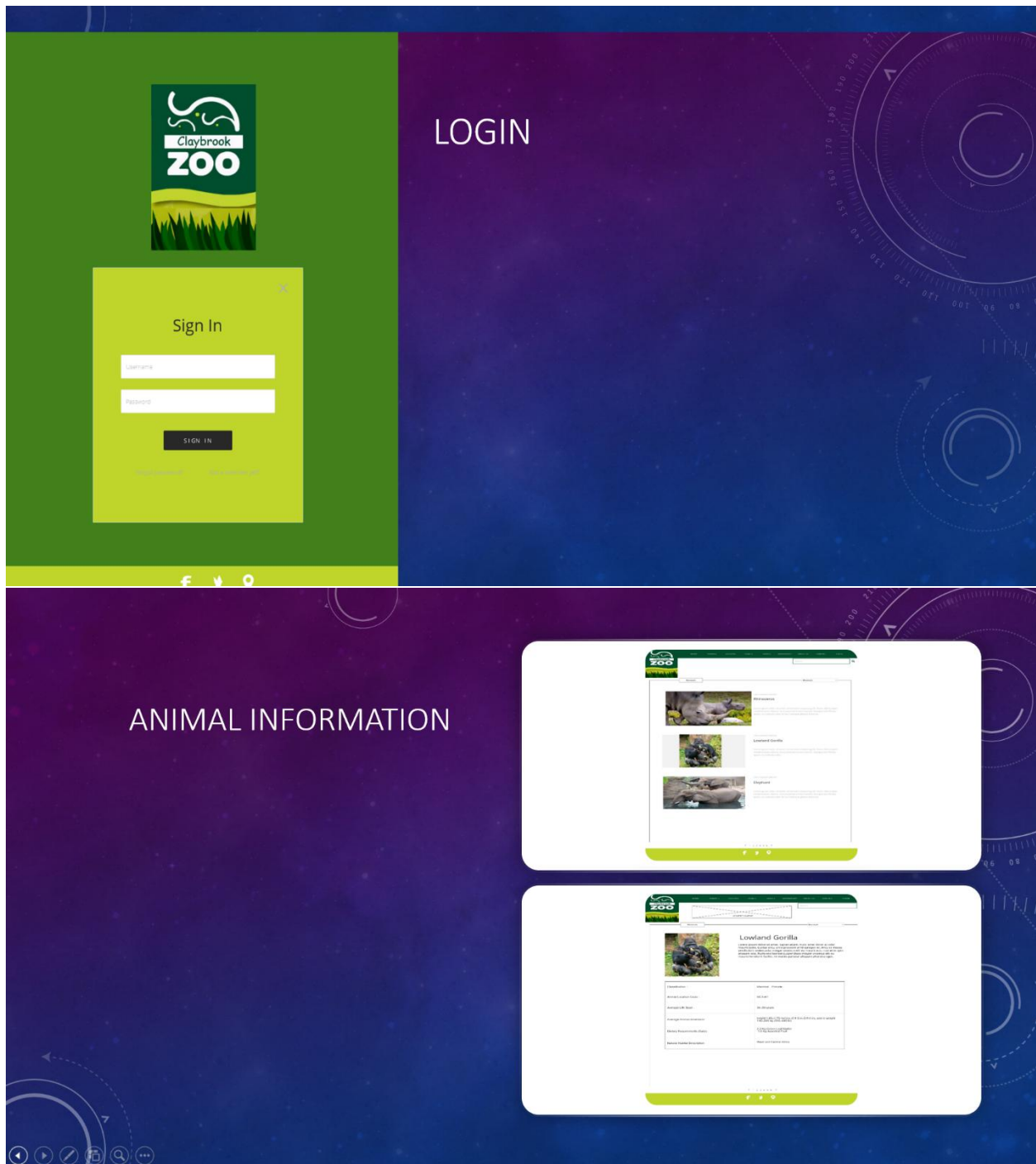
SYSTEM NAVIGATION FOR LOGIN

SCREEN MOCK UP

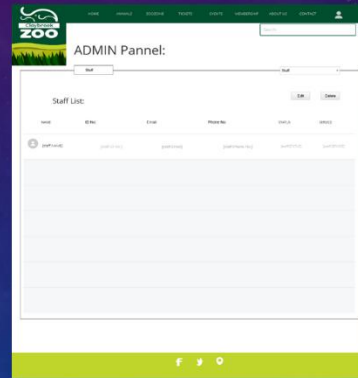
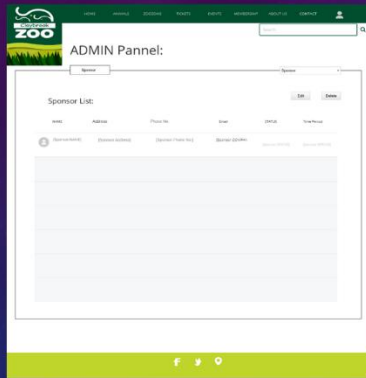
- Wireframe is the skeleton for it.
- It is the final layer(model) of user-interface design.
- Used software JustInMind



HOME PAGE



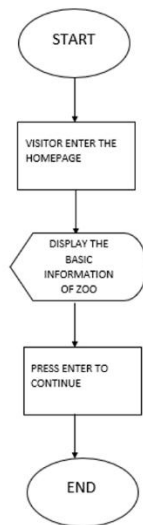
ADMIN PANEL



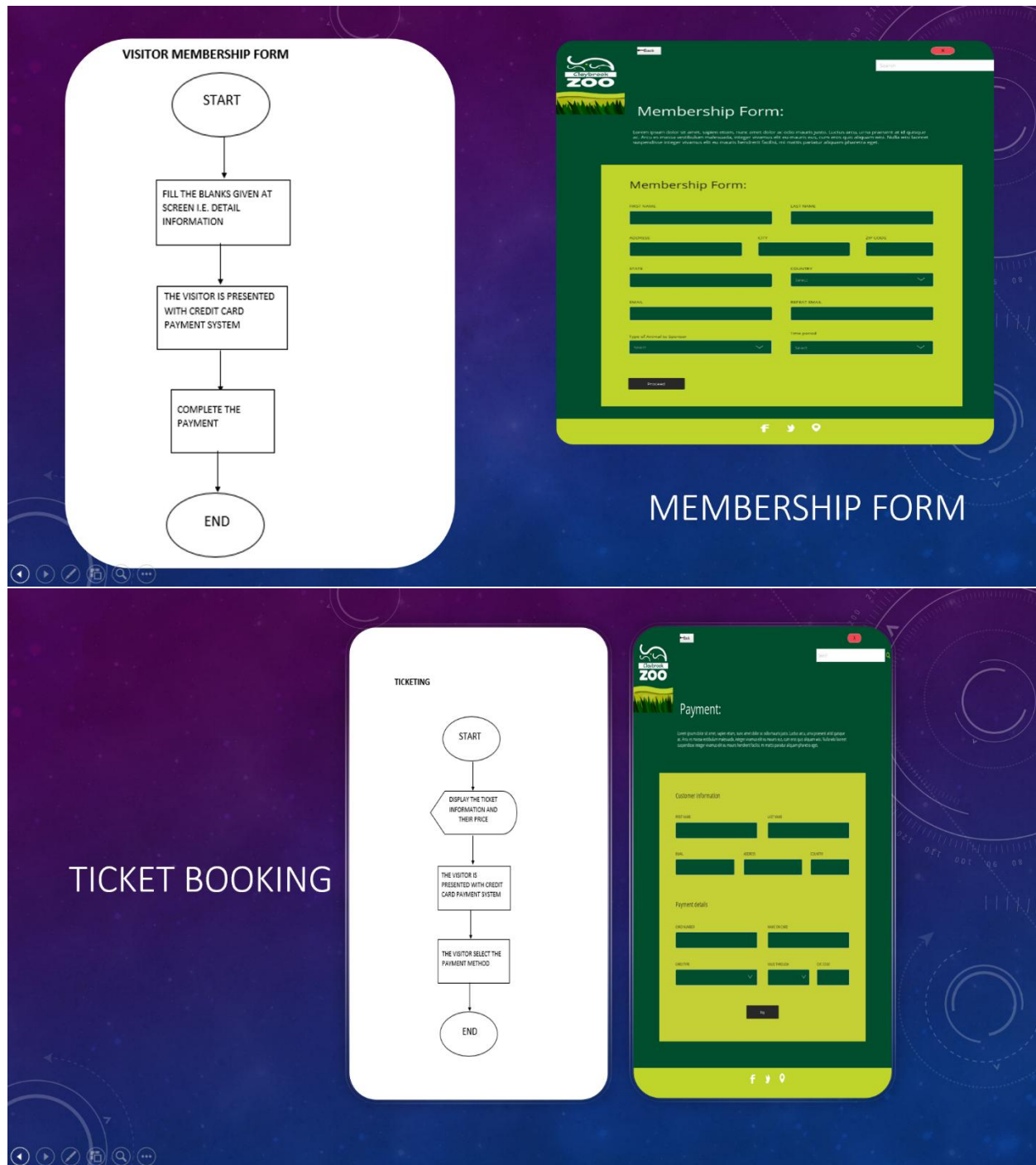
D). SYSTEM ACTIVITY EVENT DIAGRAMS

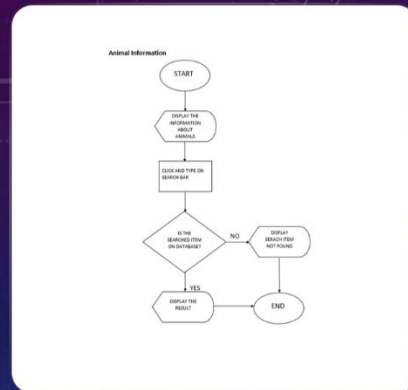
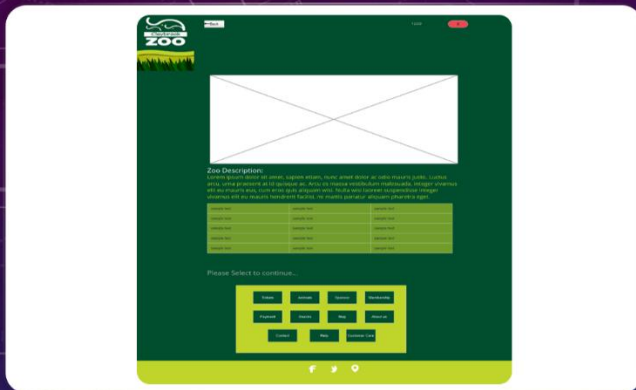
- It depicts the behavior of a system in the form of workflow.
- It portrays the control flow from a start point to a finish point.

Homepage



HOME PAGE





ANIMAL INFORMATION

COSTING OF THE WEBSITE

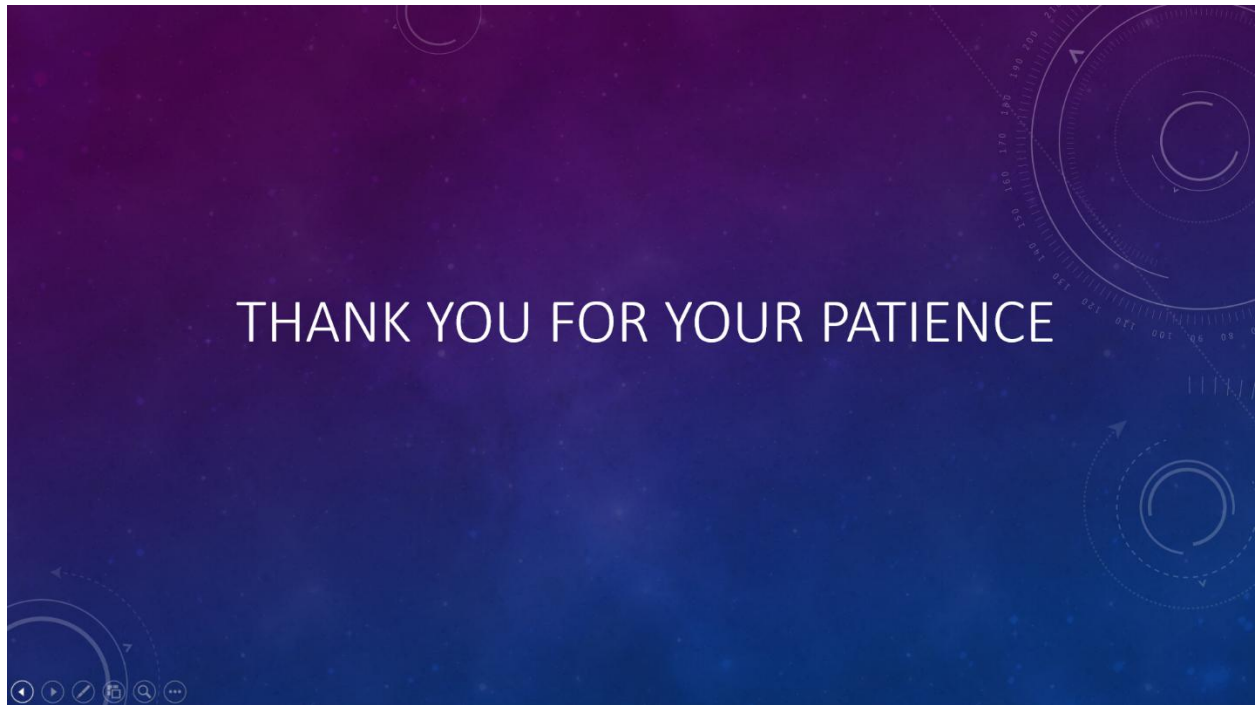
- The website cost: £ 10,567.75 sterling pound
- Testing cost : £ 1080 sterling pound
- The total build cost: £ 1,050,000 sterling pound
- Total project cost: £ 2,500,000 sterling pound

POTENTIAL BENEFITS FOR THE CLIENTS

- Better record management system
- Highly secured system
- Highly advance website and user-friendly
- Better and wide plays for advertisement and promotion of the sponsors
- Could be less time consuming and lower project cost

FUTURE RECOMMENDATION

- Virtual queuing
- Mobile food ordering inside zoo
- Upgrading website login user number
- Different language should be available
- Increase the speed and capacity
- Lastly focus on review and competitors for more advance feature



Presentation video link:

Video Title: Tokyo Manji_SE Assignment_CSY1019

Link:

https://northampton.mediaspace.kaltura.com/media/Tokyo+Manji_SE+Assignment_CSY1019/1_3339gfdg

Group Name	Student Name	ID	Contribution
Tokyo Manji	Sushila Timalina	21422893	2.2 Requirements Specification 2.2.1 Problem Domain Description 2.2.1.1 Existing Business Operation 2.2.1.1.1 Animal Life-Cycle 2.2.1.1.2 Sponsorship Life-Cycle 2.2.1.2 Summary of existing business limitations 2.2.2 Functional Requirement 2.2.2.1 Records Management System 2.2.2.2 Zoo Website 2.2.3.3 Visitor Kiosk Information System
	Rabindra Prasad Joshi	21422047	1. Introduction 1.1 Project Background 1.2 Project Aims and Objectives 1.3 Project Development Methodology 2 Requirements Engineering 2.1 Elicitation Activities 2.1.1 Interview plans 2.1.2 Interview findings 2.1.3 Other problem domain research 2.1.3.1 Comparable Software System Review 2.1.3.1.1 Records Management Systems 2.1.3.1.2 Zoo Websites 2.1.3.1.3 Visitor Kiosk/Information Systems 2.1.3.2 Development Relevant Legislation 4.2.1. First Draft BON System Architecture Diagram 5 Report Conclusion

	Rajiv Shrestha	21422063	Performance Requirements (Visitor Information System) 2.2.3.1 Speed 2.2.3.2 Capacity 2.2.3.3 Reliability 2.2.3.4 Usability 4.2.3. BON Cluster Charts
	Amir Tamang	21422018	3.3 Draft Interface Designs for Zoo Website 3.1.1 Wireframes 3.1.2 System Navigation Diagram 3.1.3 System Screen mock-ups 3.1.4 System Activity Event Diagrams 3.4 Design Revisions
	Sudarshan Thapa	21422045	Performance Requirements (Zoo Website) 2.2.4.1 Speed 2.2.4.2 Capacity 2.2.4.3 Reliability 2.2.4.4 Usability 4.2.4. BON Class Charts
	Pawan Bhusal	21422143	Performance Requirements (Records Management System) 2.2.3.1 Speed 2.2.3.2 Capacity 2.2.3.3 Reliability 2.2.3.4 Usability 4.2.2. BON System Chart
	Sudarshan Oli	21422056	4.1 Preliminary Analysis Stages 4.1.1. Textual Analysis 4.1.2. Significant Event Analysis 4.1.3. Class- Responsibility- Collaborator (CRC)
	Krishna Oli	21422057	3.5 Draft Interface Designs for Visitor Information System 3.1.1 Wireframes 3.1.2 System Navigation Diagram 3.1.3 System Screen mock-ups 3.1.4 System Activity Event Diagrams 3.6 Design Revisions
	Siddharth Yonzon Tamang	21422134	3.1 Draft Interface Designs for RMS 3.1.1 Wireframes 3.1.2 System Navigation Diagram 3.1.3 System Screen mock-ups 3.1.4 System Activity Event Diagrams 3.2 Design Revisions